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(Sultan Bayezid II Complex: One of the Earliest Medical Schools, 1488, Edirne)

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The Strong Get Stronger: The Matthew Effect in Academic Publishing

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The Matthew Effect is Merton's term for the pattern by which initial advantage builds into further advantage, as the saying goes, to those who have, more shall be given.[1] Originally applied to individual scientists, it now operates across journals and institutions, producing paths for well-resourced and under-resourced journals over time (Figure 1). Reputation builds through impact metrics and citation networks. Prestigious journals attract higher-quality submissions, receive more citations, and further elevate their status. In contrast, independent or university-based journals, often equal in rigor but lacking commercial backing, struggle to sustain operations and visibility. This asymmetry transforms academic publishing into a highly concentrated market dominated by a handful of large publishers.

Research from high-income, English-speaking countries receives more citations than comparable work from other nations, reflecting structural advantage rather than content differences.[2] This citation bias deepens disparities between scientific communities. Journals emerging from public universities or regional networks, many operating under diamond open-access models which charge neither authors nor readers, are vital but fragile.[3] Their sustainability depends on volunteer labour and minimal budgets, leaving them vulnerable to the pressures of global competition.

The experiences of independent, university-based medical and behavioural science journals illustrate these challenges. For example, the Balkan Medical Journal, an open-access publication from Türkiye operating under a diamond open-access model, has seen a surge in submissions following its rise in international ranking, despite limited financial or administrative support.[4,5] Similar pressures are reported by comparable non-commercial journals, including escalating submission volumes, static resources, and increasing editorial complexity.[6] Volunteer editors balance peer review, production, and ethics oversight with full-time academic duties, while constrained budgets prevent investment in digital infrastructure, professional copyediting, and advanced manuscript management systems.

These structural advantages are reinforced by three further mechanisms, reviewer scarcity, submission cascades, and technological asymmetry, each of which routes resources and attention disproportionately toward already-advantaged journals.

Peer review has become a systemic bottleneck. The global supply of qualified reviewers has not kept pace with submission growth, producing longer turnaround times and inconsistency in review quality.[7] Under institutional reward systems that offer little recognition for reviewing, academics reasonably prioritise their own outputs, leaving quality control to a small core of unpaid editors. This scarcity is not distributed evenly. Reviewers preferentially accept invitations from prestigious journals, where the reputational return is highest, leaving smaller titles with longer queues and weaker review pools. Slower, less rigorous review further erodes their standing, a direct feedback into the cumulative-advantage loop.

Structural strain is compounded by the "publish or perish" imperative, which encourages volume over substance. Performance metrics tied to publication count encourage authors to submit first to the highest-prestige journal plausibly within reach, moving to lower-ranked journals after rejection. Elite journals therefore enjoy first selection over nearly every competitive manuscript, while independent journals receive work only after it has been filtered through one or more rejections above them. The submission hierarchy itself becomes a sorting mechanism that channels the strongest material toward the already-strong.

The rise of generative artificial intelligence (AI) introduces new inequities into this system.[8] AI tools improve efficiency in writing, translation, and editing, but they also generate risks of undetected plagiarism, fabricated data, and synthetic manuscripts. Because AI-detection tools, integrity pipelines, and AI-assisted production systems are capital-intensive, only well-resourced publishers can deploy them at scale. The same journals that already command prestige now also command the infrastructure to police and exploit AI, accelerating, rather than merely preserving, cumulative advantage.

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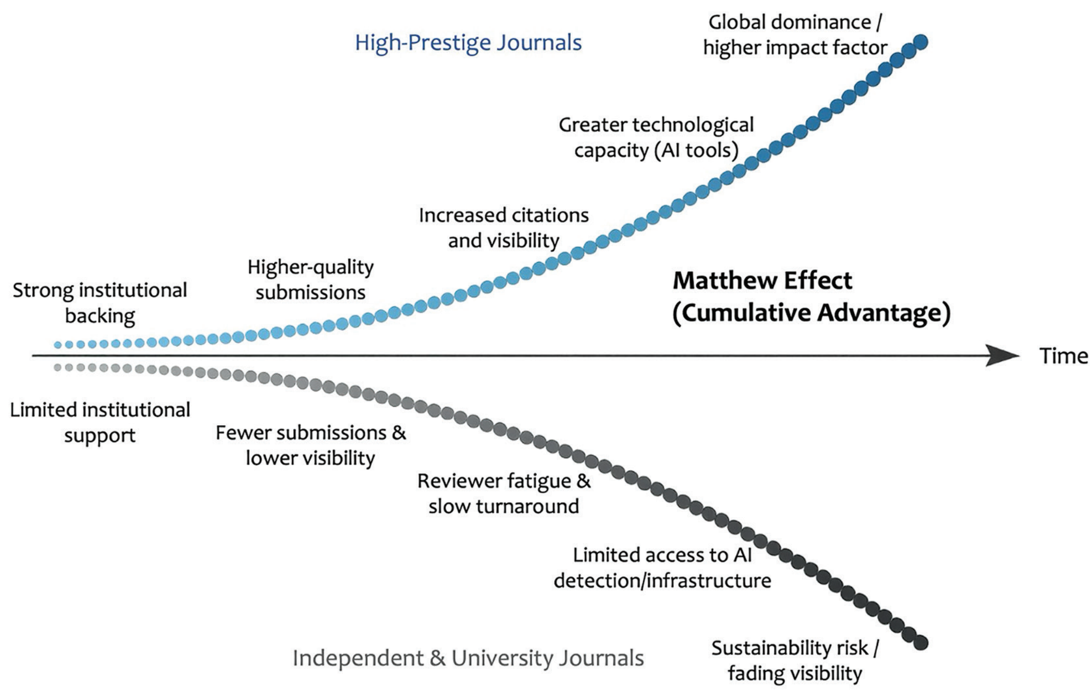

FIG. 1. Cumulative advantage and disadvantage in academic publishing. Over time, high-prestige journals (upper trajectory) accumulate reinforcing advantages, strong institutional backing, higher-quality submissions, rising citations and visibility, greater technological capacity, and global dominance reflected in impact-factor growth. Independent and university-based journals (lower trajectory) face compounding disadvantages, limited institutional support, fewer submissions and lower visibility, reviewer fatigue and slow turnaround, constrained access to AI-detection infrastructure, and ultimately sustainability risk and fading visibility. The diverging curves illustrate the Matthew Effect as two sides of a single structural mechanism. AI, artificial intelligence.

The rise of predatory journals, exploiting the author-pays model to skip peer review while preying on authors' pressure to publish, has eroded trust in open access broadly.[9] The damage spills over unevenly. Legitimate open-access journals share only the label with predatory ones, yet pay a reputational cost they did not cause.[10] Ironically, this has strengthened established commercial publishers. Authors avoiding disreputable venues increasingly favour a small circle of highly recognisable journals. Reputation becomes a proxy for legitimacy, concentrating attention and citation flow toward those already advantaged. These interconnected mechanisms, including citation bias, reviewer fatigue, AI asymmetry, and erosion of trust, together reproduce the Matthew Effect at the system level. Advantage comes not just from excellence but from structural position, such as visibility, language, and access to resources. Journals situated in low- or middle-income contexts, or dependent on public funding, face chronic sustainability challenges despite editorial excellence.

The consequences extend beyond economics. When the spread of scientific knowledge is controlled by a limited number of commercial platforms, scientific diversity becomes restricted. Bibliometric analyses show that citation networks increasingly mirror the research agendas of the Global North, even in fields where the Global South bears the greatest burden of disease or social relevance.[2] The overrepresentation of certain scientific communities distorts global research priorities, shaping what is studied, funded, and regarded as "impactful". The Matthew Effect thus becomes not only a sociological principle but a mechanism of knowledge inequality.

Sustaining a diverse and equitable publishing ecosystem requires coordinated systemic reform. Efforts by individual journals are not enough to counter the global incentives shaping the system. Four interventions could meaningfully disrupt the feedback loop of cumulative advantage.

1. Diamond open-access journals worldwide operate under similar constraints, volunteer editorial labour, minimal budgets, limited technical capacity, yet each reinvents the same compliance infrastructure in isolation. Coordinated through bodies such as the Directory of Open Access Journals (DOAJ), the Open Access Scholarly Publishing Association (OASPA), and the Committee on Publication Ethics (COPE), a Diamond Open-Access Integrity Alliance could share a cross-journal reviewer registry, negotiate collective access to AI-detection and manuscript-screening systems, and harmonise editorial and ethics standards. Collective scale would give participating journals the technological footing of the largest commercial publishers at a fraction of the cost.
2. Reviewing activity is already recorded through ORCID and Web of Science Reviewer Recognition but is rarely weighted in hiring, promotion, or tenure. National research councils, university consortia, and evaluation frameworks aligned with the

Declaration on Research Assessment (DORA) should mandate that verified peer-review contributions count in academic evaluation alongside publications and grants. Recognition that follows the reviewer across institutions would stop reviewers from favouring only top journals and help rebuild the reviewer pool that every journal depends on.

3. Major research funders, including Wellcome, cOAlition S signatories, the European Research Council, the National Institutes of Health, and UNESCO's Open Science programme, should set aside a small, defined fraction of their budgets for the operating costs of diamond open-access journals. Predictable, multi-year funding is more valuable than one-time grants, because it permits editorial planning and investment in infrastructure rather than constant crisis management.

4. Journals do not have to wait for evaluation systems to change. They can publish a short set of indicators with every article, covering reproducibility, data and code availability, and clinical or societal relevance, so that prestige reflects the work itself, not the journal.[11]

These measures would not eliminate inequality, but they would slow its cumulative acceleration and maintain pluralism in how science is produced and shared.

The current asymmetry of academic publishing is not a natural by-product of excellence. It is a behavioural and institutional construct. Addressing it demands collective responsibility across the research ecosystem. Sustaining independent, open-access journals is not an act of charity, it is an investment in the resilience of global science.

Without deliberate correction, the Matthew Effect will continue to widen disparities. The strong will grow stronger, while voices outside dominant networks will fade. Recognising and mitigating this dynamic is essential not only for fairness but for the intellectual integrity of science itself.

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Contrast Media Side Effects and Kounis Syndrome: Timeo Danaos et Dona Ferentes

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Kounis syndrome is defined as the occurrence of acute coronary events in the setting of allergic, hypersensitivity, or anaphylactic reactions. It is mediated by mast cell activation and the interaction of inflammatory cells, including T lymphocytes and macrophages. This process leads to the release of multiple inflammatory mediators, such as platelet-activating factor, histamine, neutral proteases (tryptase and chymase), arachidonic acid metabolites, cytokines, and chemokines. Kounis syndrome represents a unique form of acute vascular disorder that may involve not only the coronary arteries but also peripheral, cerebral, and mesenteric vessels as well as the venous system. Contrast media are widely used in diagnostic imaging to enhance visualization and characterization of pathological conditions. These agents can be administered via several routes, including oral, intravenous, intra-arterial, or rectal administration. Although most hypersensitivity reactions to contrast media are mild to moderate, severe complications such as anaphylaxis, cardiac arrest, and Kounis syndrome may occur. In particular, contrast media-induced Kounis syndrome has been associated with significant clinical consequences, including an increased risk of life-threatening cardiac events.

This narrative review aims to summarize current evidence regarding contrast media-related adverse effects, with a focus on hypersensitivity reactions, Kounis syndrome, and associated cardiovascular complications. Emphasis is also placed on preventive strategies, including the importance of obtaining a detailed patient history of prior hypersensitivity reactions prior to contrast administration, to reduce the risk of recurrence and severe outcomes.

INTRODUCTION

Contrast media are widely used in modern medical imaging to enhance differentiation between normal and pathological tissues, thereby improving diagnostic accuracy. These agents are administered prior to or during imaging procedures and play a critical role in modalities such as computed tomography (CT), magnetic resonance imaging (MRI), and ultrasound.[1] By altering the radiographic or magnetic properties of specific tissues, contrast agents facilitate visualization of anatomical structures and pathological processes that would otherwise be difficult to detect. Depending on the imaging modality, different types of contrast media are used, including iodinated agents and barium sulfate for X-ray and CT imaging, gadolinium-based agents for MRI, and microbubble or gas-based agents for ultrasound applications.[2] Contrast media can be administered via various routes, including oral, intravenous, intra-arterial, and rectal administration, depending on the clinical indication and the target anatomical region. While generally considered safe, their use is associated with a spectrum of adverse reactions, ranging from mild hypersensitivity responses to severe and potentially life-threatening complications. Among these, cardiovascular manifestations such as hypersensitivity-related cardiac dysfunction and Kounis syndrome have received increasing attention. Kounis syndrome represents a complex interaction between allergic reactions and acute coronary events, posing significant diagnostic and therapeutic challenges.

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This narrative review summarizes current evidence on cardiovascular hypersensitivity reactions, cardiac dysfunction, and Kounis syndrome associated with contrast media administration, with emphasis on their pathophysiology, clinical implications, and preventive strategies.

MATERIALS AND METHODS

We conducted a literature search on the PubMed, MEDLINE, Embase databases, and Google Scholar and updated it on 28 December 2025 with the keywords "contrast media," "contrast agents," "contrast chemicals," "contrast compounds," "contrast drugs," "contrast-enhanced," "contrast dye," "contrast reactions," "contrast materials," "allergy," "anaphylaxis," and "Kounis syndrome." Moreover, we searched the "brand iodine names": Xenetix, Bonorex/Omnipaque, Iomeron, Iopamiro, Ultravist, Optiray, Lugol's, Xymodine, Ioversol, Iopromide, Iopamidol, Iomeprol, Iobitridol, iodixanol, iodine tincture, iodine mild, iodine tincture decolorized, diatrizoate, strong iodine, and Ultravist; the "brand gadolinium names": Gadavist (gadobutrol), Dotarem (gadoterate meglumine), ProHance (gadoteridol), MultiHance (gadobenate dimeglumine), Eovist/Primovist (gadoxetate disodium), and Elucirem/Vueway (gadopiclenol); the "brand barium sulfate names": Entero Vu, E-Z-HD, NeuLumEx, Readi-Cat 2, Varibar Honey, Varibar Nectar, Varibar Pudding, Varibar Thin Honey, Varibar Thin Liquid, VoLumen, and Tagitol V; and the "brand microbubbles names": Definity, SonoVue, Optison, Lumasol, Luminity, perflutren, YJ Nozzle, New Spinor, and Eco-Bubble. To investigate the relationship between contrast media side effects and Kounis syndrome, a narrative review was carried out. To find pertinent publications published up to the end of January 2026, a literature search was conducted using the PubMed, Scopus, and Web of Science databases. Initially, 150 records were found. Fifty articles were eliminated for lack of relevance after titles and abstracts were screened. The eligibility of the remaining 100 articles was evaluated in full text. Fifteen of these were eliminated because they lacked clinical evidence or were not directly related to the subject. Although a rigorous systematic review technique was not strictly adhered to because this is a narrative review, attempts were made to guarantee a thorough and organized literature selection process.

CONTRAST MEDIA SUBSTANCES

There are several types of contrast media imaging, each appropriate for specific imaging techniques and diagnostic needs:

Iodine-based contrast is primarily used for X-ray and CT scans. Administered intravenously, orally, or rectally, this contrast can highlight blood vessels, kidneys, urinary tract, and the gastrointestinal tract.

Gadolinium-based contrast is employed in MRI. It is a rare-earth metal and is utilized in MRI contrast agents. Administered intravenously, it enhances the visibility of blood vessels, tumors, inflammation, and other lesions in various organs and soft tissues.

Barium sulfate contrast media are used for X-ray imaging of the gastrointestinal tract (esophagus, stomach, intestines). It is typically ingested or administered rectally.

Microbubble contrast media is used. Administered intravenously, these tiny gas bubbles enhance the visualization of blood flow in organs like the heart, liver, and kidneys.

Nanoparticles extend circulation duration, enhance bioavailability at tumor sites, and shield imaging agents from immune clearance. As contrast agents in modalities such as CT, optical imaging, MRI, and ultrasound, nanoparticles are revolutionizing biomedical imaging. They extend circulation duration, enhance bioavailability at tumor sites, and shield imaging agents from immune clearance. This leads to increased imaging sensitivity.

The contrast media categorized as ionic, non-ionic, and iodine-based agents are classified as low-osmolar and non-ionic based on their properties, such as osmolarity. The most frequently used agents are gadolinium for MRI, iodine-based contrast for CT/X-ray, and barium for gastrointestinal tract imaging. Negative agents, such as air, can also be utilized for X-ray/CT. Contrast media include substances such as iodine solutions or barium sulfate suspensions that are relatively opaque to X-rays and are injected or swallowed into the body to contrast an internal part such as the kidneys, blood vessels, or gastrointestinal tract with its surrounding tissue in radiographic visualization.[3] In medical imaging, contrast media are used to enhance the contrast of bodily fluids or structures. Unlike radiopharmaceuticals, which release radiation, contrast media absorb or modify environmental electromagnetic or ultrasound signals. Contrast compounds in X-ray imaging increase a target tissue or structure's radiodensity. Contrast drugs reduce, or sometimes lengthen, the relaxation periods of nuclei within bodily tissues in MRI to change the contrast in the picture.[4]

Since iodine is the radiopaque component of all iodinated contrast agents, the concentration of iodine in these agents determines the radiopacity resulting from their administration. The most common method of administering iodinated contrast agents is intravascular injection; however, due to the capillary permeability of contrast molecules, the material rapidly shifts to the extravascular area.[5]

Iodinated substances are occasionally used as rectal or oral contrast agents to improve imaging of the bowel and stomach. Nonetheless, the most common contrast agent used for gastrointestinal imaging is barium sulfate. The patient either receives this solution as an enema into the rectum or consumes it orally.

Certain forms of contrast media have been proven to be effective for various imaging methods. Each contrast media procedure operates at the suspected condition's natural site. For example, oral contrast media examinations may focus on the stomach or oesophagus, while rectal contrast media examinations highlight the colon or small intestine.[6] The contrast media are employed in the following manner, locations, and techniques:

Oral contrast media: Barium sulfate-containing contrast agents can be administered orally by swallowing for gastrointestinal tract CT, fluoroscopy, and X-ray imaging. Barium sulfate is sometimes replaced by iodine-based contrast media. The most common method of administering oral contrast agents is still barium sulfate, which comes in a variety of forms, including powder, liquid, paste, and pills.[7] The pharynx, oesophagus, stomach, small intestine, and large intestine (colon) are among the areas where oral contrast agents are used.

Intravenous contrast media: For intravenous injections used in CT, fluoroscopic, and X-ray imaging, contrast media are iodine-based. Gadolinium injections are used by radiologists for MRI. By altering the magnetic properties of water molecules, gadolinium increases the rate at which protons realign with the magnetic field. Thus, the image appears brighter when the proton moves more quickly. For CT scans, intravenous contrast is the most commonly used agent. In some situations, speed is necessary for the bolus to remain compact.[8] Intravenous contrast media injections are intended to highlight the following bodily parts: internal organs (brain, breasts, heart, lungs, liver, adrenal glands, kidneys, gallbladder, pancreas, uterus, and spleen), gastrointestinal system (small intestine, large intestine, and stomach), arteries and veins (brain vessels, neck, chest, abdomen, pelvis, and legs), and soft tissues (skin, muscle, and fat).

Rectal contrast media: For rectal contrast media, barium sulfate is used in an enema. To highlight the lower gastrointestinal tract and differentiate the colon and rectum regions of the body, this contrast media technique is used in X-ray, fluoroscopy, and CT scans. In some situations, iodine-based contrast agents may be used for the same purpose. Since the majority of the water-soluble contrast agents used in this procedure contain barium, rectal contrast media solutions are frequently referred to as barium enemas.[9]

SIDE EFFECTS OF CONTRAST MEDIA

Contrast media-related adverse responses can range from mild, such as nausea, a metallic taste, or skin redness, to serious, including severe allergic reactions, renal damage, or cardiac problems; these reactions typically occur during the first sixty minutes following treatment. Although these agents are often benign, there are risks associated with their use, such as thyroid disorders, contrast-induced kidney problems, and nephrogenic systemic fibrosis (NSF) from gadolinium-based drugs, as presented in Table 1. Skin rashes may develop considerably later, even up to 7 days following the treatment.[10]

a) Iodine-containing contrast can trigger adverse reactions in individuals, presenting a spectrum from mild to severe. Many people may experience mild skin irritation a few hours after the contrast agent is administered. If a patient encounters severe symptoms, seeking medical advice is crucial, as intervention may be needed. In the realm of mild reactions, symptoms may include nausea, headache, itching, skin redness, and mild skin eruptions. More severe side effects may include major skin eruptions, bronchospasm, irregular heart rate, hypertension or hypotension, or even breathlessness. Contrast agents can lead to dangerous reactions, manifesting as labored breathing, severe hypotension, bronchospasm, seizures, and Kounis syndrome.[11]

b) Gadolinium-based contrast adverse events are rather uncommon, occurring in 0.04-0.3% of administrations, of which 0.4-9% are severe. Injection-site discomfort, headache, nausea, and dizziness can occur in less than 2.5% of instances. Both acute and persistent adverse effects are possible. Patients who have previously experienced a hypersensitivity episode to gadolinium contrast agents have a 30% chance of experiencing another hypersensitivity episode. Gadolinium may cause NSF in some patients with kidney disease who undergo MRI. NSF is characterized by thickening and scarring of connective tissue. Although this is a rare complication, patients with severe kidney disease should avoid gadolinium-based contrast agents. Additionally, Kounis syndrome can be induced.[12]

c) Barium sulfate contrast can induce stomach discomfort, loose stools, nausea, vomiting, and difficulty passing stools. Should these manifestations intensify or persist, or if itching, skin redness, throat swelling, dyspnea, difficulty swallowing, hoarseness, restlessness, disorientation, elevated heart rate, or unusually light-colored skin develop, prompt medical attention is necessary. Individuals who have previously experienced conditions such as asthma, allergic reactions, cystic fibrosis, significant fluid loss due to constipation or other causes, bowel obstruction, or bowel rupture may also have adverse reactions. These individuals face an increased likelihood of experiencing unfavorable effects.[13]

d) Microbubble contrast agents are typically safe for use, although they can uncommonly result in adverse reactions ranging from minor issues such as headache, nausea, and lightheadedness to critical cardiac and pulmonary events. The majority of these reactions resolve spontaneously, but in extreme instances, individuals might experience anaphylactic shock, chest discomfort, arrhythmias, or Kounis syndrome.[14,15] In case of any adverse effects during treatment, it appears that the entire body's regions and structures are under the influence of the contrast agents being utilized. Tiny gas bubbles are injected into microbubble contrast media and are stabilized by a supporting shell. After injection of the microbubble contrast agent into the bloodstream, ultrasonography equipment is used to differentiate between the surrounding tissue and the gas bubbles. As a result, the procedure produces a clearly contrasted ultrasound image. Both targeted and untargeted microbubble contrast agents are available. The bubbles in targeted microbubble contrast agents adhere to a particular part of the body where molecules are bound to the gas surface.[16]

Nanoparticle agents have the potential to be novel agents for therapeutic and targeted imaging applications, which are currently the focus of extensive research. These include gold nanoparticles and superparamagnetic iron oxide nanoparticles that are designed to enhance MRI clarity, safety, and targeted distribution.[17] While nanoparticles provide X-ray attenuation and the possibility of targeted molecular imaging, superparamagnetic iron oxide nanoparticles work by causing T2 relaxation shortening for image contrast, providing an alternative or complementary method. Compared with conventional contrast agents, these agents offer superior safety, specificity, and multifunctionality.[18] Although anaphylaxis and Kounis syndrome have not been linked to nanoparticles used as contrast material, lipid nanoparticle-based coronavirus disease 2019 (COVID-19) vaccines and severe acute respiratory syndrome coronavirus 2 lipid-nanoparticle-based mRNA vaccination have been associated with allergic reactions and anaphylaxis.[19] It has been shown that there is an etiopathogenetic relationship between Kounis syndrome type I and COVID-19, associated with cytokine storm.[20]

TABLE 1. Iodine, Gadolinium, Barium Sulfate, Microbubble, and Nanoparticle Side Effects.

Iodine	1. Headaches, bloating, stomach pain, nausea, vomiting, and diarrhea. 2. Renal impairment, including anuria or reduced urine output after the use of contrast agents. 3. Skin eruptions, arrhythmias, dyspnea, and bronchospasm. 4. Hypersensitivity reactions, including skin rashes and itching, to severe reactions such as potentially fatal anaphylactic shock, Kounis syndrome, and rarely hyperthyroidism.
Gadolinium	1. Future gadolinium reactions are eight times more likely after a prior reaction. 2. Digestive: nausea, vomiting, and abdominal pain. 3. Neurological: paresthesia, headache, and vertigo. 4. Injection site reactions: discomfort, pain, warmth, or coldness. 5. Skin reactions: itching, rash, and urticaria. 6. Sensory/Other: fatigue, throat discomfort, and metallic taste. **Uncommon and serious adverse events** 1. NSF: An uncommon and serious disorder that mainly affects patients with severe kidney disease, causing skin thickening and joint contractures. The risk of NSF is significantly increased by impaired renal function. 2. Severe allergic reactions: About 1 in 5,000 people experience symptoms similar to anaphylaxis, such as bronchospasm, laryngospasm, or facial edema. Individuals with allergies or asthma are more vulnerable to acute hypersensitivity reactions and Kounis syndrome. 3. Gadolinium retention/toxicity: long-term deposition in the brain and bones, with unknown clinical consequences and unclear clinical implications.
Microbubbles	1. Common reactions are mild and include flushing, injection site reactions, headache, dizziness, and numbness. 2. Anaphylactic shock, back pain, chest pain, nausea and vomiting, skin allergy, and Kounis syndrome.
Barium sulfates	1. Confusion, coughing or hematemesis, dizziness, hives or welts, itching, skin rash, chest tightness, tachyarrhythmias, bradyarrhythmias, and Kounis syndrome. 2. Aspiration pneumonitis, barium impaction, melena, granuloma formation, hematuria, intravasation, embolization, and peritonitis following intestinal perforation, vasovagal and syncopal episodes. 3. Bloating, constipation, cramping, stomach pain, dyspnea, noisy breathing, and syncope. 4. Anxiety, blurred vision, bruising, cough, postural hypotension, persistent bleeding or oozing from puncture sites (mouth or nose), skin reactions, sweating, and unusual tiredness or weakness.
Nanoparticles	1. Decreased cytoskeletal disruption, decreased macrophage mobility, decreased phagocytosis, and decreased macrophage function. 2. Pro-inflammatory activity increase and the production of mediators such as cytokines 3. Negative impacts on vascular homeostasis and cardiac functioning.

NSF, nephrogenic systemic fibrosis.

CONTRAST MEDIA ANAPHYLAXIS AND KOUNIS SYNDROME

Cardiovascular symptoms associated with anaphylaxis, anaphylactoid reactions, allergy, or hypersensitivity reactions were initially categorized as acute carditis, morphologic cardiac reactions, or rheumatic carditis of unknown origin.[21] A unique type of acute vascular disease, Kounis syndrome, impacts the venous, cerebral, mesenteric, peripheral, and coronary systems.[21] Currently, Kounis syndrome is used to characterize coronary symptoms associated with conditions involving mast cell activation and interactions between inflammatory cells, such as T-lymphocytes and macrophages, which further cause allergic, hypersensitivity, anaphylactic reactions, or anaphylactic shock Figure 1. Among the inflammatory mediators that induce Kounis syndrome are platelet-activating factor, histamine, neutral proteases such as tryptase and chymase, products of arachidonic acid, and a variety of cytokines and chemokines released during the activation process.[21,22] Moreover, acute allergic reactions caused by contrast media, which act as a rapid and intense physical stressor, might cause Takotsubo syndrome.[23,24]

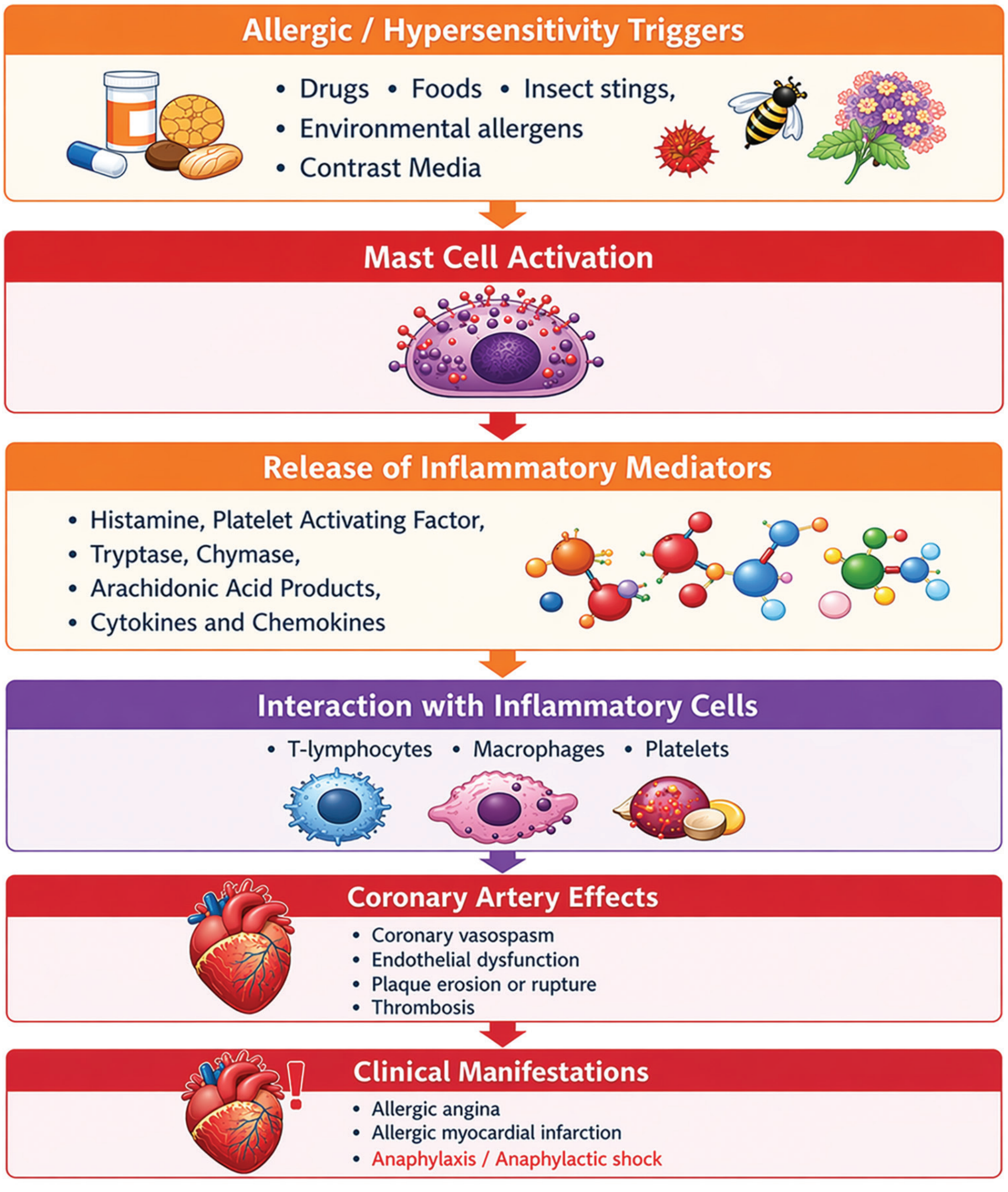

FIG. 1. The pathophysiological mechanism of Kounis syndrome with a summary of causes including contrast media, mast cell activation, inflammatory mediators, coronary artery effects and clinical manifestations.

Recent research[25] indicates that the prevalence of Kounis syndrome in those who have experienced an allergic, hypersensitive, anaphylactic, or anaphylactoid insult ranges from 1.1% to 3.4%. The National Inpatient Sample was used from 2007 to 2014 to evaluate baseline demographics and comorbidities with Kounis syndrome among patients mostly admitted for anaphylactic, hypersensitive, or allergic reactions.[26]

TABLE 2. Basic Details on Patients with Kounis Syndrome Caused by Iodines.

Country	First author	Sex/age	Trigger substance	Ref. No.	Administration
Japan	Tomida et al.	Male, 74 y	Iopamidol	**27**	Intravenous (type III KS)
Greece	Kogias et al.	Male, 48 y	Ultravist	**28**	Intravenous (during excretory urography)
Switzerland	Zmolik et al.	Female, 60 y	Ultravist	**29**	Intravenous (during CT scan → type I Kounis)
Japan	Kurokawa et al.	Male, 77 y	Iopamidol	**30**	Intravenous (during CT → cardiac arrest; subsequent CAG showed coronary spasm (type I Kounis), resolved with intracoronary nitrates)
France	Darwish et al.	Male, 57 y	Visipaque	**31**	Intra-arterial during coronary angioplasty → type II Kounis (anaphylactic shock + diffuse vasospasm)
Romania	Benchea et al.	Male, 68 y	Iodine	**32**	Intra-coronary (during angiography→asystole, coronary spasm)
South Korea	Seong et al.	Pediatric patients (≤ 18 y, n = 13,172)	Xenetix, Bonorex/ Omnipaque, Iomeron, Iopamiro, Ultravist, Optiray	**33**	Intravenous (during contrast-enhanced CT)
China	Sun and Zhang	Male, 59 y	Iodophorol	**34**	Intra-coronary (during coronary angiography → type II Kounis with cardiogenic shock)
Japan	Okuda et al.	Male, 60 y	Iopamidol	**35**	Intravenous during contrast-enhanced CT → type III Kounis (stent thrombosis)
USA	Nasrollahi et al.	Male, 63 y	Iodine	**36**	Intravenous (during contrast-enhanced CT → type II Kounis)
USA	Singh et al.	Male, 54 y	Omnipaque	**37**	Intravenous [during contrast-enhanced CT → type II Kounis (NSTEMI)]
Japan	Nagasawa et al.	Female, 73 y	Iopamidol	**38**	Intravenous (during contrast-enhanced CT → possible Kounis syndrome with anaphylactic shock)
China	Wang et al.	Male, 50 y	Ioversol	**39**	Intra-coronary (during coronary angiography → type II Kounis with cardiogenic shock)
Malaysia	Wong and Ahmad Hatib	Male, 56 y	Ultravist®	**40**	Intravenous (during CT coronary angiography → type II Kounis with ST-elevation and shock)
Switzerland	Bonnet et al.	Male, 46 y	Iodine	**41**	Intra-coronary [during PCI → type III Kounis (acute stent thrombosis + coronary spasm)]

TABLE 2. Continued.

Country	First author	Sex/age	Trigger substance	Ref. No.	Administration
Multinational (Asia/Europe/USA/Chile/Australia)	Wang et al.	7 males/1 female	Iodinated Contrast Media (8)	**42**	5 IV / 3 intraarterial
		Male, 48 y	Ultravist		Intravenous
		Male, 79 y	Visipaque		Intraarterial
		Male, 62 y	Iohexol		Intravenous
		Male, 71 y	Iodine or dextran-40		Intraarterial
		Male, 74 y	Iomeron		Intravenous
		Male, 59 y	Visipaque		Intraarterial
		Male, 78 y	Iohexol		Intravenous
		Female, 72 y	Iopromide gadoterate		Intravenous
China	Wang and Zhang	Male, 63 y	Iopromide	**43**	Intravenous (during coronary CT angiography → Kounis syndrome following anaphylactic shock)
USA	Prisco et al.	Female, 53 y	Iodine	**44**	Intra-coronary (during coronary angiography (FFR) → type I Kounis with cardiac arrest)
USA	Shahab and Sangha	Male, 66 y	Omnipaque	**45**	Intravenous (during contrast-enhanced imaging → type II Kounis syndrome)
Poland	Dorniak et al.	Female, 59 y	Iodine	**46**	Intravenous (during coronary CT angiography → type I Kounis (vasospasm with severe LVOT obstruction)
Tain	Chien et al.	Female, 81 y	Iodine	**47**	Intravenous (during contrast-enhanced CT → type I Kounis syndrome)
Japan	Shibuya et al.	Male, 60 y	Iopamiron	**48**	Intravenous [during contrast-enhanced CT → type III Kounis (stent thrombosis)]
Australia	Bhaskaran et al.	Male, 83 y	Omnipaque	**49**	Intra-coronary (during coronary angiography → type II Kounis)
China	Zhang et al.	Female, 77 y	Iodixanol	**50**	Intra-arterial during coronary angiography → type II Kounis syndrome
China	Xu et al.	Male, 38 y	Visipaque	**51**	Intravenous (during CT → delayed type I Kounis syndrome)

y, years old; IV, intravenous; CT, computed tomography; CAG, coronary angiography; KS, Kounis syndrome; STEMI, ST- elevation myocardial infarction; PCI, percutaneous coronary intervention; DES, drug eluting stent; MRI, magnetic resonance imaging; NSTEMI, non-ST-elevation myocardial infarction; CM, contrast media; GBCAs, Gadolinium-based contrast agents; SF, sulfur hexafluoride; FFR, fractional flow reserve; LVOT, left ventricular outflow tract; OCT, optical coherence tomography.

TABLE 3. Basic Details on Patients with Kounis Syndrome Caused by Gadolinium.

Country	First author	Sex/age	Trigger substance	Ref. No.	Administration
Croatia	Zlojtro et al.	Female, 46 y	Dotarem®	**52**	Intravenous (during brain MRI → type I Kounis syndrome)
Mexico	Macías et al.	Female, 57 y	Gadovist	**53**	Intravenous (during MRI → anaphylaxis leading to Kounis syndrome)
China	Wang et al.	Mixed cases: 4 cases (induced by Gadolinium)	Gadolinium-based contrast agent (4 cases) + 5 cases (unknown contrast media induced)	**42**	Intravenous (administration during imaging procedures)
		Male, 57 y	MultiHance		Intravenous
		Female, 46 y	Dotarem		Intravenous
		Male, 30 y	MultiHance		Intravenous
		Male, 45 y	Dotarem		Intravenous
USA	Abusnina et al.	Female, 53 y	Gadolinium	**54**	Intravenous (brain MRI → type I variant of Kounis syndrome)
Japan	Tanaka et al.	Male, 78 y	Magnevist	**55**	Intravenous (during MRI → ST-segment elevation)
USA	Amro et al.	Female, 52 y	MultiHance	**56**	Intravenous (during brain MRI → Kounis syndrome)
France	Demoulin et al.	(sex not specified), 81 y	Dotarem	**57**	Intravenous (during MRI → inferior MI + heart block, type II Kounis)

y, years old; IV, intravenous; MRI, magnetic resonance imaging; ST-segment, ST-segment (electrocardiographic finding); MI, myocardial infarction.

TABLE 4. Basic Details on Patients with Kounis Syndrome Caused by Microbubbles.

Country	First author	Sex/age	Trigger substance	Ref. No.	Administration
China	Wang et al.	Mixed cases; 1 case (induced by microbubbles)	Perflutren	42	Intravenous
		Male, 76 y			
USA	Levy et al.	Female, 39 y	Lumason	58	IV (during TTE→ anaphylaxis with cardiogenic shock)
USA	Corsi et al.	Male, 62 y	Perflutren	59	IV (during TTE → cardiac arrest)
USA	Yopes et al.	Male, 47 y	Lumason	60	IV (during TTE → Kounis syndrome type I/II)
Italy	Marchi et al.	Male, 44 y	SonoVue	61	IV [during stress perfusion echocardiography → type III Kounis (stent thrombosis)]
USA	Sagalov et al.	Male, 51 y	Lumason	62	IV (during TTE → ST-segment elevation/ type I Kounis Syndrome)
USA	Steinhauer et al.	Female, 49 y	Lumason	63	IV (during TTE → anaphylaxis + ACS)
The Netherlands	van Ginkel A. et al.	Male, 60 y	SonoVue	64	IV (bolus before dobutamine/echocardiographic contrast → allergic reaction + ST-segment elevation)
USA (multicenter)	Wei et al.	Large cohort study Male, 63 y	Perflutren	65	Retrospective analysis of 78,383 IV contrast administrations (used in TTE)
Spain	Portero-Portaz et al.	Female, 72 y	SonoVue	66	IV (during dobutamine stress echocardiography → type III Kounis syndrome/DES thrombosis)

y, years old; IV, intravenous; TTE, transthoracic echocardiography; ACS, acute coronary syndrome; DES, drug-eluting stent; Definity, perflutren lipid microsphere injectable suspension; Optison, perflutren protein-type A microspheres injectable suspension; Lumason, sulfur hexafluoride lipid-type A microspheres (known as SonoVue in Europe); SonoVue, Sulfur hexafluoride microbubble contrast agent.

The degranulation of mast cells and other interacting and related cells, including T-lymphocytes, macrophages, eosinophils, and platelets, produces a range of inflammatory mediators following an anaphylactic or allergic reaction or insult, which are the main causes of Kounis syndrome. In the Kounis syndrome cascade, acute ischemia, coronary spasm, atheromatous plaque erosion/rupture, and platelet activation can all be caused by derivatives of histamine, tryptase, and arachidonic acid as well as chymase, which is a converting enzyme. Kounis syndrome may be triggered[67] by medications, Hymenoptera stings, metals, foods, environmental exposures, diseases, and immunizations. Histamine, chymase, or arachidonic acid derivatives (leukotrienes, platelet-activating factor) can induce myocardial infarction type I, also known as ischemia with no obstructive coronary arteries, which affects 76.6% of patients with normal or near-normal coronary arteries. Type II Kounis syndrome, occurring in 22.3% of individuals with quiescent prior coronary disease, can also be brought on by acute myocardial infarction with platelet activation and similar conditions that cause type I. Because of stent polymers, stent metals, eluted medications, dual antiplatelet therapy, and environmental exposures, 5.1% of patients may exhibit type III stent thrombosis (subtype IIIa) or stent restenosis (subtype IIIb) Figure 2.

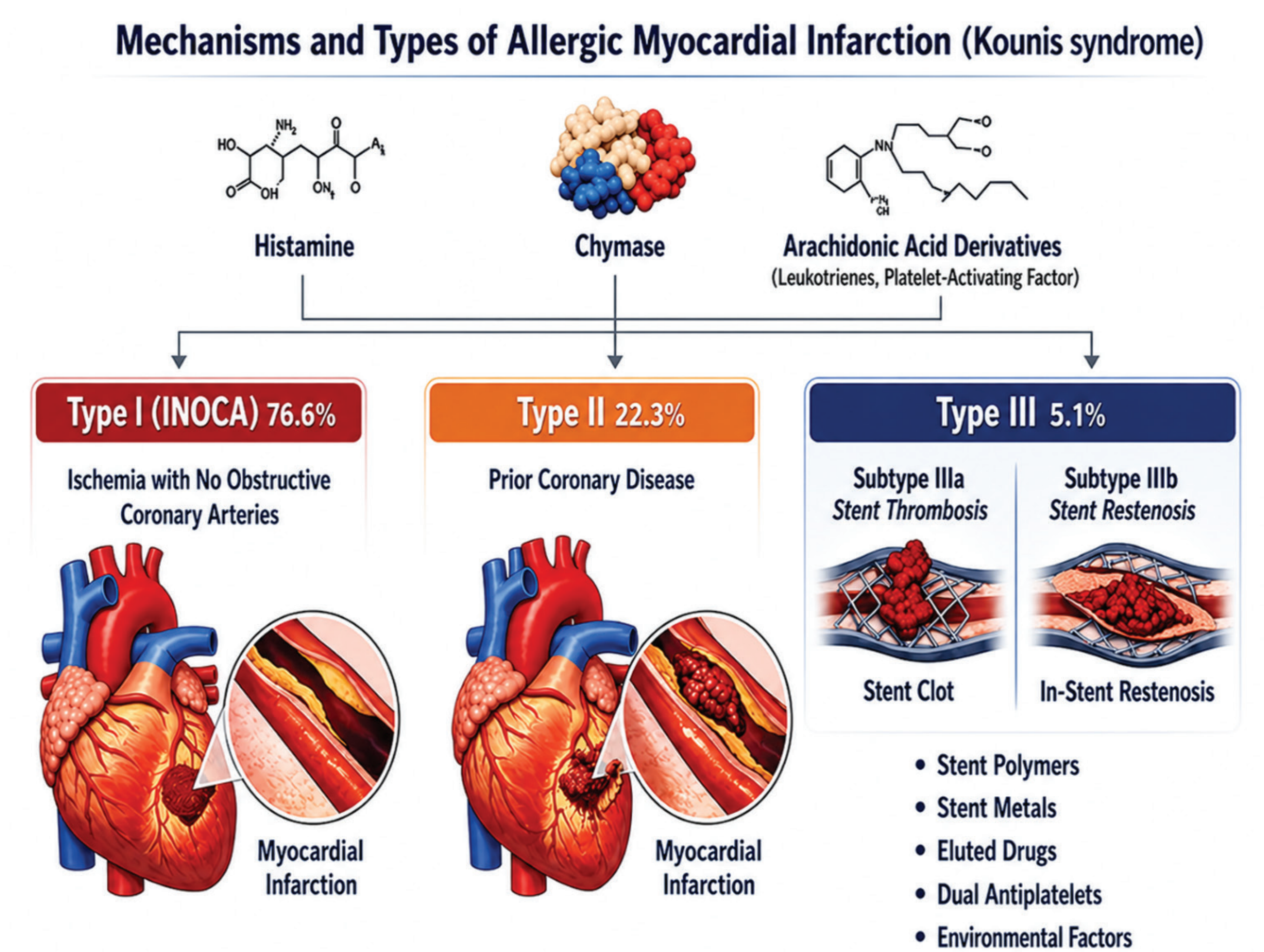

FIG. 2. The mechanism of type I, type II and type III of Kounis syndrome.
INOCA, ischemia with no obstructive coronary arteries.

DISCUSSION

Kounis syndrome is a cardiac hypersensitivity illness that has been reported across a wide age range, from two to eighty years. It encompasses a broad spectrum of mast cell activation disorders and has several, steadily increasing causes with expanding clinical manifestations. The cerebral and mesenteric arteries have been reported to be affected by Kounis-like syndromes, and the heart and entire arterial system are believed to be susceptible to allergic, hypersensitive, anaphylactic, and/or anaphylactoid events. Additionally, this syndrome may exacerbate mast cell activation disorders, and the affected areas appear to be the main locations and targets of hypersensitivity. Inflammatory mediators released either locally or in the systemic circulation after mast cell degranulation cause this condition. The activation of high- and low-affinity FcγRI, FcγRII, FcεRI, and FcεRII receptors on the platelet surface by mast cell mediators may also trigger the allergic thrombotic process. It has been shown that mast cells degranulate when relevant antigens bridge 2,000 adjacent antibodies that are attached to the cell surface to create the critical 1,000 bridges. Antibodies with distinct specificities can accomplish this process, and the severity of the disease, particularly in children, is correlated with the number of exposures within a specific time period.[68]

A recent study[69] discovered that contrast media-induced Kounis syndrome can result in major complications, such as a 7.7% mortality rate and a 23.1% cardiac arrest rate. Multiple organ failure and recurrent cardiac arrest still claimed the lives of two patients. When glucocorticoids and H1 (cetirizine and loratadine) and H2 (cimetidine and famotidine) receptor blockers are used to treat allergies, the symptoms of Kounis syndrome type I can be completely resolved in individuals.[70]

Corticosteroids and antihistamines should be used in conjunction with an acute coronary event protocol to initiate treatment for patients with type II Kounis syndrome.[70] They can inhibit the release of arachidonic acid from mast cell membranes by blocking phospholipase A2 and eicosanoid synthesis. Corticosteroids can also induce cell death and promote the synthesis of annexin or lipocortin, which are substances that regulate adhesion molecule expression, inflammatory cell activation, and transmigration and phagocytic processes. Drug exposure, antigen-antibody interactions, and hapten formation are some of the factors that have been implicated.

Systemic corticosteroids can trigger allergic responses through the following routes:[71]

a) Anaphylactic responses involving IgE antibodies to methylprednisolone have been documented.

b) Drugs that most frequently result in type 1 (immediate) adverse reactions include methylprednisolone and succinate esters of hydrocortisone.

c) Because succinate esters are more water-soluble and have a higher affinity for serum proteins, they facilitate hapten formation and act as complete antigens.

d) Methylprednisolone and hydrocortisone succinate esters are the drugs that most frequently result in type 1 (immediate) adverse reactions.

The current approach for acute myocardial infarction in patients with type III variants of Kounis syndrome includes prompt intrastent thrombus aspiration and histological analysis of the aspirated material. Moreover, nitrates and calcium channel blockers can prevent allergic coronary artery spasm. Beta-blockers should be avoided since they can exacerbate coronary artery spasm.

Life-saving drug in anaphylactic cases, can also trigger anaphylaxis on its own. A preservative present in all commercially available forms of adrenaline is sodium metabisulfite, according to Drug Facts and Comparisons, a standard pharmacy reference published by Wolters Kluwer and updated monthly. The food and pharmaceutical sectors frequently use sodium metabisulfite as an antioxidant. Metabisulfite, an additive in local anesthetics containing adrenaline, has been linked to cases of anaphylactic shock occurring during epidural anesthesia for caesarean sections. The occurrence of anaphylactic shock in sulfite-sensitive patients poses a therapeutic dilemma.

Fortunately, American Regent Inc., a pharmaceutical company located in New Albany, OH, USA, is now able to provide sulfite-free adrenaline to patients who are sensitive to it.[72] A possible alternative in this situation is glucagon, which has been successfully used to treat anaphylaxis in patients on β-blockers. International recommendations specifically suggest administering exogenous adrenaline intramuscularly at a dose of 0.01 mg/kg of a 1:1000 (1 mg/mL) solution11, with an adult maximum dose of 0.5 mg. This process is life-saving. On the other hand, dilute intravenous solutions [1:10,000 (0.1 mg/mL) or 1:100,000 (0.01 mg/mL)] may exacerbate coronary spasm.

Regarding contrast media, a recent study on gadolinium-based contrast agents-which are commonly used in MRI-raised concerns about their safety due to potential cytotoxic and genotoxic effects, especially at high doses or with repeated exposure. The cytogenotoxic potential of gadoteric acid in NIH3T3 fibroblast cells was assessed in this study using both *in vitro* and *in silico* methods. NIH3T3 cells were treated with increasing doses of gadoteric acid. The fibroblast cell line NIH/3T3 is derived from a mouse NIH/Swiss embryo. Studies on DNA transfection have found this cell line to be highly useful. Gadoteric acid generally showed dose-dependent cytotoxicity and moderate genotoxicity at high concentrations, indicating that DNA damage is caused by cytotoxic effects rather than a direct genotoxic mechanism.[73]

Examining the crucial but frequently overlooked issue of documenting hypersensitivity reactions to contrast media is important since incorrect documentation affects other medical specialties in addition to future imaging procedures. It is recommended to maintain accurate and up-to-date records of hypersensitivity episodes in order to ensure quality practice and improve patient outcomes.[74]

PERSPECTIVES

The European Society of Urogenital Radiology's Contrast Media Safety Committee revised its recommendations for preventing hypersensitivity reactions to intravascular, intravenous, or intra-arterial contrast media delivery.[75] To prevent a repeated hypersensitivity reaction and the resulting Kounis syndrome, it is crucial to determine whether the patient has ever had a reaction after a contrast-enhanced test with confirmed hypersensitivity reaction symptoms. Determining the precise type of contrast material used is also crucial. Unfortunately, information about past hypersensitivity reactions is often missing or not well documented. In order to ensure that data regarding prior adverse reactions is appropriately collected, among other potential risk factors, a questionnaire should be used. Choosing the best preventive treatment to prevent additional reactions is one of the main issues in managing hypersensitivity reactions to contrast media. Premedication is no longer considered protective, and new research suggests using a different contrast agent.[76] The best course of action is to choose the safest contrast agent based on an evaluation of allergies and cardiology. Patients in urgent need of contrast media administration who have a history of severe hypersensitivity to an unknown contrast medium may avoid complicated scenarios by being referred promptly to allergologists and/or cardiologists. Given the complexity of these circumstances, a customized approach based on the patient's features, the indication for administering contrast media, and local preferences is advised.

CONCLUSION

Kounis syndrome caused by contrast media is not a common side effect. To ensure prompt diagnosis and appropriate treatment, cardiologists and radiologists should be aware of this rare but serious condition. To prevent a repeated hypersensitivity reaction and the resulting Kounis syndrome, physicians need to inquire about any past history of hypersensitivity or reaction after a contrast-enhanced test; a questionnaire should be used. A customized approach based on the patient's features, the indication for administering contrast media, and local preferences is advised. There are currently no universally accepted recommendations to help select an alternative contrast agent in these situations. This requirement is particularly important in cases of Kounis syndrome, where choosing an alternative contrast agent can be challenging.

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The Global Burden of Chronic Kidney Disease Attributable to Hypertension in Young Adults From 1990 to 2021 and Projections to 2050

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Background: Chronic kidney disease (CKD) attributable to hypertension represents a major global public health challenge. This study evaluates the global burden of CKD statistically attributable to high systolic blood pressure (SBP), with a focus on young adults aged 25–49 years.

Aims: To analyze trends in mortality and disability-adjusted life years (DALYs) attributable to high SBP among individuals with CKD aged 25–49 years from 1990 to 2021 and to project the future attributable burden through 2050.

Study Design: Observational study.

Methods: Data were obtained from the Global Burden of Disease Study 2021. The analysis estimated the burden of CKD attributable to high SBP using a comparative risk assessment framework. Temporal trends were quantified using the estimated annual percentage change (EAPC). The association between attributable burden and the sociodemographic index (SDI) was also examined. Projections to 2050 were generated using autoregressive integrated moving average models.

Results: From 1990 to 2021, global mortality and DALY rates for CKD attributable to high SBP among young adults increased significantly (mortality EAPC, 1.75%; DALYs EAPC, 1.6%). The attributable burden remained consistently higher in males and increased with age, peaking in the 45–49-year age group. Overall, low- and low-middle-SDI regions experienced the greatest relative increases, although substantial heterogeneity was observed within SDI strata. At the national level, Ukraine showed the largest increase in mortality rate (EAPC, 13.21%), whereas the Republic of Korea exhibited the largest decline (EAPC, −4.68%). Model-based projections suggest a continued increase in both attributable mortality and DALYs through 2050, with persistent disparities by sex, age, and geographic region.

Conclusion: The burden of CKD attributable to high SBP among young adults has increased substantially over the past three decades and is projected to rise further, reflecting persistent global inequalities. These findings highlight the urgent need for strengthened and targeted strategies for hypertension prevention and management in younger populations worldwide.

INTRODUCTION

In contemporary society, hypertension has emerged as a major global public health problem and a leading modifiable risk factor for numerous cardiovascular and renal diseases, posing a substantial threat to population health and quality of life.[1] A considerable proportion of the associated renal burden is reflected in hypertension-related chronic kidney disease (HRCKD)—the component of chronic kidney disease (CKD) that is statistically attributable to elevated systolic blood pressure (SBP). Among many health conditions, the burden associated with CKD is particularly concerning.[2,3] CKD, including its end-stage form requiring dialysis or kidney transplantation, causes significant patient suffering and imposes substantial socioeconomic burdens.

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Within the public health research framework of the Global Burden of Disease (GBD) study, the proportion of CKD burden that can be statistically attributed to high SBP is quantified as "HRCKD".[4] However, the term "HRCKD" may be misinterpreted as a distinct clinical co-diagnosis rather than an attribution metric used within the GBD comparative risk assessment (CRA) framework. Throughout this manuscript, to maintain conceptual clarity and minimize ambiguity, we therefore use the explicit term "CKD burden attributable to high SBP" and avoid the abbreviated term "HRCKD." Over recent decades, alongside lifestyle changes and population aging, the global prevalence of hypertension has increased markedly.[5] Notably, the age of hypertension onset appears to be declining, prolonging cumulative exposure and potentially increasing the burden of CKD attributable to high SBP among younger adults (aged 25–49 years).[6] As the primary workforce supporting social and economic development, the health of this demographic group is critically important. Poor health in this population may result in reduced productivity and increased healthcare expenditures, with conditions such as CKD attributable to high SBP contributing substantially to this societal burden.

Although previous studies have examined the burden of CKD, comprehensive analyses specifically quantifying the evolving burden of CKD attributable to high SBP among young adults (aged 25–49 years), evaluating long-term temporal trends, and providing future projections remain limited.[7] Therefore, we used data from the GBD database to analyze trends in the burden of CKD attributable to high SBP among young adults aged 25–49 years from 1990 to 2021. We further conducted a detailed analysis of mortality and disability-adjusted life years (DALYs) attributable to CKD attributable to high SBP across global regions during the same period. In addition, we examined the influence of different sociodemographic index (SDI) levels on mortality and DALYs and projected future trends through 2050 to inform public health planning. Understanding these patterns of CKD attributable to high SBP burden is essential for informing public health strategies, guiding resource allocation, and prioritizing prevention efforts aimed at reducing the blood pressure–mediated burden of CKD.

MATERIALS AND METHODS

Data source and definition

Data for this analysis were obtained from the GBD 2021 study through the Global Health Data Exchange and were accessed on May 20, 2025. The dataset included annual estimates of mortality, DALYs, and their corresponding 95% uncertainty intervals (UIs) attributable to CKD attributable to high SBP from 1990 to 2021 (http://ghdx.healthdata.org/).[8,9] According to GBD definitions, data on mortality and DALYs attributable to CKD attributable to high SBP were available for both sexes across 21 global regions and five SDI categories.[10] These regions were defined based on geographic proximity and epidemiological similarity. Data on the burden of CKD attributable to high SBP were subsequently screened for the present analysis. Specifically, we selected data for young adults aged 25–49 years for detailed investigation and further stratified this population into five age groups (25–29, 30–34, 35–39, 40–44, and 45–49 years). Subsequently, we conducted a more detailed analysis of the epidemiological characteristics of this hypertension-attributable burden.

Definition of CKD burden attributable to high SBP

CKD was defined and classified according to the standardized protocols used in the GBD study. The primary outcome of this analysis was the burden of CKD attributable to high SBP.[4] This metric represents the estimated proportion of the total CKD burden that is attributable to elevated SBP, derived using the GBD CRA framework. It does not represent a distinct clinical entity with a specific International Classification of Diseases code but instead serves as a population-level metric for quantifying risk attribution. The underlying CKD case definition was based on permanent loss of kidney function, indicated by estimated glomerular filtration rate and urinary albumin-to-creatinine ratio.[8] CKD cases were mapped to the following ICD-10 codes: N18.1–N18.6, N18.8, and N18.9.

Following the GBD CRA framework, high SBP was defined as SBP ≥ 115 mmHg. The theoretical minimum risk exposure level (TMREL) was set at 110–115 mmHg.[3] The attributable burden was calculated as the proportion of total CKD burden associated with SBP levels exceeding this TMREL. This estimate represents a counterfactual measure of potentially preventable burden rather than the number of clinically diagnosed cases of hypertensive nephropathy. Consistent with GBD methodology, SBP was used as the primary exposure metric because of its stronger epidemiological association with cardiovascular and renal outcomes compared with diastolic blood pressure.

Statistical analysis

Both mortality and DALY rates as well as case numbers, were used to assess the burden of CKD attributable to high SBP. Rates were expressed as estimated values per 100,000 population to reflect the relative burden of disease, whereas case numbers represented the absolute burden. All corresponding estimates are presented with 95% UIs.

Temporal trends were quantified by calculating the estimated annual percentage change (EAPC) and its 95% confidence interval (CI). Prior to estimating the EAPC, long-term temporal patterns for all key outcomes (e.g., global, regional, and sex-specific rates) were visually examined to determine whether the overall trend could be reasonably approximated using a log-linear model. The EAPC was derived from a log-linear regression model [where y = ln(rates) and x = calendar year] and expressed as follows:

$$y = \alpha + \beta x + \varepsilon$$

$$\text{EAPC} = 100 \times [\exp(\beta) - 1]$$

The EAPC provides a widely accepted summary measure of the average annual rate of change over a specified period and offers a useful approximation of long-term trends for comparative purposes in epidemiological studies.[11-13] A statistically significant decreasing trend was defined when the upper bound of the 95% CI of the EAPC was < 0, whereas a statistically significant increasing trend was defined when the lower bound of the 95% CI was > 0. Trends were considered stable when the 95% CI included 0.

Spearman correlation analyses were performed to examine the relationships between the EAPC and the number of CKD cases attributable to high SBP in 1990, between the EAPC and SDI in 2021, and between DALYs attributable to CKD attributable to high SBP and SDI across countries. The resulting ρ coefficients and *p* values were used to evaluate the strength and statistical significance of these associations. Additionally, global maps were generated to illustrate country-level mortality and DALY rates attributable to CKD attributable to high SBP among young adults aged 25–49 years in 2021 as well as EAPC trends in this hypertension-attributable burden from 1990 to 2021. These visualizations provide insight into the geographic distribution and temporal patterns of the burden of CKD attributable to high SBP among young adults across different regions. Finally, autoregressive integrated moving average (ARIMA) models were used to forecast mortality and DALYs through 2050. Time-series forecasting was conducted using ARIMA models implemented in R (forecast package).[14] The ARIMA model includes three principal parameters: *p*, *d*, and *q*, where *p* represents the order of the autoregressive term, *d* represents the order of differencing, and *q* represents the order of the moving average term. Optimal ARIMA (*p, d, q*) parameters were automatically selected using the auto.arima function based on minimization of the corrected Akaike Information Criterion (AICc).[15] Model adequacy was assessed through residual diagnostics, including inspection of autocorrelation function plots and Ljung–Box Q tests to confirm the absence of residual autocorrelation.[16] Stationarity of the time series was achieved through differencing as determined by the ARIMA algorithm.[17] To evaluate forecasting performance, an in-sample validation procedure was applied. Models were initially fitted using data from 1990 to 2014 and subsequently used to generate forecasts for the period 2015–2019. Predicted values were compared with the corresponding GBD estimates, and forecasting accuracy was quantified using the mean absolute error and root mean square error.[18] Following validation, final ARIMA models were fitted to the complete dataset (1990–2021) to generate projections through 2050. All primary parameters and selected ARIMA model parameters (*p*, *d*, *q*) are presented in Supplementary Table 1. All statistical analyses and visualizations were performed using R software (version 3.5.2) and GraphPad Prism (version 8.02).

RESULTS

Global burden of CKD attributable to high SBP

Globally, the annual number of deaths from CKD attributable to high SBP among young adults increased substantially, from 8,584 (95% UI, 3,742–14,034) in 1990 to 21,766 (95% UI, 9,395–36,267) in 2021, representing an absolute increase of 153.56% (Figure 1a; Table 1). Mortality attributable to high SBP was consistently higher in males than in females. In 1990, the crude death rate was 0.38 (95% UI, 0.16–0.62) per 100,000 population for males and 0.25 (95% UI, 0.11–0.41) for females, rising to 0.68 (95% UI, 0.30–1.11) for males and 0.42 (95% UI, 0.17–0.71) for females in 2021 (Figure 1a, b; Table 1).

The global trend in DALYs due to CKD attributable to high SBP paralleled that of attributable deaths. In 2021, these DALYs reached 1,321,478 (95% UI, 627,899–2,144,240) among young adults aged 25–49 years (Figure 1e; Table 2). The crude attributable DALY rate was approximately 33.47 (95% UI, 15.90–54.30) per 100,000 population (Figure 1f; Table 2). Attributable DALY rates increased significantly over the study period, with an EAPC of 1.6% (95% CI, 1.56–1.64). The attributable burden was higher in males than in females. In 2021, attributable DALY rates were 40.82 (95% UI, 19.17–64.87) per 100,000 population among males and 25.93 (95% UI, 11.46–42.53) among females (Figure 1f; Table 2).

Age-specific trends in the attributable burden

From 1990 to 2021, the global burden of CKD attributable to high SBP showed an upward trend across all age groups among young adults aged 25–49 years. The attributable burden exhibited a clear age-dependent gradient, with the highest mortality (Figure 1 c, d; Table 1) and attributable DALY rates (Figure 1g, h; Table 2) consistently observed in the 45–49-year subgroup.

Among younger adults, both the attributable mortality rate and DALYs continued to rise over time, with the smallest increase seen in the 25–29 age group. The absolute number of deaths from CKD attributable to high SBP increased from 684 (95% UI, 273–1,259) in 1990 to 1,529 (95% UI, 646–2,694) in 2021, while the corresponding mortality rate rose from 0.15 (95% UI, 0.06–0.28) to 0.26 (95% UI, 0.11–0.46) per 100,000 population (Figure 1c, d; Table 1). Similarly, the absolute number of DALYs due to CKD attributable to high SBP increased from 57,834 (95% UI, 25,140–100,087) to 120,981 (95% UI, 55,640–207,705), and the attributable DALY rate increased from 13.07 (95% UI, 5.68–22.61) to 20.56 (95% UI, 9.46–35.30) per 100,000 population (Figure 1g, h; Table 2).

Variations in attributable burden by SDI and region

From 1990 to 2021, substantial variation was observed in the mortality and DALY burden of CKD attributable to high SBP across SDI regions. However, trends were not uniform within each SDI category. At the aggregate SDI level, the high-middle SDI region exhibited the smallest relative increases in both attributable death counts (41.99%; 95% UI, 15.20–72.34) and attributable DALYs (40.66%; 95% UI, 19.19–67.24), with corresponding EAPCs of 0.29% and 0.41%, respectively (Figure 2a-d; Tables 1, 2). In contrast, low and low-middle SDI regions showed the largest percentage increases in attributable death counts (190.65% and 212.41%) and attributable DALYs (190.22% and 203.32%) on average (Figure 2a-d; Tables 1, 2).

Marked geographic disparities were evident, underscoring heterogeneity within broader SDI strata. The most pronounced increases in attributable mortality and DALYs occurred in Oceania, Central Latin America, high-income North America, and Western Sub-Saharan Africa. For example, in Oceania, attributable death counts increased by 393.34% (95% UI, 214.77–704.53) and attributable DALYs by 384.67% (95% UI, 229.83–615.26), with EAPCs of 2.67% and 2.62%, respectively (Tables 1–2; Figure 2e, f). Notably, several regions within lower SDI categories demonstrated flat or declining trends, while divergent patterns were observed among high-SDI regions. For instance, high-income Asia Pacific experienced substantial decreases, with a 61.93% reduction in attributable death counts and a 47.02% reduction in attributable DALYs (Table 1; Figure 2e, f). These complex patterns suggest that the trajectory of CKD burden attributable to high SBP is influenced by factors beyond national SDI level alone.

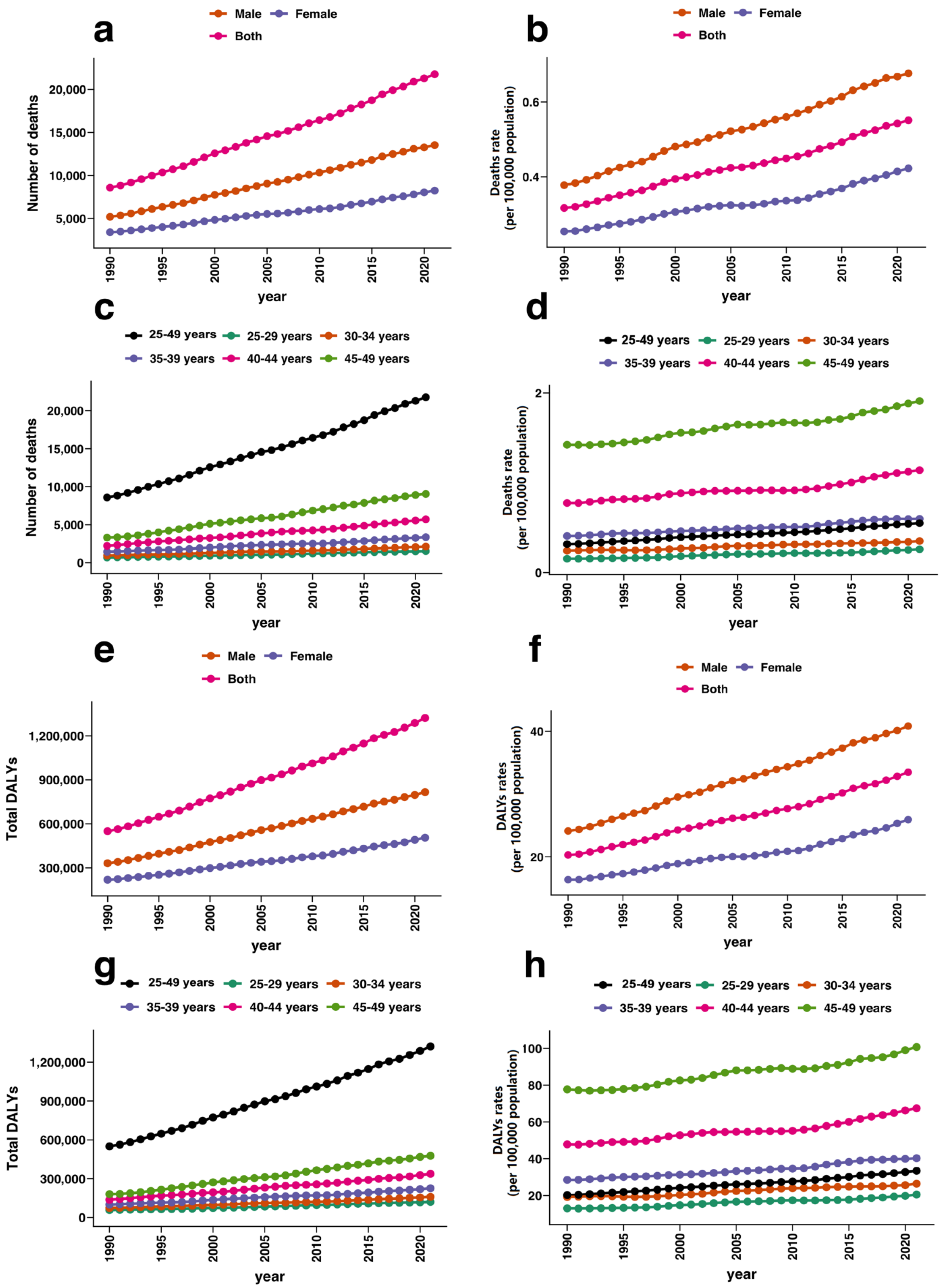

FIG. 1. Global trends in the number of deaths and DALYs from CKD attributable to high SBP among young adults aged 25–49 years from 1990 to 2021. (a, b) The number of attributable deaths and sex-specific attributable death rates over time. (c, d) The number of attributable deaths and age-specific attributable death rates by age group over time. (e, f) Number of attributable DALYs and sex-specific attributable DALY rates over time. (g, h) Number of attributable DALYs and age-specific attributable DALY rates by age group over time.

SBP, systolic blood pressure; CKD, chronic kidney disease; DALYs, disability-adjusted life years.

TABLE 1. Global and Regional Trends in Attributable Deaths from CKD due to High SBP Among Young Adults Aged 25–49 Years (1990–2021).

Feature	Cases_1990 (per 100,000 population, 95% UI)	Rates_1990 (95% UI)	Cases_2021 (per 100,000 population, 95% UI)	Rates_2021 (95% UI)	Cases_change (95% UI)	EAPC (95% CI)
Global	8584 (3742 to 14034)	0.32 (0.14 to 0.52)	21766 (9395 to 36267)	0.55 (0.24 to 0.92)	153.56 (121.45 to 186.3)	1.75 (1.69 to 1.81)
Sex group						
Female	3397 (1418 to 5494)	0.25 (0.11 to 0.41)	8237 (3402 to 13833)	0.42 (0.17 to 0.71)	142.49 (99.33 to 186.98)	1.54 (1.45 to 1.62)
Male	5188 (2210 to 8550)	0.38 (0.16 to 0.62)	13529 (6035 to 22101)	0.68 (0.3 to 1.11)	160.8 (117.9 to 200.09)	1.88 (1.83 to 1.94)
Age group						
25–29 years	684 (273 to 1259)	0.15 (0.06 to 0.28)	1529 (646 to 2694)	0.26 (0.11 to 0.46)	123.46 (83.85 to 176.51)	1.69 (1.59 to 1.79)
30–34 years	939 (400 to 1623)	0.24 (0.1 to 0.42)	2125 (919 to 3633)	0.35 (0.15 to 0.6)	126.23 (88.27 to 170.17)	1.25 (1.18 to 1.33)
35–39 years	1436 (557 to 2499)	0.41 (0.16 to 0.71)	3357 (1344 to 5729)	0.6 (0.24 to 1.02)	133.84 (90.99 to 180.73)	1.29 (1.23 to 1.34)
40–44 years	2220 (854 to 3953)	0.77 (0.3 to 1.38)	5709 (2211 to 10362)	1.14 (0.44 to 2.07)	157.16 (117.59 to 202.3)	1.13 (1.02 to 1.23)
45–49 years	3305 (1345 to 5777)	1.42 (0.58 to 2.49)	9047 (3943 to 15847)	1.91 (0.83 to 3.35)	173.7 (133.62 to 217.14)	0.93 (0.88 to 0.99)
SDI regions						
High SDI	662 (267 to 1094)	0.14 (0.06 to 0.24)	1428 (618 to 2410)	0.28 (0.12 to 0.48)	115.71 (70.19 to 169.76)	2.28 (1.89 to 2.67)
High-middle SDI	1238 (524 to 2037)	0.22 (0.09 to 0.36)	1758 (795 to 2931)	0.28 (0.13 to 0.47)	41.99 (15.2 to 72.34)	0.29 (0.09 to 0.49)
Middle SDI	3253 (1427 to 5347)	0.36 (0.16 to 0.59)	8127 (3614 to 13757)	0.65 (0.29 to 1.1)	149.82 (109 to 194.7)	1.91 (1.83 to 1.99)
Low-middle SDI	2249 (971 to 3704)	0.41 (0.18 to 0.67)	7027 (3078 to 11500)	0.69 (0.3 to 1.13)	212.41 (158.18 to 269.42)	1.79 (1.73 to 1.84)
Low SDI	1171 (501 to 1918)	0.53 (0.23 to 0.87)	3402 (1506 to 5640)	0.63 (0.28 to 1.04)	190.65 (148.7 to 242.11)	0.53 (0.49 to 0.57)
Geographical regions						
Andean Latin America	61 (21 to 118)	0.33 (0.11 to 0.63)	243 (99 to 423)	0.7 (0.28 to 1.21)	299.39 (162.99 to 559.18)	2.75 (2.55 to 2.96)
Australasia	5 (2 to 10)	0.05 (0.01 to 0.09)	10 (3 to 19)	0.07 (0.02 to 0.13)	80.95 (20.88 to 152.11)	1.24 (0.91 to 1.57)
Caribbean	121 (56 to 195)	0.66 (0.31 to 1.07)	348 (181 to 543)	1.45 (0.76 to 2.27)	187.33 (121.65 to 284.13)	3.06 (2.92 to 3.2)
Central Asia	95 (37 to 151)	0.28 (0.11 to 0.45)	311 (121 to 505)	0.64 (0.25 to 1.04)	228.24 (157.76 to 332.84)	2.06 (1.65 to 2.47)
Central Europe	237 (95 to 387)	0.38 (0.15 to 0.62)	139 (59 to 225)	0.26 (0.11 to 0.43)	–41.31 (–51.16 to –26.88)	–1.14 (–1.32 to –0.95)
Central Latin America	383 (173 to 630)	0.47 (0.21 to 0.77)	1693 (717 to 2870)	1.27 (0.54 to 2.16)	342.2 (215.23 to 518.15)	3.54 (3.31 to 3.77)
Central Sub-Saharan Africa	241 (100 to 416)	0.99 (0.41 to 1.7)	604 (252 to 1105)	0.93 (0.39 to 1.7)	150.35 (65.68 to 274.77)	–0.47 (-0.58 to -0.37)
East Asia	902 (330 to 1782)	0.13 (0.05 to 0.26)	1248 (473 to 2425)	0.18 (0.07 to 0.35)	38.31 (–24.95 to 138.45)	0.75 (0.5 to 1.01)
Eastern Europe	371 (163 to 588)	0.34 (0.15 to 0.53)	273 (115 to 436)	0.28 (0.12 to 0.45)	–26.42 (–35.85 to –14.42)	–2 (–2.48 to –1.52)
Eastern Sub-Saharan Africa	385 (156 to 663)	0.46 (0.19 to 0.8)	1347 (590 to 2239)	0.64 (0.28 to 1.07)	249.98 (186.3 to 339.55)	1.09 (1.02 to 1.15)
High-income Asia Pacific	162 (72 to 263)	0.17 (0.08 to 0.28)	62 (24 to 110)	0.08 (0.03 to 0.14)	–61.93 (–73.59 to –47.83)	–3.44 (-3.96 to –2.93)
High-income North America	177 (72 to 310)	0.12 (0.05 to 0.21)	741 (323 to 1217)	0.44 (0.19 to 0.72)	319.16 (204.57 to 522.2)	4.63 (4.07 to 5.2)
North Africa and Middle East	716 (305 to 1232)	0.45 (0.19 to 0.77)	2392 (994 to 4230)	0.72 (0.3 to 1.27)	234.03 (112.47 to 352.61)	1.7 (1.5 to 1.9)
Oceania	2 (1 to 3)	0.05 (0.02 to 0.11)	9 (3 to 17)	0.12 (0.05 to 0.24)	393.34 (214.77 to 704.53)	2.67 (2.4 to 2.94)
South Asia	1852 (797 to 3182)	0.35 (0.15 to 0.6)	4735 (1910 to 8363)	0.47 (0.19 to 0.83)	155.66 (90.86 to 215.5)	0.78 (0.69 to 0.88)
Southeast Asia	1378 (589 to 2360)	0.58 (0.25 to 1)	3861 (1722 to 6824)	1.04 (0.46 to 1.84)	180.14 (122.5 to 244.46)	2.25 (2.14 to 2.35)
Southern Latin America	68 (26 to 130)	0.28 (0.11 to 0.53)	109 (42 to 192)	0.32 (0.12 to 0.55)	61.14 (9.24 to 137.21)	0.8 (0.59 to 1.01)
Southern Sub-Saharan Africa	209 (90 to 355)	0.81 (0.35 to 1.38)	620 (270 to 1032)	1.44 (0.63 to 2.39)	196.82 (123.74 to 305.97)	1.33 (0.89 to 1.76)
Tropical Latin America	554 (272 to 829)	0.71 (0.35 to 1.06)	796 (385 to 1188)	0.66 (0.32 to 0.99)	43.7 (27.42 to 62.54)	–0.58 (–0.76 to –0.4)
Western Europe	151 (57 to 254)	0.08 (0.03 to 0.13)	109 (44 to 181)	0.06 (0.02 to 0.1)	–28 (-38.17 to –13.81)	–0.85 (–0.99 to –0.71)
Western Sub-Saharan Africa	513 (210 to 903)	0.6 (0.25 to 1.06)	2116 (974 to 3515)	0.92 (0.42 to 1.53)	312.43 (221.01 to 446.38)	1.63 (1.46 to 1.81)

CI, confidence interval; EAPC, estimated annual percentage change; UI, uncertainty intervals; SDI, sociodemographic index; SBP, systolic blood pressure; CKD, chronic kidney disease.

TABLE 2. Global and Regional Trends in Attributable DALYs from CKD due to High SBP Among Young Adults Aged 25–49 years (1990–2021).

Feature	Cases_1990 (per 100,000 population, 95% UI)	Rates_1990 (95% UI)	Cases_2021 (per 100,000 population, 95% UI)	Rates_2021 (95% UI)	Cases_change (95% UI)	EAPC (95% CI)
Global	549888 (250984 to 881977)	20.29 (9.26 to 32.54)	1321478 (627899 to 2144240)	33.47 (15.9 to 54.3)	140.32 (113.68 to 167.4)	1.6 (1.56 to 1.64)
Sex group						
Female	218822 (100337 to 350516)	16.36 (7.5 to 26.21)	505258 (223419 to 828799)	25.93 (11.46 to 42.53)	130.9 (96 to 167.17)	1.42 (1.35 to 1.48)
Male	331066 (149731 to 530608)	24.11 (10.9 to 38.64)	816220 (383415 to 1297203)	40.82 (19.17 to 64.87)	146.54 (113.24 to 177.7)	1.72 (1.67 to 1.76)
Age group						
25-29 years	57834 (25140 to 100087)	13.07 (5.68 to 22.61)	120981 (55640 to 207705)	20.56 (9.46 to 35.3)	109.19 (74.85 to 150.84)	1.49 (1.39 to 1.58)
30-34 years	74053 (33185 to 122342)	19.21 (8.61 to 31.74)	159971 (69435 to 263543)	26.46 (11.49 to 43.6)	116.02 (82.26 to 154.45)	1.13 (1.05 to 1.22)
35-39 years	100464 (40150 to 169083)	28.52 (11.4 to 48)	226117 (94183 to 373040)	40.32 (16.79 to 66.51)	125.07 (90.13 to 162.88)	1.18 (1.12 to 1.23)
40-44 years	137079 (58110 to 238386)	47.85 (20.28 to 83.21)	337478 (140920 to 592469)	67.46 (28.17 to 118.43)	146.19 (112.7 to 183.47)	1.04 (0.95 to 1.12)
45-49 years	180458 (77769 to 306447)	77.72 (33.49 to 131.98)	476931 (213189 to 808785)	100.72 (45.02 to 170.81)	164.29 (129.68 to 200.66)	0.84 (0.79 to 0.89)
SDI regions						
High SDI	60346 (28159 to 96128)	13.09 (6.11 to 20.86)	102065 (45020 to 165689)	20.32 (8.96 to 32.99)	69.13 (40.55 to 105.79)	1.35 (1.02 to 1.68)
High-middle SDI	85523 (39405 to 134880)	15.15 (6.98 to 23.9)	120298 (56895 to 192550)	19.11 (9.04 to 30.58)	40.66 (19.19 to 67.24)	0.41 (0.27 to 0.56)
Middle SDI	195469 (89029 to 315129)	21.47 (9.78 to 34.61)	476204 (221602 to 788245)	37.94 (17.66 to 62.81)	143.62 (111.78 to 183.47)	1.84 (1.77 to 1.91)
Low-middle SDI	139387 (62996 to 229400)	25.29 (11.43 to 41.63)	422794 (192146 to 673762)	41.6 (18.91 to 66.3)	203.32 (157.96 to 251.79)	1.73 (1.68 to 1.77)
Low SDI	68493 (29544 to 113180)	30.99 (13.37 to 51.2)	198781 (89475 to 323342)	36.65 (16.5 to 59.62)	190.22 (151.4 to 236.41)	0.54 (0.51 to 0.57)
Geographical regions						
Andean Latin America	3328 (1207 to 6274)	17.86 (6.48 to 33.67)	13228 (5512 to 22211)	37.82 (15.76 to 63.5)	297.46 (167.67 to 538.93)	2.81 (2.61 to 3.01)
Australasia	697 (311 to 1188)	6.46 (2.88 to 11.01)	1185 (494 to 2069)	8.2 (3.42 to 14.33)	69.97 (27.47 to 119.76)	0.32 (0.03 to 0.61)
Caribbean	6914 (3180 to 10924)	37.85 (17.41 to 59.8)	19035 (10100 to 29089)	79.5 (42.18 to 121.48)	175.3 (116 to 261.54)	2.87 (2.75 to 2.99)
Central Asia	9566 (4771 to 14678)	28.69 (14.31 to 44.02)	26102 (12129 to 40863)	53.53 (24.88 to 83.81)	172.87 (127.89 to 232.85)	1.73 (1.53 to 1.92)
Central Europe	16341 (7386 to 25831)	26.32 (11.89 to 41.6)	11762 (5513 to 17991)	22.32 (10.46 to 34.15)	–28.02 (-38.15 to –14.58)	–0.34 (-0.44 to –0.23)
Central Latin America	25033 (11575 to 40731)	30.67 (14.18 to 49.9)	96323 (42439 to 159265)	72.35 (31.88 to 119.63)	284.79 (176 to 421.01)	3.09 (2.93 to 3.24)
Central Sub-Saharan Africa	13769 (5961 to 23665)	56.39 (24.42 to 96.92)	33591 (14414 to 60514)	51.52 (22.11 to 92.81)	143.96 (65.24 to 249.24)	–0.59 (-0.7 to –0.48)
East Asia	53058 (19770 to 101677)	7.7 (2.87 to 14.76)	79303 (31269 to 146177)	11.52 (4.54 to 21.23)	49.46 (-15.93 to 153.98)	1.02 (0.78 to 1.26)
Eastern Europe	27017 (12801 to 41211)	24.5 (11.61 to 37.37)	21032 (10069 to 32427)	21.86 (10.46 to 33.7)	–22.15 (-30.74 to –11.9)	–1.46 (-1.81 to –1.11)
Eastern Sub-Saharan Africa	21735 (9248 to 36529)	26.06 (11.09 to 43.79)	75970 (34463 to 124968)	36.28 (16.46 to 59.68)	249.52 (189.89 to 325.68)	1.11 (1.06 to 1.17)
High-income Asia Pacific	13566 (6586 to 21528)	14.61 (7.1 to 23.19)	7187 (3196 to 11798)	9.19 (4.09 to 15.08)	–47.02 (-60.65 to –32.51)	–2.06 (–2.55 to –1.57)
High-income North America	15727 (6906 to 26119)	10.55 (4.63 to 17.53)	47697 (21176 to 76522)	28.28 (12.55 to 45.37)	203.29 (130.85 to 316.6)	3.36 (2.88 to 3.85)

TABLE 2. Continued.

Feature	Cases_1990 (per 100,000 population, 95% UI)	Rates_1990 (95% UI)	Cases_2021 (per 100,000 population, 95% UI)	Rates_2021 (95% UI)	Cases_change (95% UI)	EAPC (95% CI)
North Africa and Middle East	41772 (18113 to 69517)	26.06 (11.3 to 43.37)	134720 (58607 to 229225)	40.3 (17.53 to 68.56)	222.51 (119.17 to 324.59)	1.62 (1.46 to 1.78)
Oceania	122 (49 to 222)	3.82 (1.54 to 6.96)	592 (256 to 1099)	8.37 (3.62 to 15.53)	384.67 (229.83 to 615.26)	2.62 (2.36 to 2.88)
South Asia	125332 (54601 to 210815)	23.69 (10.32 to 39.85)	319550 (140762 to 536372)	31.74 (13.98 to 53.28)	154.96 (104.51 to 199.03)	0.87 (0.8 to 0.94)
Southeast Asia	74364 (32471 to 123604)	31.43 (13.72 to 52.24)	202127 (91546 to 351008)	54.51 (24.69 to 94.66)	171.81 (118.67 to 229.56)	2.11 (2.02 to 2.21)
Southern Latin America	4089 (1683 to 7625)	16.69 (6.87 to 31.13)	7357 (3075 to 12546)	21.21 (8.87 to 36.17)	79.93 (23.9 to 157.14)	1.25 (1.04 to 1.46)
Southern Sub-Saharan Africa	12260 (5436 to 20770)	47.6 (21.1 to 80.64)	33978 (14784 to 56659)	78.71 (34.25 to 131.25)	177.15 (110.78 to 279.3)	1.13 (0.71 to 1.54)
Tropical Latin America	33274 (16871 to 50233)	42.38 (21.49 to 63.98)	49198 (25185 to 73019)	41.06 (21.02 to 60.94)	47.86 (31.48 to 67.4)	−0.39 (−0.52 to −0.26)
Western Europe	23371 (11069 to 37844)	12.08 (5.72 to 19.57)	20570 (9534 to 32811)	10.91 (5.06 to 17.41)	−11.99 (−21.48 to −1.03)	−0.43 (-0.53 to −0.33)
Western Sub-Saharan Africa	28552 (11413 to 50010)	33.35 (13.33 to 58.42)	120970 (55057 to 201273)	52.75 (24.01 to 87.77)	323.68 (233.98 to 449.41)	1.76 (1.58 to 1.94)

CI, confidence interval; EAPC, estimated annual percentage change; UI, uncertainty intervals; SDI, sociodemographic index; SBP, systolic blood pressure; CKD, chronic kidney disease; DALYs, disability-adjusted life years.

National variations in the attributable burden across 204 countries

From 1990 to 2021, the mortality burden of CKD attributable to high SBP varied substantially among 204 countries and territories. The absolute number of attributable deaths increased by more than 600% in 10 countries, including Ukraine, Uzbekistan, Guatemala, Cameroon, Belize, Papua New Guinea, Oman, Zambia, Saudi Arabia, and Libya (Supplementary Figure 1a, b; Supplementary Table 1). In contrast, the number of attributable deaths decreased in several countries, with reductions greater than 50% in seven countries, including the Republic of Korea, Czechia, Germany, Poland, Japan, Romania, and Hungary (Supplementary Figure 1a, b; Supplementary Table 1).

Regarding the attributable mortality rate in 2021, the highest value was observed in Mauritius (3.08 per 100,000 population; 95% UI, 1.35–5.51), whereas the lowest value was found in Sweden (0.02 per 100,000 population; 95% UI, 0.01–0.05) (Supplementary Figure 1a; Supplementary Table 1). Among trends, Ukraine had the largest annual increase in the attributable mortality rate (EAPC, 13.21%; 95% CI, 11.20–15.25), while the Republic of Korea achieved the largest annual decline (EAPC, −4.68%; 95% UI, −5.14 to −4.22) (Supplementary Figure 1c; Supplementary Table 2).

From 1990 to 2021, DALYs due to CKD attributable to high SBP showed substantial variation among 204 countries and territories. Guatemala exhibited the largest increase, with attributable DALYs rising by 1,033.96% (95% UI, 618.78–1,918.25) (Supplementary Figure 1d, e; Supplementary Table 3). In contrast, the Republic of Korea reported the largest reduction in attributable DALYs. Guatemala also had the fastest annual growth rate in attributable DALY rates (EAPC, 6.5%; 95% CI, 5.73–7.27), whereas the Republic of Korea experienced the largest declines (Supplementary Figure 1f; Supplementary Table 3).

Correlations between the attributable burden and SDI

Globally, we assessed the correlation between SDI and two aspects of the CKD burden attributable to high SBP: its temporal trend and its absolute level in 2021. Regarding the temporal trend, the EAPC in mortality attributable to high SBP showed no significant correlation with the baseline attributable mortality rate in 1990 ($\rho = -0.017$, $p = 0.812$) (Figure 3a), nor with national SDI levels ($\rho = -0.005$, $p = 0.972$) (Figure 3b). These results suggest that the rate of change in attributable mortality over the study period was not systematically associated with a country's initial attributable burden or its level of socioeconomic development. In contrast, when examining the absolute burden level, a significant negative correlation was observed between SDI and the attributable mortality rate in 2021 ($\rho = -0.372$, $p < 0.001$) (Figure 3c), indicating that regions with higher SDI generally exhibited lower attributable mortality rates.

A parallel pattern was observed for DALYs. The EAPC for attributable DALYs was not significantly correlated with baseline DALYs in 1990 ($\rho = -0.029$, $p = 0.682$) (Figure 3d) or with SDI levels ($\rho = -0.031$, $p = 0.829$) (Figure 3e). This suggests that the pace of change in attributable DALYs over time was similar across countries, regardless of their initial burden or development status. However, the attributable DALY rate in 2021 showed a significant negative correlation with SDI ($\rho = -0.32$, $p < 0.001$) (Figure 3f), indicating that regions with higher sociodemographic development tended to have a lower absolute attributable burden.

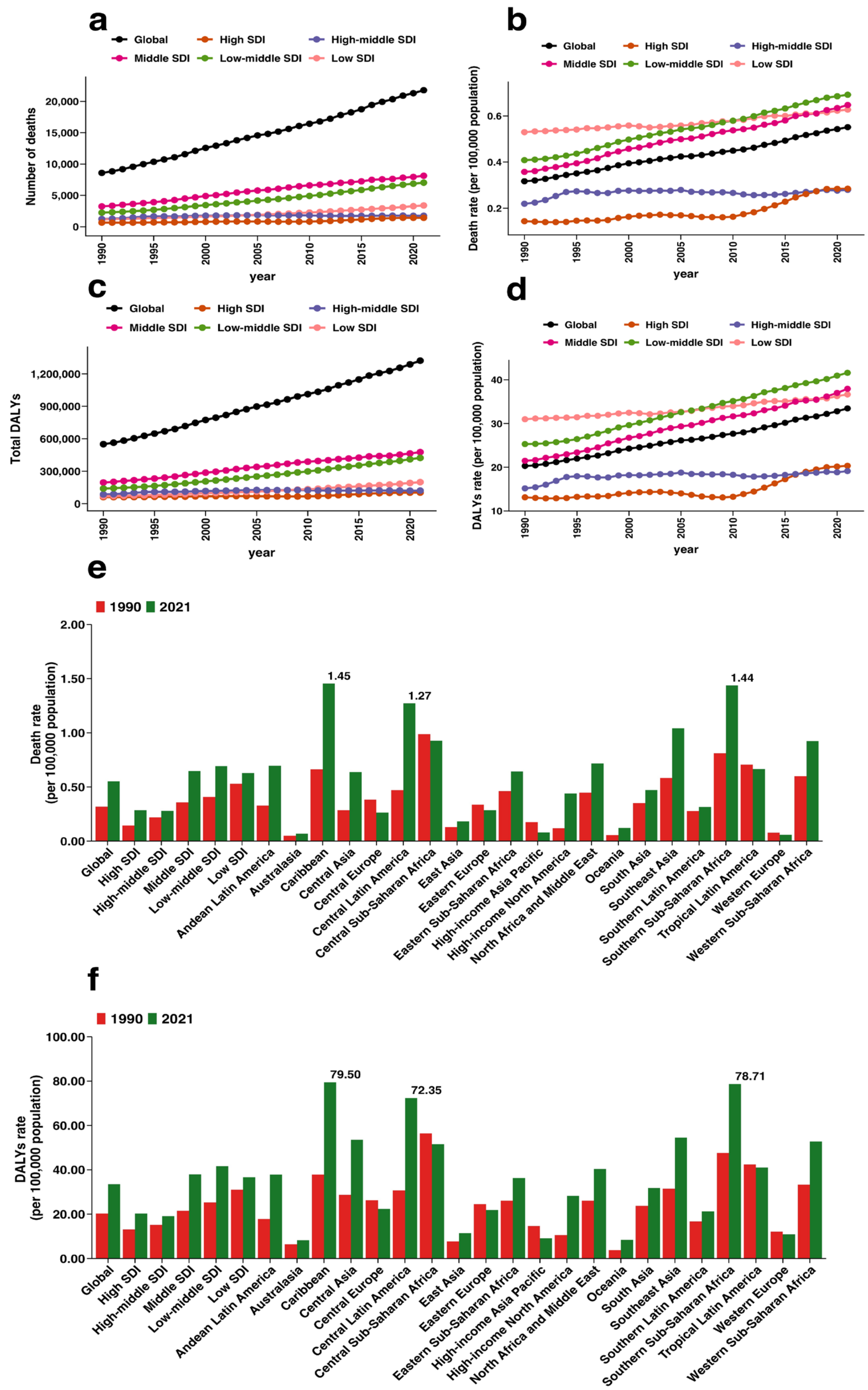

FIG. 2. Regional differences in the number of deaths and DALYs from CKD attributable to high SBP among young adults aged 25–49 years in 1990 and 2021. (a, b) The number of attributable deaths and attributable death rates over time at SDI region and globally. (c, d) Temporal trend of attributable DALYs and attributable DALY rates at SDI region and globally. (e, f) Attributable death and DALY rates at the global, SDI, and regional levels.

SDI, sociodemographic index; SBP, systolic blood pressure; CKD, chronic kidney disease; DALYs, disability-adjusted life years.

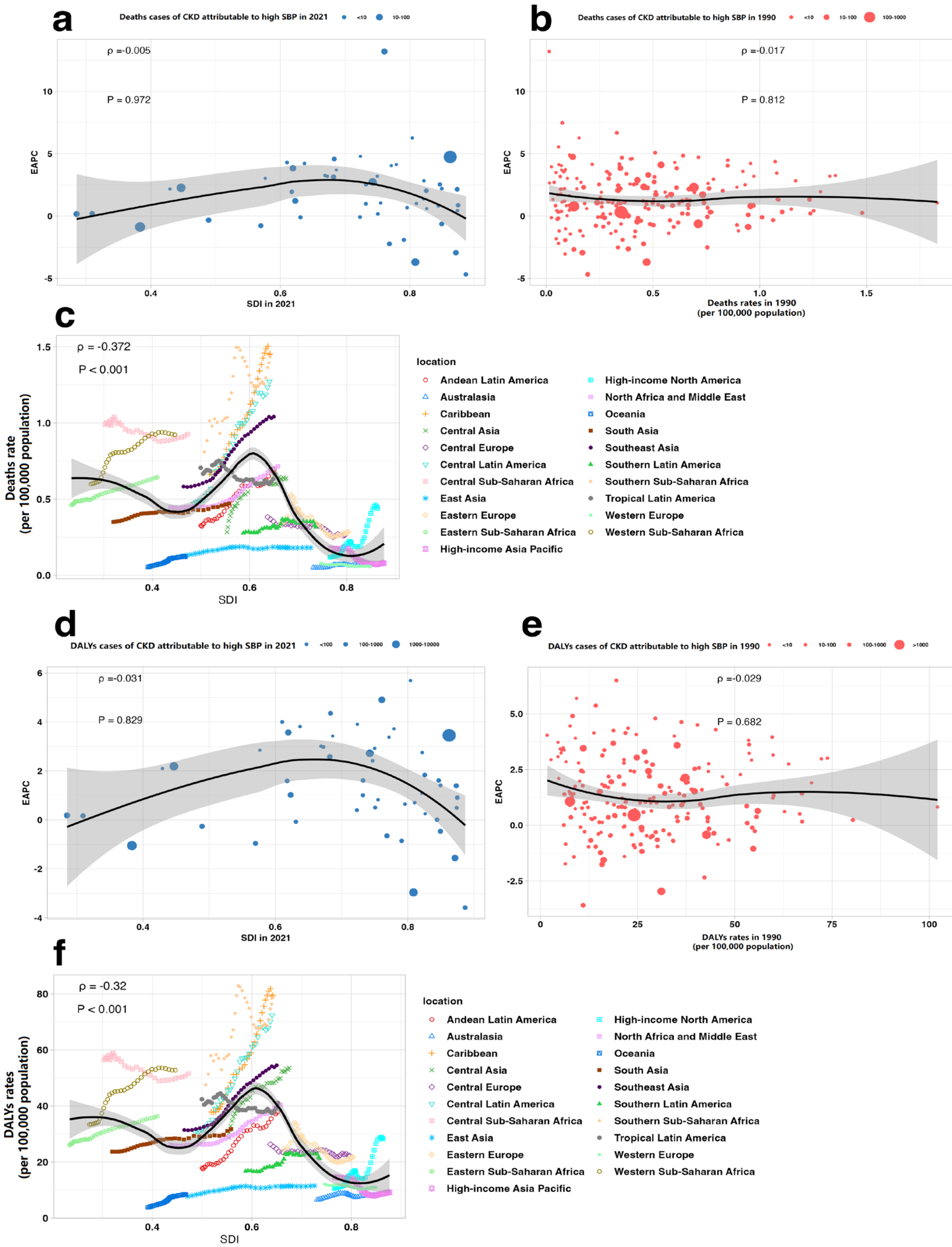

FIG. 3. Correlations between the attributable burden, its temporal trends, and SDI among young adults aged 25-49 years. (a) Association between the SDI in 2021 and the EAPC in attributable death rates from 1990 to 2021. (b) Association between attributable death rates in 1990 and EAPC in attributable death rates from 1990 to 2021. (c) Relationship between SDI and attributable death rates from 1990 to 2021 across global regions. (d) Association between the SDI in 2021 and the EAPC in attributable DALY rates from 1990 to 2021. (e) Association between attributable DALY rates in 1990 and EAPC in attributable DALY rates from 1990 to 2021. (f) Relationship between SDI and attributable DALY rates from 1990 to 2021 across global regions. Note: Spearman rank correlation coefficients (ρ) were calculated to assess associations between variables. The solid curves represent locally weighted regression (LOESS) smoothing to illustrate potential non-linear trends, with shaded areas indicating 95% confidence intervals.

SDI, sociodemographic index; DALYs, disability-adjusted life years; EAPC, estimated annual percentage change.

Projected trends in attributable burden to 2050

The projected trend for mortality from CKD burden attributable to high SBP among young adults aged 25–49 years suggests a potential global increase, with substantial variation across sex, age, and region. According to model projections, the attributable mortality rate for males is expected to reach approximately 0.96 (95% UI, 0.92–0.99) per 100,000 population in 2050, whereas for females, it is projected to reach approximately 0.66 (95% UI, 0.19–1.13) per 100,000 population (Figure 4a; Supplementary Table 4). The projections indicate a notably higher potential increase in attributable mortality among the 45–49-year age group compared with other age groups (Figure 4b; Supplementary Table 5). Globally, the attributable mortality rate is projected to rise to 0.77 (95% UI, 0.73–0.81) per 100,000 population in 2050 (Figure 4a-c; Supplementary Tables 4-6). Model results suggest that mortality from high SBP–related CKD may increase to varying degrees across SDI regions, although a slight decrease is projected in high-middle SDI regions (Figure 4c; Supplementary Table 6).

The projected trend for DALYs due to CKD attributable to high SBP among the 25–49 age group indicates a potential global increase, with persistent disparities by sex, age, and SDI level. Based on projections, attributable DALYs for both males and females are expected to increase by 2050 (Figure 4d; Supplementary Table 7). Age-specific predictions suggest that older age groups may experience a greater rate of increase, with the 40–44-year age group showing the largest projected rise, from 72.31 (95% UI, 68.54–76.08) in 2025 to 102.59 (95% UI, 38.91–166.28) per 100,000 population in 2050 (Figure 4e; Supplementary Table 8). The global attributable DALY rate is projected to increase from 35.68 (95% UI, 34.77–36.59) in 2025 to 49.23 (95% UI, 41.30–57.16) per 100,000 population in 2050 (Figure 4d-f; Supplementary Tables 7-9). The model projects that middle and low-middle SDI regions will experience the greatest increases in the attributable DALY rate, from 40.20 (95% UI, 39.14–41.26) in 2025 to 53.47 (95% UI, 50.45–56.49) and from 43.80 (95% UI, 42.75–44.86) to 56.62 (95% UI, 52.85–60.38) per 100,000 population, respectively, by 2050 (Figure 4f; Supplementary Table 9).

DISCUSSION

The burden of CKD attributable to high SBP represents a significant and growing public health concern globally. Our findings indicate an increasing trend in both mortality and DALYs attributable to this risk factor among young adults aged 25–49 years between 1990 and 2021. Epidemiological studies suggest a trend toward earlier onset of hypertension, which may contribute to the rising attributable burden in this age group.[19,20] This trend may be influenced by multiple factors, including lifestyle behaviors (such as obesity and alcohol consumption), genetic susceptibility, and environmental changes.[21,22] Young and middle-aged adults often face considerable life and work pressures, which can lead to emotional tension. Coupled with irregular diet and rest, this may result in abnormal sympathetic nervous system activation, thereby increasing the risk of high blood pressure.[23] However, hypertension control in this demographic remains suboptimal. For example, data from the China Health and Nutrition Survey (2011) showed that among individuals aged 40–59, the awareness, treatment, and control rates of hypertension were 50.9%, 40.9%, and 18.8%, respectively.[24] In contrast, corresponding rates in the United States were notably higher at 83.0%, 73.7%, and 57.8%.[25] Furthermore, renal damage caused by hypertension is cumulative over time and often asymptomatic in early stages. This insidious nature may foster complacency in blood pressure control, ultimately contributing to CKD development. These findings underscore the need for enhanced public health efforts focused on improving hypertension awareness, detection, and management among young adults, which could serve as a foundational strategy for mitigating the future burden of CKD attributable to this risk factor.

The estimated burden of CKD attributable to high SBP varied substantially by sex, with a consistently higher burden observed in males than in females. This disparity may partly reflect a higher prevalence of modifiable risk factors commonly reported among men, including smoking, harmful alcohol use, and potentially lower rates of adherence to hypertension treatment.[26-28] In addition, the Global Status Report on Alcohol and Health highlights that, except in the WHO European Region, global alcohol consumption is increasing, particularly among young men.[29] Simultaneously, differences exist in the proportion of attributable burden across age groups. Generally, the burden increases with age, with the greatest rise observed in adults aged 45–49 years. This pattern is likely due to the longer duration of illness and higher prevalence of comorbidities in this age group, including diabetes,[30] dyslipidemia,[31] and hyperuricemia.[32] Therefore, these sex- and age-specific risk factors should be considered when developing targeted policies for young adults.

Our analysis revealed significant disparities in the attributable burden across regions and SDI levels. Both attributable mortality and DALYs were generally higher in regions with lower SDI and lower in high-SDI regions, a pattern consistent with trends observed for many non-communicable diseases.[33] These disparities may be influenced by differences in health policies, public awareness, and chronic disease management capacity. Our results underscore the importance of equity-oriented approaches in global health. To address these disparities, future efforts could focus on strengthening health systems, improving access to care, and implementing context-specific hypertension management programs, particularly in low-resource settings. International collaboration and investment may be crucial to support such efforts in low-SDI regions.

Notably, our forecasting model indicates that the mortality and DALYs attributable to CKD from high SBP are projected to continue rising globally over the next three decades. Although these projections are uncertain, they suggest a persistent and growing public health challenge associated with this risk factor. The forecasts also reveal heterogeneous trends across development strata, with minimal projected change in high-SDI regions. This disparity may reflect advances in healthcare systems, broader access to effective management, and the implementation of comprehensive non-communicable disease policies in these settings. Therefore, the international community and governments can leverage the

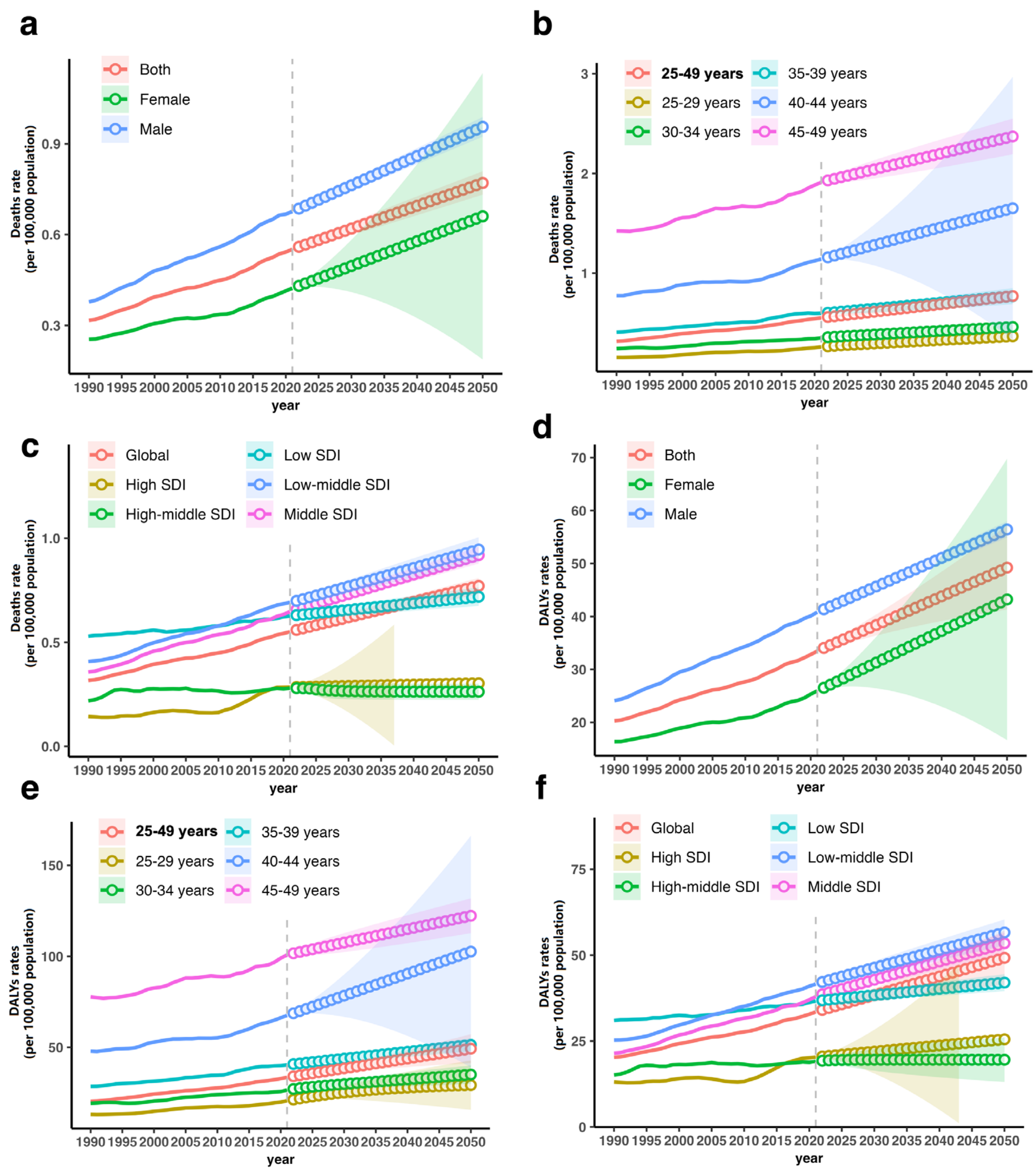

FIG. 4. Projected trends in attributable death and DALY rates by sex, age and SDI region. (a-c) Trends and projections in attributable death rate by sex, age, and SDI region. (d-f) Trends and projections in attributable DALY rates by sex, age, and SDI region.
SDI, sociodemographic index; DALYs, disability-adjusted life years.

valuable experience of high-SDI countries to support under-resourced regions and achieve truly global health.

Several limitations of this study should be acknowledged to ensure appropriate interpretation of our findings. First, the analysis relies on GBD data and its methodological framework. As a secondary analysis, the estimates depend on the quality, coverage, and modeling assumptions of the underlying source data, which vary across regions and over time. Observed trends may also be influenced by methodological updates across GBD cycles, not solely by true epidemiological changes. Second, our use of the EAPC to quantify temporal trends assumes an approximately log-linear pattern over the study period (1990–2021). While trends were visually inspected before analysis and EAPC provides a robust summary measure of net change, it may not fully capture sub-period accelerations, decelerations, or more complex non-linear fluctuations, particularly at the regional level. A single EAPC estimate could smooth over important temporal variations. More nuanced methods, such as segmented regression, could be considered in future studies to identify potential turning points in specific contexts. Third, comparisons may be biased by differences in population age structure. Although this study focused on adults aged 25–49 years, rates were not age-standardized within this range. Given the strong age gradient shown in the results, differences in population age structure across countries could affect the comparability of crude rates. Fourth, the study assesses attributable burden, not direct causation. This population-level observational analysis quantifies the statistical association between high SBP and CKD within a counterfactual framework. It does not establish individual-level clinical causation or evaluate the effectiveness of specific interventions. Fifth, estimates for some subgroups have greater uncertainty. UIs were wider for groups with low event rates or high internal heterogeneity, reflecting data sparsity. Finally, long-term projections are inherently uncertain. Future changes in demographics, risk factors, health policies, or treatments could alter the projected burden. Despite these limitations, this study provides valuable evidence to guide interventions and policy planning aimed at reducing the CKD burden attributable to high SBP.

In conclusion, our analysis reveals a rising and inequitably distributed burden of CKD attributable to high SBP among young adults globally, with pronounced disparities by sex and SDI. These findings highlight the potential need for enhanced, equity-focused public health strategies that prioritize hypertension prevention and management within broader non-communicable disease agendas. They also underscore the importance of sustained surveillance and research to track trends and evaluate the impact of interventions. Ultimately, this study provides a population-level evidence base to inform health policy planning and guide future investigations aimed at reducing the blood pressure–mediated burden of kidney disease.

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Informed Consent: Not applicable.

Data Sharing Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Conflict of Interest: The authors declared that they have no conflict of interest.

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Supplementary Tables: https://balkanmedicaljournal.org/img/files/Supplemental-Tables%281%29.pdf

Supplementary Figure: https://balkanmedicaljournal.org/img/files/Supplemental-Figure.pdf

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Rheumatoid Arthritis Burden in Middle-Aged Adults (40–59 Years): Evidence from the Global Burden of Disease 2021

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Background: Rheumatoid arthritis (RA) is a leading cause of disability worldwide. Although global assessments of RA have been reported, age-specific estimates focusing on middle-aged adults [(MAA); 40–59 years] remain limited.

Aims: To quantify the global, regional, and temporal burden of RA among MAA from 1990 to 2021 and project age-standardized trends through 2050.

Study Design: A population-based descriptive epidemiological study using Global Burden of Disease (GBD) 2021 estimates.

Methods: GBD 2021 data (1990–2021) were used to estimate age-standardized incidence (ASIR), prevalence (ASPR), mortality (ASMR), and disability-adjusted life-year rates (ASDR) per 100,000 population among MAA. Temporal trends were assessed using joinpoint regression and expressed as the average annual percentage change (AAPC). Projections to 2050 were generated using a Bayesian age–period–cohort model. Burden patterns were summarized across sociodemographic index (SDI) quintiles and 21 GBD regions.

Results: In 2021, the global ASIR, ASPR, ASMR, and ASDR among MAA were 19.53, 368.73, 0.19, and 56.74 per 100,000 population, respectively. From 1990 to 2021, ASIR and ASPR increased (AAPC = 0.24% and 0.43%), ASMR decreased (AAPC = -1.70%), and ASDR showed minimal net change (AAPC = 0.05%). Substantial variation was observed across SDI quintiles and GBD regions; SDI was positively correlated with ASIR and ASPR and negatively correlated with ASMR. Projections to 2050 indicated an ASIR of 19.22, an ASPR of 367.47, an ASMR of 0.111, and an ASDR of 52.18 per 100,000, with widening credible intervals over time.

Conclusion: Between 1990 and 2021, the RA burden among MAA was characterized by increasing ASIR and ASPR, declining ASMR, and relatively stable ASDR. Projections suggest that ASIR and ASPR will remain near recent levels, whereas ASMR and ASDR will show lower median values over time.

INTRODUCTION

Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease characterized by persistent synovial inflammation, progressive joint destruction, and long-term functional impairment.[1] It is a major contributor to years lived with disability globally and represents a sustained burden on individuals, healthcare systems, and societies.[1] The clinical course of RA is typically prolonged, and its cumulative impact is influenced not only by disease severity but also by the age at onset and the duration of disease.

Although RA can occur at any age, epidemiological evidence indicates that incidence rates increase markedly during midlife, with a substantial proportion of diagnoses occurring between 40 and 59 years of age.[1] This stage of life represents a critical period in which individuals often experience peak occupational engagement and social responsibilities. Consequently, the onset of RA in middle-aged adults (MAA) may lead to prolonged health service utilization, productivity loss, and sustained disability over subsequent decades.[2] Despite this, most global epidemiological assessments have focused on the general population or older adults, and age-specific analyses targeting MAA remain limited.[3]

Over the past decades, advances in early diagnosis and treatment—including treat-to-target strategies and the expanded use of biologic and targeted synthetic disease-modifying antirheumatic drugs—

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have improved disease control and survival.[4] Nevertheless, the global number of RA cases has continued to increase, largely driven by demographic transitions, such as population growth and aging.[4] Substantial regional and sociodemographic disparities also persist.[5] However, comprehensive evaluations of long-term temporal trends and future projections specifically among MAA remain limited, constraining age-targeted planning and accurate assessment of disease burden in this economically and socially active population.

In this study, we conducted a comprehensive assessment of the burden of RA among MAA using data from the Global Burden of Disease (GBD) 2021 study,[6,7] which provides standardized estimates for 369 diseases and injuries across 204 countries and territories from 1990 to 2021. Joinpoint regression was used to quantify temporal trends in age-standardized incidence (ASIR), prevalence (ASPR), mortality (ASMR), and disability-adjusted life years (DALY) rates (ASDR),[8] and a Bayesian age–period–cohort (BAPC) model was applied to project these trends through 2050. This analysis provides an updated global overview of the RA burden among MAA and offers age-specific evidence to support epidemiological monitoring and health planning.

MATERIALS AND METHODS

Data source

Data were obtained from the GBD 2021 study conducted by the Institute for Health Metrics and Evaluation, University of Washington. GBD 2021 provides comprehensive epidemiological estimates for 369 diseases and injuries across 204 countries and territories from 1990 to 2021. RA was defined according to the GBD cause hierarchy based on ICD-10 codes M05–M06. Age-standardized rates (ASRs) were calculated using the GBD standard world population to allow comparability across time and regions. In this study, incidence, prevalence, mortality, and DALYs for individuals aged 40–59 years were extracted using the GBD Results Tool. Estimates were reported as counts and ASRs per 100,000 population, with corresponding 95% uncertainty intervals (UIs). Countries were categorized into five sociodemographic index (SDI) quintiles and 21 GBD regions.

Joinpoint regression analysis

Joinpoint regression was used to evaluate temporal trends in ASRs from 1990 to 2021. Analyses were conducted using the Joinpoint Regression Program (National Cancer Institute, USA). A maximum of five joinpoints was allowed, and the optimal model was identified using Monte Carlo permutation tests (overall two-sided $\alpha = 0.05$). Log-transformed ASRs were modeled under the assumption of a log-linear relationship. Variance was estimated using the default settings of the Joinpoint software for rate data. The analysis followed the default independent error assumption implemented in the Joinpoint Regression Program, and no additional autoregressive error structure was specified. For each time segment, the annual percentage change (APC) and its 95% confidence interval (CI) were estimated as APC = $(e^{\beta} - 1) \times 100\%$, where β is the slope coefficient. The average annual percentage change (AAPC) was calculated as the weighted geometric mean of the segment-specific APCs across the entire study period, with weights proportional to the length of each segment.

Bayesian age–period–cohort models

Future projections of the RA burden through 2050 were estimated using a BAPC model applied to ASRs, implemented using the BAPC package. The model incorporates temporal effects to capture historical patterns and generate future projections. A Poisson likelihood was assumed, and parameter estimation was conducted within the Integrated Nested Laplace Approximation framework implemented in the BAPC package. Future rates were projected by extrapolating the estimated temporal effects beyond the observed period. Uncertainty in the projections was quantified using 95% credible intervals (CrIs) derived from the posterior distributions.

Statistical analysis

Temporal trends in ASIR, prevalence, mortality, and DALY rates were summarized using the APC and AAPC derived from joinpoint regression analysis. A temporal trend was considered statistically significant when the 95% CI for the APC or AAPC did not include zero. Descriptive comparisons of the burden of RA were conducted across SDI quintiles and GBD regions. Associations between SDI and ASRs were assessed using Pearson correlation coefficients and interpreted as exploratory ecological analyses. Data management was performed using Microsoft Excel, and statistical analyses and visualizations were conducted in R (version 4.3.3) using appropriate packages.

RESULTS

Global burden and overall trends, 1990–2021

In 2021, the global ASIR of RA among MAA was 19.53 (95% UI: 12.60–27.98) per 100,000 population. The ASPR was 368.73 (95% UI: 298.90–446.08) per 100,000, the ASMR was 0.19 (95% UI: 0.16–0.22) per 100,000, and the ASDR was 56.74 (95% UI: 39.44–78.91) per 100,000 population.

Between 1990 and 2021, ASIR increased from 18.14 (95% UI: 11.61–26.26) per 100,000 in 1990 to 19.53 (95% UI: 12.60–27.98) per 100,000 in 2021 (AAPC = 0.24%, 95% CI: 0.23–0.25). The ASPR increased from 323.99 (95% UI: 260.33–395.49) to 368.73 (95% UI: 298.90–446.08) (AAPC = 0.43%, 95% CI: 0.42–0.44). In contrast, the ASMR decreased from 0.33 (95% UI: 0.29–0.37) to 0.19 (95% UI: 0.16–0.22) during the same period [AAPC = -1.70%, 95% CI: -1.75-(-1.65)]. Meanwhile, the ASDR changed from 55.85 (95% UI: 40.49–75.76) to 56.74 (95% UI: 39.44–78.91) (AAPC = 0.05%, 95% CI: 0.04–0.06).

Sex-specific burden and temporal trends, 1990–2021

In 2021, all ASRs of RA among MAA were higher in females than in males (Table 1). The global ASIR was 27.22 (95% UI: 17.63–38.72) per 100,000 in females and 11.83 (95% UI: 7.51–17.27) in males. The ASPR was 535.42 (95% UI: 437.92–640.43) in females compared with 201.41 (95% UI: 157.66–250.30) in males. The ASMR was 0.26 (95% UI: 0.23–0.30) in females and 0.13 (95% UI: 0.08–0.16) in males. Likewise, the ASDR was higher in females at 81.16 (95% UI: 56.41–113.43) than in males at 32.22 (95% UI: 22.21–45.17).

Between 1990 and 2021, ASIR increased in both sexes, with an AAPC of 0.28% (95% CI: 0.27–0.29) in males and 0.20% (95% CI: 0.19–0.20)

TABLE 1. Sex-Specific ASRs and Temporal Trends of RA Among Adults Aged 40–59 Years Globally, 1990–2021.

Measure	Male	Female
1990		
ASIR (per 100, 000 population, 95% UI)	10.86 (6.81 to 16.08)	25.64 (16.51 to 36.86)
ASPR (per 100, 000 population, 95% UI)	173.16 (133.73 to 217.74)	479.04 (388.27 to 578.1)
ASMR (per 100, 000 population, 95% UI)	0.18 (0.13 to 0.22)	0.47 (0.4 to 0.55)
ASDR (per 100, 000 population, 95% UI)	30.58 (22.01 to 41.98)	81.8 (59.13 to 110.5)
2021		
ASIR (per 100, 000 population, 95% UI)	11.83 (7.51 to 17.27)	27.22 (17.63 to 38.72)
ASPR (per 100, 000 population, 95% UI)	201.41 (157.66 to 250.3)	535.42 (437.92 to 640.43)
ASMR (per 100, 000 population, 95% UI)	0.13 (0.08 to 0.16)	0.26 (0.23 to 0.3)
ASDR (per 100, 000 population, 95% UI)	32.22 (22.21 to 45.17)	81.16 (56.41 to 113.43)
1990-2021		
AAPC of ASIR (95% CI)	0.28* (0.27 to 0.29)	0.20* (0.19 to 0.20)
AAPC of ASPR (95% CI)	0.50* (0.49 to 0.51)	0.37* (0.36 to 0.38)
AAPC of ASMR (95% CI)	-1.26* (-1.33 to -1.19)	-1.93* (-1.98 to -1.88)
AAPC of ASDR (95% CI)	0.17* (0.16 to 0.19)	-0.03* (-0.04 to -0.01)

ASIR, age-standardized incidence rate; ASMR, age-standardized mortality rate; ASPR, age-standardized prevalence rate; ASDR, age-standardized disability-adjusted life years rate; AAPC, average annual percentage change; UI, uncertainty interval; CI, confidence interval; ASR, age-standardized rate; RA, rheumatoid arthritis. *Significantly different from 0 at alpha = 0.05 ($p < 0.05$).

in females. ASPR also increased in both sexes, with an AAPC of 0.50% (95% CI: 0.49–0.51) in males and an AAPC of 0.37% (95% CI: 0.36–0.38) in females. In contrast, ASMR declined in both sexes, with a greater decrease observed in females (AAPC = -1.93%, 95% CI: -1.98 to -1.88) than in males (AAPC = -1.26%, 95% CI: -1.33 to -1.19). ASDR showed divergent trends, increasing slightly in males (AAPC = 0.17%, 95% CI: 0.16–0.19) but decreasing marginally in females [AAPC = -0.03%, 95% CI: -0.04-(-0.01)].

Regional variation by SDI quintiles and GBD regions, 1990–2021

In 2021, ASRs varied across SDI quintiles. The ASIR of RA among MAA was 30.93 (95% UI: 20.95–42.29) per 100,000 in high-SDI regions and 11.10 (95% UI: 7.06–16.11) per 100,000 in low-SDI regions. The ASPR ranged from 488.09 per 100,000 in high-SDI regions to 185.92 per 100,000 in low-SDI regions (Supplementary Table 1).

Between 1990 and 2021, ASIR remained stable in high-SDI regions [31.15–30.93; AAPC = -0.02%, 95% CI: -0.06–(-0.01)], whereas it increased in low-middle SDI regions (11.43–15.00; AAPC = 0.90%, 95% CI: 0.89–0.91). ASPR increased across SDI quintiles, including in low-middle SDI regions (AAPC = 0.89%, 95% CI: 0.88–0.91). In contrast, ASMR declined in all SDI categories, including high-SDI regions [0.37–0.15; AAPC = -2.71%, 95% CI: -2.96-(-2.52)]. ASDR increased in several lower SDI quintiles but decreased in high-SDI regions (AAPC = -0.18%, 95% CI: -0.20 to -0.15).

Across the 21 GBD regions, considerable variation was observed in both levels and trends. ASIR increased in Andean Latin America (AAPC = 2.11%, 95% CI: 2.06 to 2.16), Southern Latin America (AAPC = 1.80%, 95% CI: 1.78–1.81), Central Asia (AAPC = 1.35%, 95% CI: 1.32–1.37), and North Africa and the Middle East (AAPC = 1.38%, 95% CI: 1.36–1.39), but decreased in high-income Asia Pacific (AAPC = -0.65%, 95% CI: -0.71-(-0.61)]. ASMR declined in most regions, including high-income Asia Pacific (AAPC = -4.16%, 95% CI: -4.43 to -3.92), but increased in Central Asia (AAPC = 5.66%, 95% CI: 4.54 to 6.92). ASDR increased in Andean Latin America (AAPC = 1.32%, 95% CI: 1.26–1.38), while it decreased in Southern Sub-Saharan Africa [AAPC = -0.69%, 95% CI: -0.76-(-0.63)] and high-income Asia Pacific [AAPC = -0.62%, 95% CI: -0.66-(-0.59)].

Country-level variation in RA burden, 1990–2021

Marked variation was observed across countries in both levels and temporal trends of RA burden (Figure 1). In 2021, Ireland reported the highest ASIR (71.33 per 100,000; 95% UI: 44.90–100.62), whereas Kiribati recorded the lowest (3.15 per 100,000; 95% UI: 1.79–4.88). Similar disparities were observed in prevalence, mortality, and DALYs.

Between 1990 and 2021, substantial heterogeneity was identified in country-level trends. The largest increase in ASIR was observed in Guatemala (AAPC = 2.18%, 95% CI: 2.16–2.20), and Guatemala, which also showed the greatest increase in ASPR (AAPC = 2.01%, 95% CI: 2.00–2.03). For mortality, Guyana exhibited the largest increase in ASMR (AAPC = 8.16%, 95% CI: 7.42–8.82), whereas Norway experienced one of the most pronounced declines [AAPC = -4.99%, 95% CI: -5.40-(-4.43)]. Regarding DALYs, Mauritius showed the highest increase in ASDR (AAPC = 2.08%, 95% CI: 1.57–2.74). Detailed country-level estimates are provided in Supplementary Table 2.

Temporal trends identified by joinpoint regression

Joinpoint regression identified distinct phase-specific changes in the global RA burden among MAA from 1990 to 2021 (Figure 2). The ASIR remained relatively stable during 1990–1996 (APC = 0.06), increased during 1996–2000 (APC = 0.29%), rose more markedly

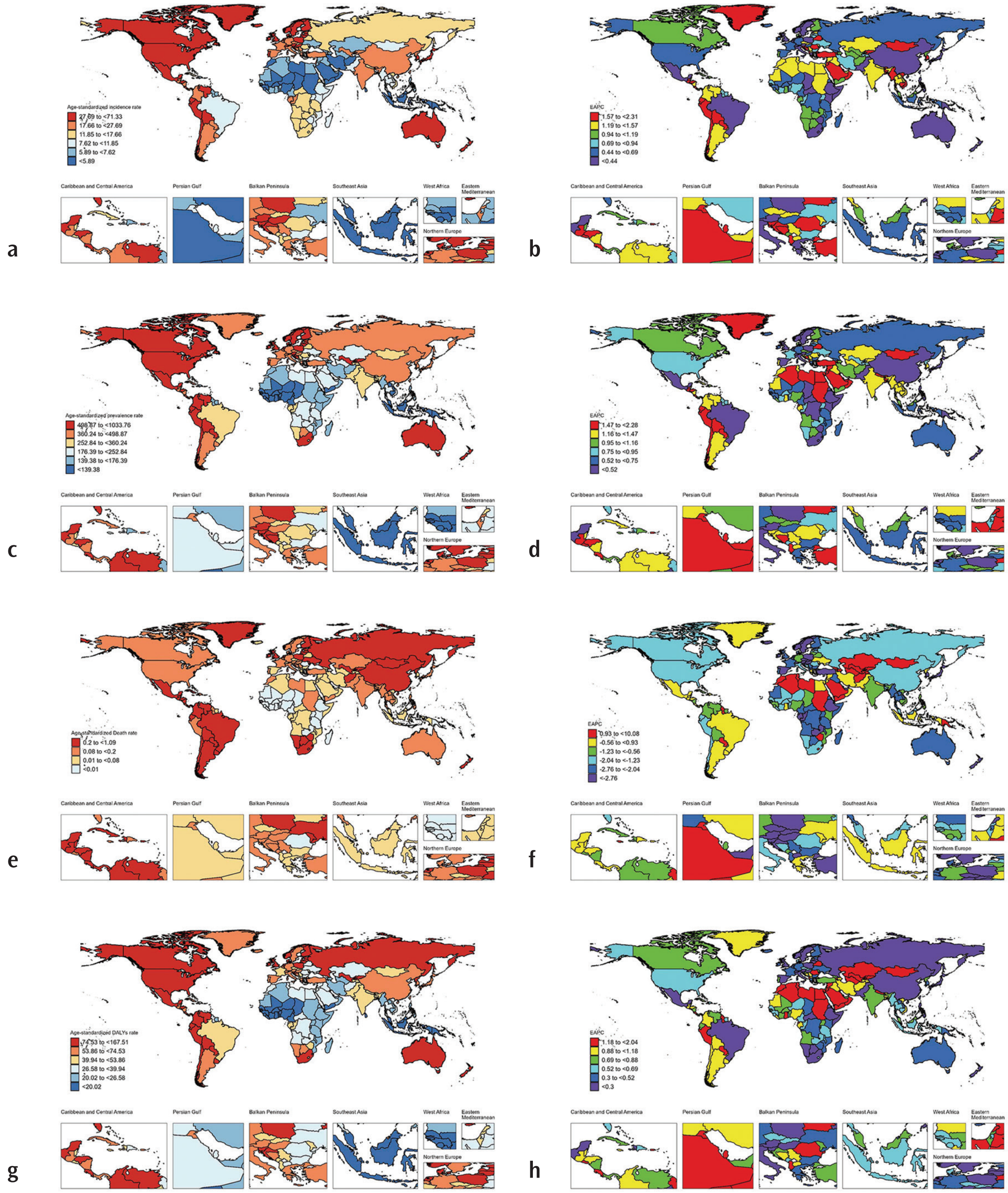

FIG. 1. Global distribution and temporal trends of age-standardized incidence (ASIR), prevalence (ASPR), death (ASMR), and disability-adjusted life years rates (ASDR) of rheumatoid arthritis among middle-aged adults in 2021 and their estimated annual percentage change (AAPC), 1990–2021. (a) ASIR; (b) AAPC of ASIR; (c) ASPR; (d) AAPC of ASPR; (e) ASMR; (f) AAPC of ASMR; (g) ASDR; (h) AAPC of ASDR.

during 2000–2009 (APC = 0.56%), and continued to increase at a slower rate during 2009–2021 (APC = 0.08%). The ASPR showed a similar pattern, with modest growth during 1990–1995 (APC = 0.11%), accelerated increases during 1995–2000 (APC = 0.49%) and 2000–2009 (APC = 0.80%), followed by continued but slower growth during 2009–2021 (APC = 0.25%). In contrast, ASMR declined during 1990–1998 (APC = -2.31%), increased briefly during 1998–2004 (APC = 0.83%), and then decreased substantially during 2004–2013 (APC = -2.82%) and 2013–2021 (APC = -1.70%). ASDR decreased during 1990–1992 (APC = -0.62%) and 1992–1998 (APC = -0.23%), increased during 1998–2005 (APC = 0.77%), and subsequently declined slightly during 2005–2021 (APC = -0.07%).

Association between disease burden and SDI

According to the GBD classification, countries and territories were grouped into five SDI quintiles (low, low-middle, middle, high-middle, and high). Across regions, ASIR and ASPR were positively correlated with SDI (ASIR: R = 0.60, $p < 0.001$; ASPR: R = 0.62, $p < 0.001$), with higher rates observed in high-SDI regions, such as Western Europe, high-income North America, and Australasia, and lower rates in Sub-Saharan Africa and South Asia. Andean Latin America and Central Latin America were located above the fitted regression line, whereas high-income Asia Pacific was below the fitted line. ASMR was negatively correlated with SDI (R = 0.27, $p <$ 0.001), with Central Asia and Eastern Sub-Saharan Africa positioned above the fitted line. ASDR showed a positive correlation with SDI (R = 0.56, $p < 0.001$), with higher values observed in Andean, Southern, and Central Latin America, whereas high-income Asia Pacific remained below the fitted line (Figure 3).

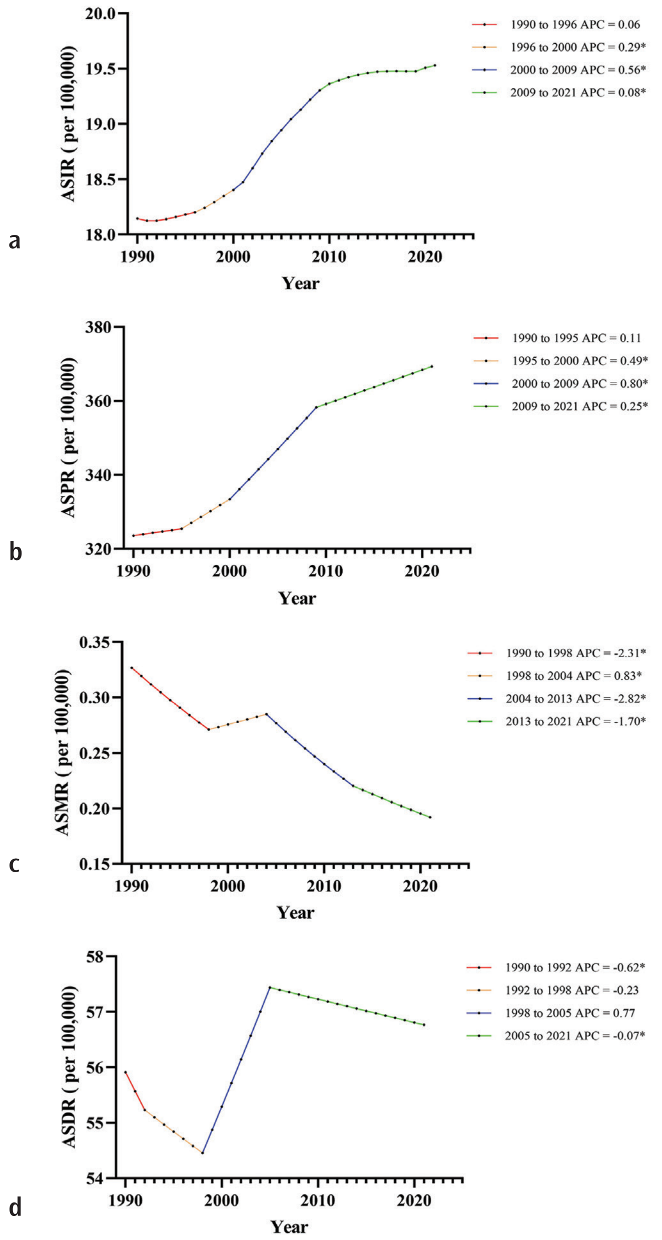

FIG. 2. Joinpoint regression analysis of global age-standardized incidence (ASIR), prevalence (ASPR), mortality (ASMR), and disability-adjusted life years rates (ASDR) of rheumatoid arthritis among middle-aged adults, 1990–2021. (a) ASIR; (b) ASPR; (c) ASMR; (d) ASDR. Solid lines indicate fitted trends, and segments are labeled with annual percentage change (APC, $p < 0.05$). APC, annual percentage change.

Projected disease burden through 2050

Using the BAPC model, projected global ASRs of RA among MAA from 2022 to 2050 are presented in Figure 4. The projected ASIR was 19.53 (95% CrI: 19.47–19.59) per 100,000 in 2021 and is expected to reach 19.22 (95% CrI: 3.76–34.67) per 100,000 by 2050. The projected ASPR was 368.73 (95% CrI: 368.45–369.00) per 100,000 in 2021 and 367.47 (95% CrI: 88.31–646.63) per 100,000 in 2050. The projected ASMR was 0.193 (95% CrI: 0.187 to 0.198) per 100,000 in 2021 to 0.111 (95% CrI: 0.001–0.313) per 100,000 by 2050. Similarly, the projected ASDR was 56.74 (95% CrI: 56.63–56.85) per 100,000 in 2021 and is projected to decrease to 52.18 (95% CrI: 9.06–95.31) per 100,000 in 2050. The CrIs increased over the projection period (Figure 4).

DISCUSSION

This study quantified the burden of RA among MAA (40–59 years) using GBD 2021 estimates from 1990 to 2021. It examined ASIR, prevalence, mortality, and DALY rates across locations and over time, and provided projections to 2050. By focusing on the 40–59-year age range—a life stage associated with high work and family responsibilities—our findings provide age-specific evidence that complements prior GBD-based assessments of RA across all ages.

At the global level, the ASIR and ASPR increased between 1990 and 2021, whereas the ASMR decreased and the ASDR changed only slightly. Similar patterns have been reported in recent global analyses of RA using GBD data, which consistently describe rising ASIR and ASPR alongside declining ASMR.[1] These divergent trends should be interpreted descriptively within the context of GBD model-based estimates. Changes in observed incidence and prevalence may reflect multiple factors, including temporal variations in case ascertainment, diagnostic access, and classification practices, in addition to underlying disease occurrence. In this context, the periods of accelerated increases in incidence and prevalence beginning in the early 2000s and extending into subsequent years coincide with major changes in RA management and classification frameworks. During this time, treat-to-target strategies were formalized and widely disseminated, and the 2010 ACR/EULAR classification criteria were introduced to facilitate earlier classification of RA compared with the 1987 criteria.[9,10] These contemporaneous developments may have contributed to changing epidemiological patterns; however, within the scope of the current analysis, they should be interpreted as potential contextual factors rather than established causal explanations.

Substantial geographic heterogeneity was observed across SDI quintiles and GBD regions. Incidence and prevalence were higher

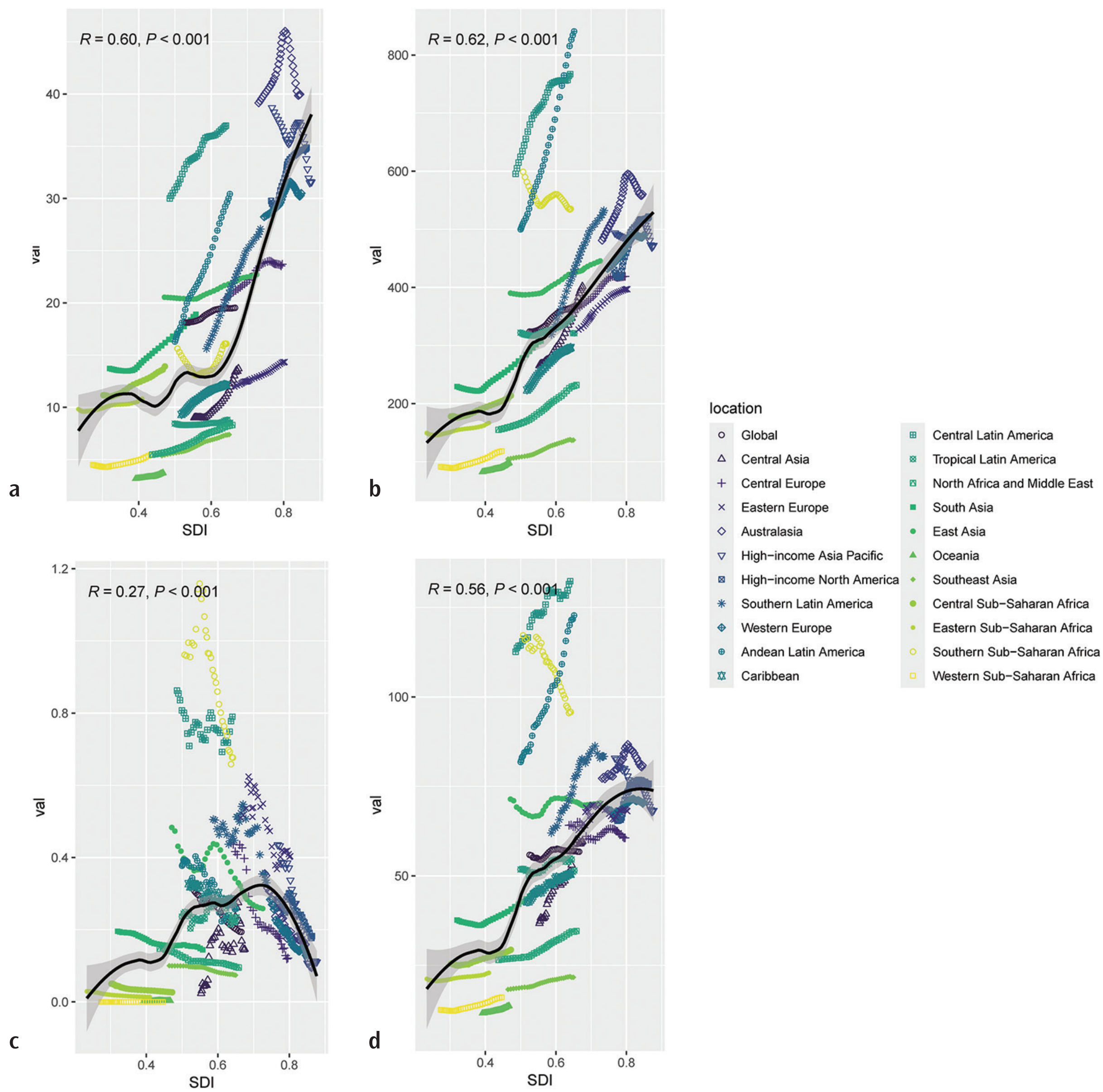

FIG. 3. Association between sociodemographic index and age-standardized incidence (ASIR), prevalence (ASPR), mortality (ASMR), and disability-adjusted life years rates (ASDR) of rheumatoid arthritis among middle-aged adults, 1990–2021. (a) ASIR across 21 Global Burden of Disease regions; (b) ASPR across 21 regions; (c) ASMR across 21 regions; (d) ASDR across 21 regions. Black lines represent fitted non-linear relationships; R indicates Pearson correlation coefficient. SDI, sociodemographic index.

in high-SDI settings, whereas mortality tended to be lower at higher SDI levels. This SDI pattern is consistent with the use of SDI in GBD studies as a composite indicator of development and with the broader observation that detection and reporting capacity, access to rheumatology services, and the availability of diagnostic tools often vary with health-system resources.[11] However, the SDI comparisons in this study are ecological and descriptive and therefore do not provide explanations for between-region differences. The country- and region-specific outliers shown in the SDI plots further highlight that substantial variation exists even among locations with similar SDI, supporting the value of presenting results at multiple geographic levels.

Given the strong sex-dimorphism of RA, sex-stratified results are essential for interpretability. Prior epidemiological studies consistently report a higher RA burden in females, with female-to-male ratios commonly exceeding 2:1.[1] Consistent with this established pattern, our sex-stratified estimates demonstrate substantially higher incidence and prevalence among females than males in the 40–59-year group, along with differences in mortality and DALYs trends. These findings underscore that analyses restricted

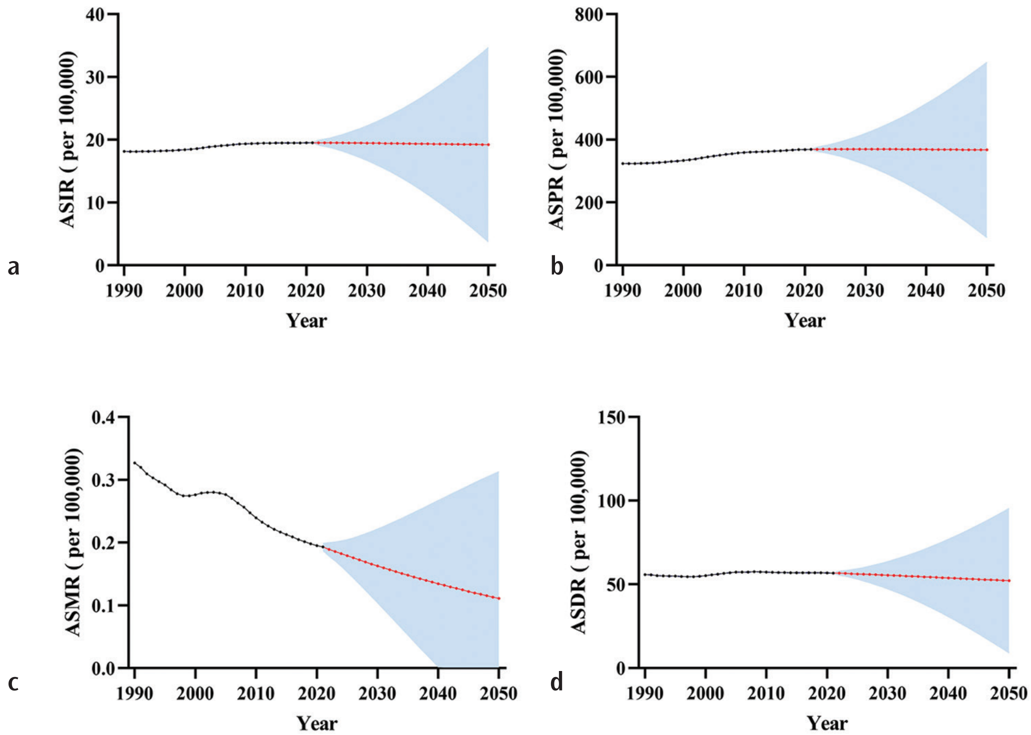

FIG. 4. Projected trends in age-standardized incidence (ASIR), prevalence (ASPR), mortality (ASMR), and disability-adjusted life years rates (ASDR) of rheumatoid arthritis among middle-aged adults through 2050. (a) ASIR; (b) ASPR; (c) ASMR; (d) ASDR. Red lines represent model-based projections from the Bayesian age–period–cohort model; shaded areas represent 95% uncertainty intervals.

to MAA remain clinically relevant only when sex-specific differences are explicitly presented.

Projections to 2050 suggested broadly stable incidence and prevalence, accompanied by declining mortality and DALYs among MAA. However, UIs widened over time. Such widening intervals are common in long-term projections and reflect the accumulation of uncertainty arising from both model structure and unobserved future conditions. The BAPC framework is widely used for epidemiological forecasting and can provide useful scenario-like estimates; nevertheless, projections should be interpreted with caution, particularly when UIs become very wide.[12,13] In this study, the large uncertainty bounds projected for 2050 should therefore be interpreted as indicating limited precision rather than exact predictions of future levels.

Several limitations should be considered. First, GBD 2021 estimates are model-based syntheses that rely on the availability and quality of underlying data sources as well as the assumptions embedded within the GBD modeling framework. Second, the ecological SDI analyses do not capture within-country heterogeneity and do not incorporate treatment coverage or health-system variables. Therefore, interpretations related to differences in diagnosis or access to therapy should be considered hypotheses rather than definitive inferences. Third, RA is a heterogeneous condition (e.g., seropositive vs. seronegative phenotypes), but the GBD outputs do not stratify RA by serostatus; future studies integrating subtype-specific data could enable more refined comparisons across settings. Finally, projections to 2050 inherently rely on modeling assumptions and should be interpreted alongside their UIs.

Overall, this GBD 2021–based analysis characterizes the long-term burden of RA among adults aged 40–59 years. The findings highlight a persistent burden, substantial geographic variation, clear sex differences, and substantial uncertainty in long-term projections. Together, these descriptive results complement existing global RA assessments by providing an age-specific perspective and may support comparative monitoring across regions and over time.

Using GBD 2021 estimates, this study quantified the global burden of RA among MAA (40–59 years) from 1990 to 2021 and examined projected trends through 2050. During the study period, the ASIR and ASPR increased, whereas the ASMR declined. The ASDR showed limited overall change. Substantial heterogeneity was observed across SDI quintiles, GBD regions, countries, and sexes. Projections to 2050 indicate that median ASIR and ASPR remain close to recent levels, whereas median ASMR and ASDR decline over time, with CrIs widening in later years. These age-specific findings complement existing all-age RA assessments and support comparative epidemiological monitoring across locations and over time.

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Complementary ROC-Derived Indices for Screening Improper Expression Profiles in RNA-Seq Differential Expression Analysis

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Background: Differential expression (DE) analysis of RNA sequencing (RNA-Seq) data are cornerstone of transcriptomic research. Widely used statistical frameworks are primarily optimized to detect monotonic mean shifts between conditions and may therefore overlook genes or microRNAs whose disease association arises at both low and high expression levels. Such non-monotonic patterns, referred to here as improper expression profiles, may reflect biologically relevant heterogeneity but remain difficult to identify using standard tools.

Aims: To evaluated whether receiver operating characteristic (ROC)-based indices, specifically the generalized area under the curve (gAUC) and the length of the ROC curve (LROC), can support exploratory screening and prioritization of improper expression profiles in RNA-Seq data, as a complement to conventional DE methods.

Study Design: Methodological study.

Methods: Using simulated negative binomial count data, we compared DESeq2, classical AUC (cAUC), gAUC, and LROC across varying sample sizes and dispersion levels, focusing on improper expression profiles. Performance was summarized using true positive rate and positive predictive value under ranking-based feature selection, including a one-shot benchmark operating point (available only in simulations) and sensitivity analyses across selection sizes. The methods were also applied to a publicly available CC miRNA dataset using heuristic post-hoc screening rules informed by simulation diagnostics.

Results: cAUC was largely insensitive to improper expression patterns. DESeq2 performed robustly for conventionally differentially expressed features but recovered a smaller fraction of simulated improper profiles under ranking-based selection. Across simulation scenarios, gAUC showed the highest and most stable recovery of improper profiles, whereas LROC provided complementary signal under low-to-moderate dispersion but degraded under extreme overdispersion. In the CC dataset, ROC-derived indices identified candidate improper miRNAs that were not prioritized by DESeq2, and several top candidates had literature support consistent with biological plausibility.

Conclusion: gAUC, supported by LROC as an auxiliary index, provides a practical ROC-based screening extension to standard RNA-Seq workflows. Because these indices are applied using heuristic thresholds without controlled error rates, the resulting candidates should be interpreted as exploratory prioritization and require independent validation.

INTRODUCTION

Gene expression reflects the transcription of DNA into RNA and, in many cases, its subsequent translation into proteins that regulate cellular function. In contrast, the collection of all expressed RNA molecules defines the transcriptome. Transcriptomic profiling provides a dynamic snapshot of cellular activity and is essential for understanding how gene regulation changes across biological conditions and disease states.[1] Disruptions in transcriptomic patterns are central to disease mechanisms, including cancer development and therapeutic resistance, thereby positioning transcriptomic analysis as a foundation for precision medicine and biomarker discovery.[2,3] While early transcriptome studies relied on microarray technologies,[4-6] RNA sequencing (RNA-Seq) has become the dominant platform due to its higher sensitivity, broader dynamic range, and ability to identify novel transcripts.[1,7] A major application of RNA-Seq

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is differential expression (DE) analysis, which detects differences in gene expression across biological conditions and links transcriptional changes to disease mechanisms and treatment response.[8,9]

To perform DE analysis of RNA-Seq data, several statistical frameworks are widely used, including DESeq2, edgeR, and limma-voom. DESeq2 and edgeR model count data using negative binomial generalized linear models with dispersion estimation and shrinkage.[8,10] These methods perform reliably even with small numbers of replicates and are effective in resource-limited experimental designs. In contrast, limma-voom uses mean–variance modeling to derive precision weights for linear modeling of log-scale expression values. This approach offers computational efficiency and stability across large gene sets.[11] Alternative approaches include baySeq,[12] PoissonSeq,[13] and SAMseq,[14] and a broader overview of DE analysis tools is provided by Rosati et al.[15] These methods provide robust inferential testing for features whose association with condition is well captured by group-level mean differences (i.e., consistent up- or downregulation). However, because their primary hypotheses target mean shifts, they may be less sensitive to non-monotonic relationships in which both unusually low and unusually high expression levels are informative. In this study, we refer to such features as improper genes (IGs), also described as bimodal patterns in the literature.

The presence of genes or microRNAs (miRNAs) with improper expression profiles is now well recognized in biomedical research, as both very low and very high expression levels can contribute to disease. This “Goldilocks” phenomenon, in which either loss-of-function or gain-of-function of the same gene can lead to pathology, has been observed in diverse contexts.[16-19] Similarly, many miRNAs may play dual roles as tumor suppressors or oncogenes depending on their expression levels, underscoring the importance of balanced expression in normal physiology.[20] Detecting these patterns is important because they may indicate hidden biological variability, distinct disease subtypes, or multiple mechanisms operating within the same condition.

Several statistical and computational frameworks have been proposed to detect bimodal or multimodal expression patterns in microarray data. Early efforts, such as the Cancer Outlier Profile Analysis, aimed to identify oncogenes with extreme expression in a subset of tumor samples and successfully uncovered events such as the TMPRSS2–ERG fusion in prostate cancer.[21,22] Extensions include methods based on outlier detection and subgroup identification statistics,[23-25] clustering approaches,[26] and bimodality filtering techniques.[27] However, these methods were primarily developed for microarray platforms, which assume approximately Gaussian distributions, thereby limiting their direct applicability to RNA-Seq data characterized by discrete and overdispersed count distributions.

To address this limitation, two general strategies have emerged: (i) adapting existing workflows through RNA-Seq-specific extensions, such as SIBER[28] and GMMchi,[29] or enabling the use of microarray-based methods by applying variance-stabilizing transformations (VSTs) that render count data approximately normal; and (ii) employing distribution-free approaches, including receiver operating characteristic (ROC)-based frameworks, which evaluate discriminatory performance independently of the underlying expression distribution. ROC-derived indices typically provide descriptive scores and rankings rather than formal hypothesis tests with *p* values or explicit error-rate control.

In this study, we evaluated ROC-derived indices alongside DESeq2, treating DESeq2 as the primary hypothesis-testing framework for DE and ROC-based indices as complementary exploratory screening tools. Specifically, ROC-based rankings were used to prioritize candidate genes and miRNAs with improper expression profiles that may be missed by mean-shift–focused testing. This evaluation was conducted through simulation studies and illustrated using a cervical cancer (CC) miRNA-Seq dataset, with literature-based plausibility assessments for selected candidates.

MATERIALS AND METHODS

DE analysis

Several frameworks exist for RNA-Seq DE analysis; see Rosati et al.[15] for an overview. In this study, DESeq2 was used as the primary hypothesis-testing benchmark for DE inference.[8] ROC-derived indices were used only as complementary screening scores, providing rankings without *p* values or formal error-rate control.

IG expression profiles

IGs, introduced conceptually in the Introduction, exhibit non-monotonic relationships with phenotypes, where both unusually low and high expression levels may be associated with disease. In this study, these profiles were defined using the toy distributions shown in Figure 1. These simplified scenarios illustrate two generic mechanisms that can generate improper expression profiles: (i) a “high-low” mixture, in which diseased samples occupy both extremes of the expression distribution (Figure 1, middle panel), and (ii) group-specific variance inflation, in which one group shows markedly higher dispersion without a corresponding mean-shift (Figure 1, right panel).

In the simulation study, IGs were generated using the high-low mixture mechanism. A separate variance-inflation-only mechanism was not explicitly simulated, which limits the generalizability of conclusions to other forms of improper expression profiles. The variance-based scenario is retained here to highlight an alternative pathway through which non-monotonic signals may arise in practice and which ROC-based methods are also expected to capture.

Data preprocessing

All count-based datasets, including simulated and real RNA-Seq data, were processed using a standard pipeline implemented in the DESeq2 framework. Raw counts were first filtered by removing low-quality genes (i.e., retaining genes with counts ≥10 in at least three samples), followed by near-zero-variance filtering using the nearZeroVar function from the caret R package.[30] The remaining counts were normalized using DESeq2’s median-of-ratios method to correct for library size and compositional effects, after which a VST was applied to obtain continuous-scale expression values with approximately stabilized variance across the dynamic range. Unless

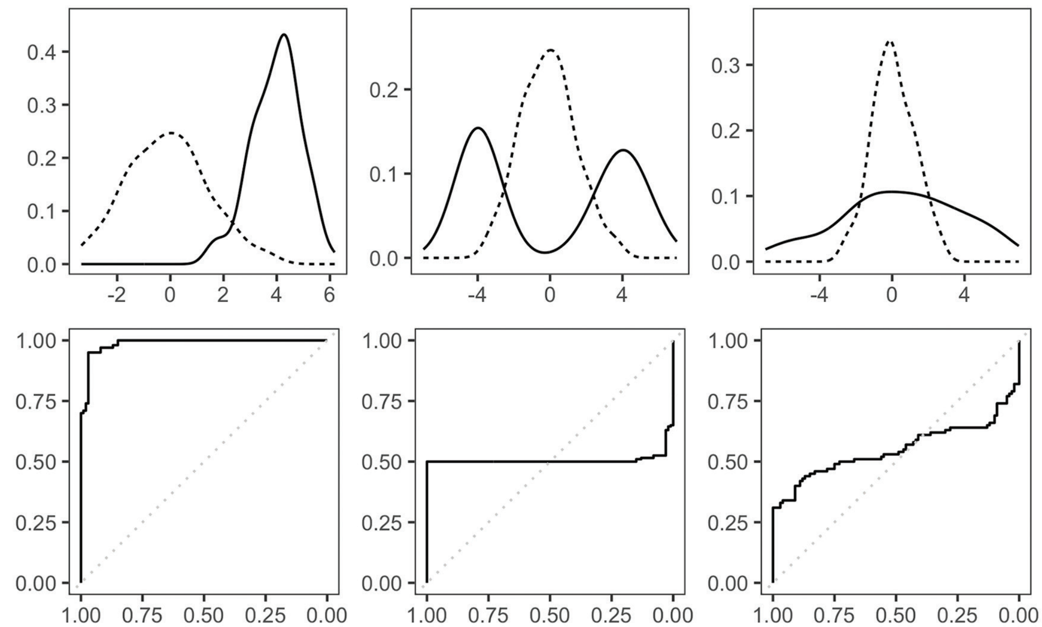

FIG. 1. Distribution of proper and improper gene expression profiles and corresponding ROC curves—(left) proper, (middle/right) improper; dashed lines correspond healthy group. ROC, receiver operating characteristic.

otherwise stated, all ROC-based analyses and visualizations were performed using these variance-stabilized expression values.

Detecting DEGs with improper expression profiles

DE analysis in this study was performed using the DESeq2 workflow. In addition to DESeq2's hypothesis-testing-based strategy for identifying differentially expressed genes (DEGs), we applied three ROC-based methods as supporting tools for gene selection and downstream performance evaluation, specifically targeting DEGs with improper expression profiles: (i) the classical area under the ROC curve (cAUC), (ii) the generalized area under the curve (gAUC), and (iii) the length of the ROC curve (LROC). Thus, cAUC, gAUC, and LROC were not used as standalone DE analysis methods; rather, they were integrated into the DESeq2-based workflow and applied to the preprocessed, variance-stabilized expression values described in the "Data preprocessing" subsection. The gAUC extends the conventional AUC by allowing dual thresholds, thereby capturing cases in which both high and low expression levels are informative.[31] In contrast, LROC quantifies discriminatory ability based on the LROC and may capture signals that conventional AUC underestimates.[32] Here, cAUC, gAUC, and LROC were used as descriptive screening scores for VST-transformed expression values. These methods provide rankings without p values or type I error control; therefore, all ROC-based results are interpreted as exploratory prioritization rather than formal DE inference.

Simulation study and real-data analysis

comprehensive simulation study was designed to evaluate the performance of DESeq2, cAUC, gAUC, and LROC in detecting IGs under realistic RNA-Seq conditions. We evaluated each method across 18 simulation scenarios defined by sample size (n), number of features (p), proportions of DEGs (π_{DEGs}) and DEGs with improper profiles (π_{IR}), overdispersion (ϕ), and the dispersion of log-fold-change values. For each scenario, 1,000 datasets were generated. The overall analysis pipeline is shown in Figure 2.

Performance evaluation and sensitivity analysis: Method performance was summarized using the true positive rate (TPR) and positive predictive value (PPV), computed for DEGs and IGs (a subset of true DEGs). For each dataset replicate, each method produced a ranked feature list. DESeq2 features were ranked by Benjamini-Hochberg-adjusted p values ($\log_2$ fold change was not used for ranking), whereas cAUC, gAUC, and LROC features were ranked by their respective ROC-derived scores. Because the nominal number of DEGs is fixed by design at, we first report a reference "one-shot" benchmark at to summarize ranking performance at the design operating point. Due to prefiltering, the number of retained true DEGs may vary around across replicates. We then performed a sensitivity analysis by selecting the top-K features for and evaluating how TPR and PPV changed with K, thereby reflecting the trade-off between true and false positives under ranking-based screening.

CC dataset: In addition to the simulation study, we analyzed a real RNA-Seq dataset originally published by Witten et al.[33] The dataset comprises small RNA-Seq measurements from 58 human cervical tissue samples, including 29 tumor and 29 matched non-tumor controls. Sequencing was performed using the Illumina (Solexa) platform, yielding expression profiles for 714 distinct miRNAs. The data are publicly available in the Gene Expression Omnibus under accession number GSE20592.

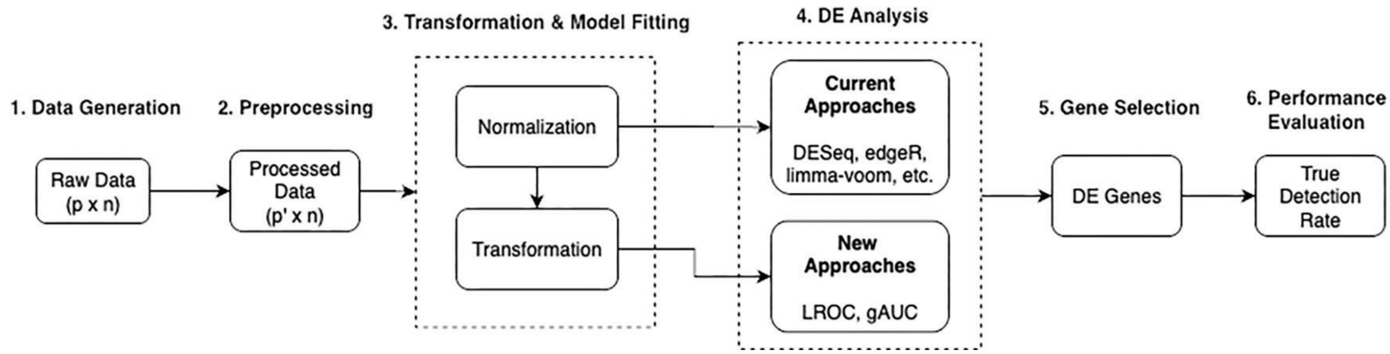

FIG. 2. Simulation workflow. Number of total features (p) is fixed at 3,000. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve; DE, differential expression.

In the CC analysis, candidate improper miRNAs were identified using heuristic cut-offs informed by simulation-based diagnostic plots: combined with either or . Because these thresholds are post-hoc screening rules rather than formal inferential tests, findings based on ROC-derived indices are considered exploratory and are interpreted alongside the DESeq2-based results.

For detailed information, including the mathematical background of the ROC-based approaches, simulation workflow, and dataset description, see [Supplementary Materials (Section A)].

RESULTS

This section summarizes the comparative performance of DESeq2, cAUC, gAUC, and LROC based on simulation studies and presents an illustrative application to a CC miRNA dataset, accompanied by a literature-based plausibility assessment.

Performance evaluation

Using the simulation framework described in the Materials and Methods section, we evaluated the performance of DESeq2 and three ROC-derived screening indices (cAUC, gAUC, and LROC) under varying RNA-Seq conditions. Performance metrics were computed from 1000 simulated datasets and summarized across levels of overdispersion (very low, moderate, and very high) and sample sizes (100, 300, and 500) for both DEGs and IGs. Figure 3 presents results for the low-DE scenario (π_{DEGs}=0.05), in which differences among methods are most pronounced and comparative interpretation is most informative. Corresponding high-DE results (π_{DEGs}=0.30) are provided in the [Supplementary Materials (Section B)]. We focus on the low-DE setting because it yields conclusions consistent with the high-DE case while providing clearer contrasts among methods.

Across all scenarios, increasing sample size improved detection performance for all methods, although the magnitude of improvement varied. The cAUC performed worst overall, particularly for IGs, with TPR values often below 0.25, indicating weak discriminatory ability. DESeq2 showed strong and stable performance for DEGs but lower sensitivity for IGs. At the benchmark level, gAUC and LROC achieved higher recovery of IGs than DESeq2 under very low and moderate overdispersion, reflecting their ability to capture non-monotonic expression patterns.

Under very high overdispersion, particularly for IGs, LROC performance declined substantially and, in some cases, fell below that of DESeq2, whereas gAUC remained robust, with median detection rates exceeding 0.75 except at small sample sizes. Increasing overdispersion also increased variability in detection performance across all methods, while low-dispersion settings produced more stable results. Overall, the one-shot results summarize ranking behavior at the design operating point, whereas practical trade-offs are further examined in the sensitivity analysis below.

Sensitivity analysis

To evaluate the sensitivity of the methods to the choice of DEG set size, we examined both TPR and PPV for DEGs and IGs across a range of selection sizes. Although the simulation design specifies a nominal DEG count of 150 per dataset, prefiltering can result in varying numbers of retained true DEGs around this value across replicates. This empirical range, which may vary slightly across simulation scenarios, is indicated by the gray-shaded band in Figures 4-7. Because PPV directly reflects the proportion of false positives among selected features, these curves summarize the sensitivity analysis, that is, the trade-off between true detection and false positives under ranking-based screening. We did not calibrate a nominal false-positive rate (FPR) for the ROC-derived indices using an inferential procedure; therefore, false positives are summarized empirically via PPV rather than through a controlled error rate, while empirical FPR curves are reported in the [Supplementary Materials (Section B)] as a specificity benchmark.

For DEGs, the sensitivity analysis indicated that all methods achieved broadly similar TPR values when only a small number of features were selected. TPR increased as the selection size grew; however, once the selected set size exceeded the retained number of true DEGs (i.e., the gray-shaded band), marginal gains diminished and method-specific differences became more apparent. Selecting additional features beyond this range primarily increased false positives rather than improving true detection (Figures 4 and 5). In this setting, DESeq2 showed a modest advantage under small sample sizes and very high overdispersion.

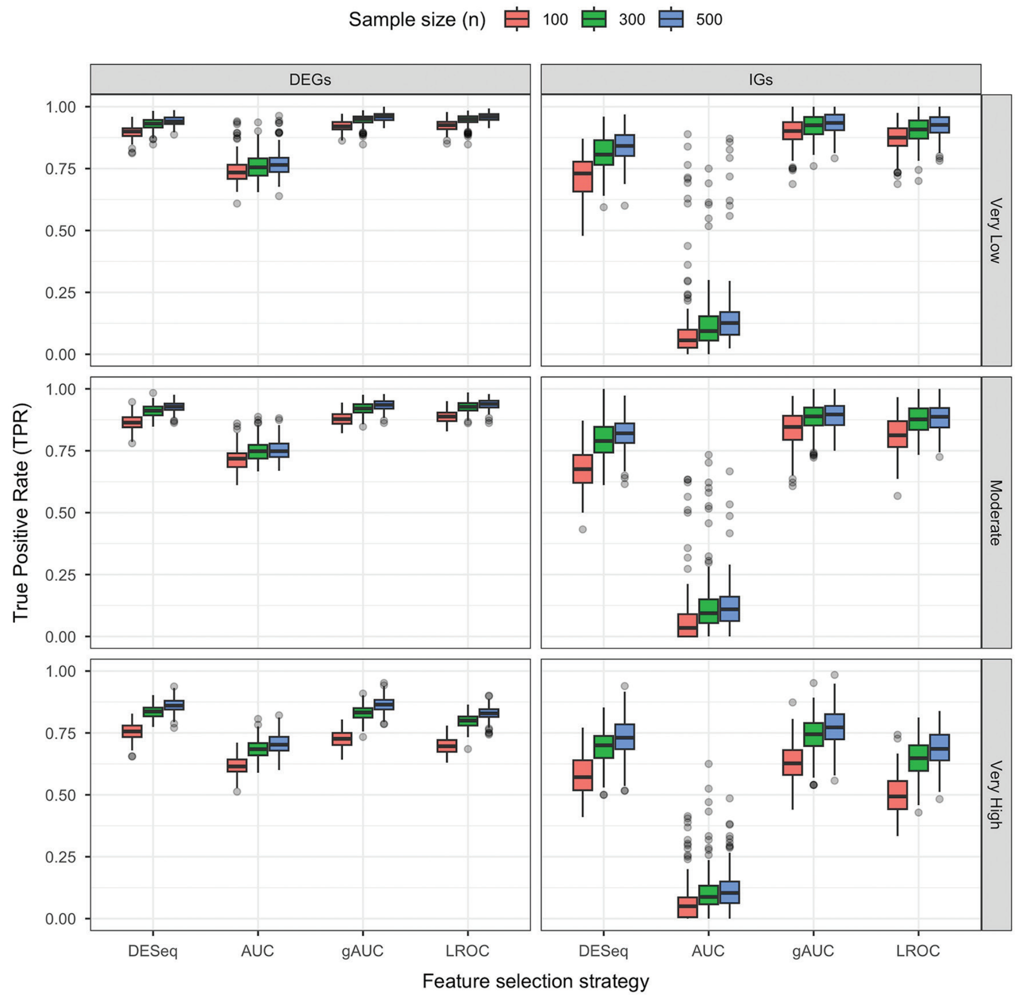

FIG. 3. Simulation results at the one-shot benchmark. The figure reports TPRs across simulated datasets when the selected set size is fixed at the nominal design value K=150 (i.e., p × π_{DEGs}) under the low-DE scenario π_{DEGs} = 0.05). This operating point is available only in simulations and is used here as a descriptive ranking benchmark. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve; AUC, area under the curve; TPRs, true positive rates; DE, differential expression.

For IGs, gAUC consistently achieved the highest TPR and PPV. LROC generally ranked second, except under very high overdispersion (Figure 6). Notably, gAUC began recovering IGs even at small selection sizes, whereas other methods showed limited sensitivity in this range. Under very high overdispersion, DESeq2 also began recovering IGs at smaller selection sizes than in low-to-moderate overdispersion settings and generally ranked second after gAUC (Figure 7).

Visual inspection of improper expression profiles

To further characterize the behavior of ROC-based metrics, we visually examined their joint distributions in a simulated dataset under the very high overdispersion scenario (i.e., and). Figure 8 provides a diagnostic view of how the ROC-derived indices behave across different simulated feature types. IGs tend to cluster at low cAUC values (approximately 0.5-0.65) but exhibit higher LROC values and often elevated gAUC despite low cAUC, reflecting non-monotonic expression patterns. In this high-dimensional setting, visual separation is descriptive and does not quantify statistical uncertainty; therefore, these plots are interpreted as exploratory diagnostics that may guide screening and threshold tuning rather than as formal evidence of signal.

CC results

As a real-data counterpart to the simulation analyses, we applied the proposed methods to a CC miRNA dataset. After filtering low-quality reads and near-zero-variance features, 416 miRNAs were retained for downstream analysis. DE analysis was first performed using DESeq2, with significance defined by an absolute $\log_2$ fold-change greater than 0.6 and a Benjamini-Hochberg-adjusted *p* value

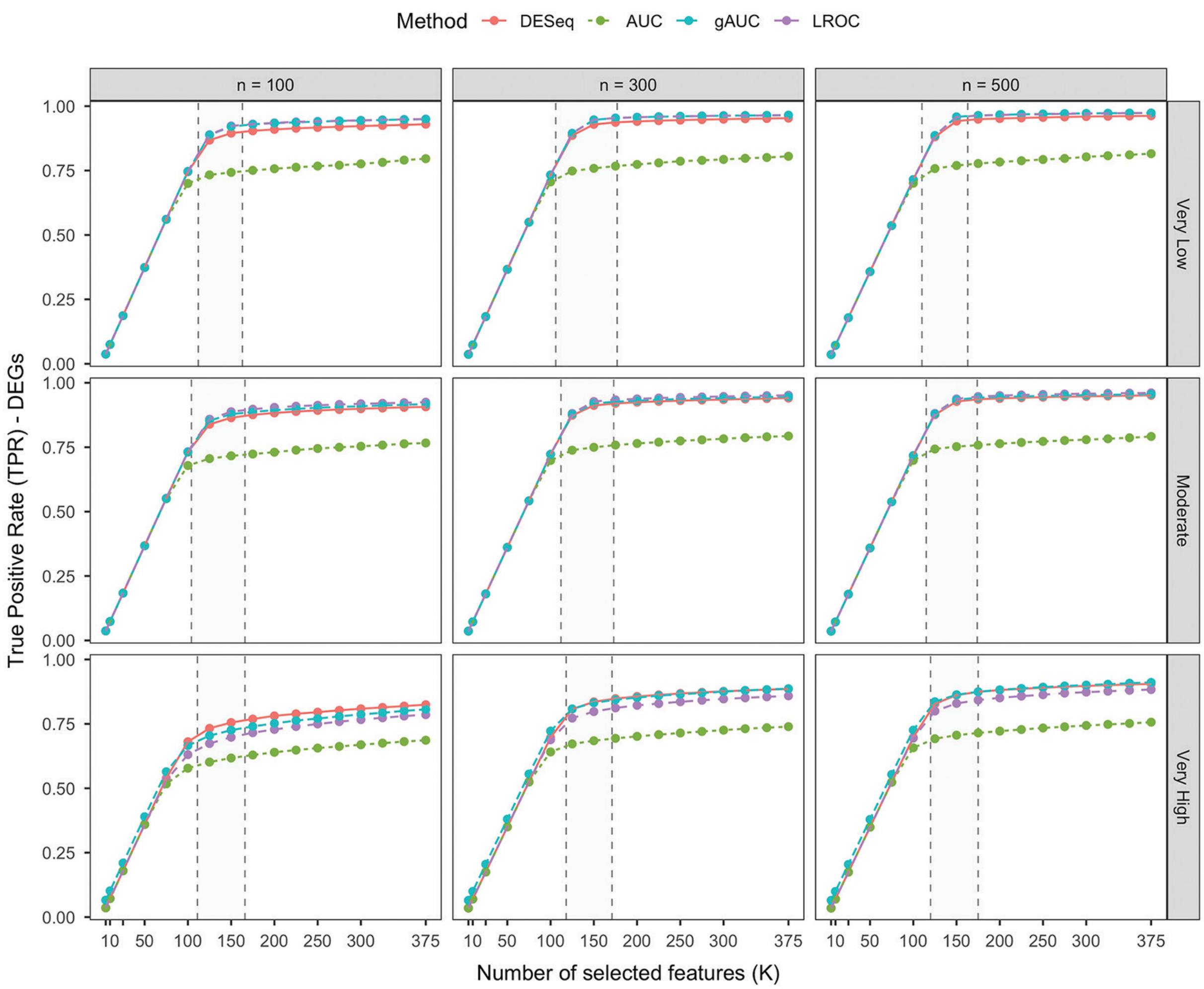

FIG. 4. Sensitivity analysis of the methods across simulations under the low-differential expression scenario—true positive rate for DEGs. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve; AUC, area under the curve; DEGs, differentially expressed genes.

below 0.05. Subsequently, gAUC and LROC were applied as screening scores to identify candidate improper miRNA profiles that may not be prioritized by DESeq2. Visual inspection of the ROC-based indices (Figure 9) revealed patterns similar to those observed in the simulation diagnostics (Figure 8).

The volcano plot (Figure 10) summarizes the DESeq2 results, with slight vertical jitter applied to improve visual clarity near the significance boundary. miRNAs flagged by the heuristic screening rule but not detected by DESeq2 were considered candidate improper miRNAs and were primarily located in the non-significant region of the volcano plot. Using this rule, 56 miRNAs were identified as candidate improper miRNAs, of which 34 were not detected by DESeq2 (Figure 11).[34] ROC curves for the top candidates (Figure 12) exhibit non-monotonic patterns, suggesting potential associations between both low and high expression levels and tumor status. Collectively, these results illustrate how ROC-derived screening indices can complement DESeq2 by prioritizing candidate non-monotonic miRNA profiles for downstream follow-up, such as literature-based plausibility assessment.

DISCUSSION

In this study, we evaluated ROC-derived indices (cAUC, gAUC, and LROC) as complementary screening tools alongside DESeq2. Because DESeq2 primarily targets mean-shift signals, it may deprioritize genes and miRNAs with IG profiles. Our results suggest that ROC-derived rankings can help flag candidate improper features for follow-up analysis; however, these indices do not provide *p* values or formal error control and should therefore be interpreted as exploratory diagnostics.

Importance of IG expression profiles: IG expression profiles have important biological relevance beyond their statistical characterization. For example, in breast cancer, the estrogen receptor gene exhibits a bimodal distribution that distinguishes ER-

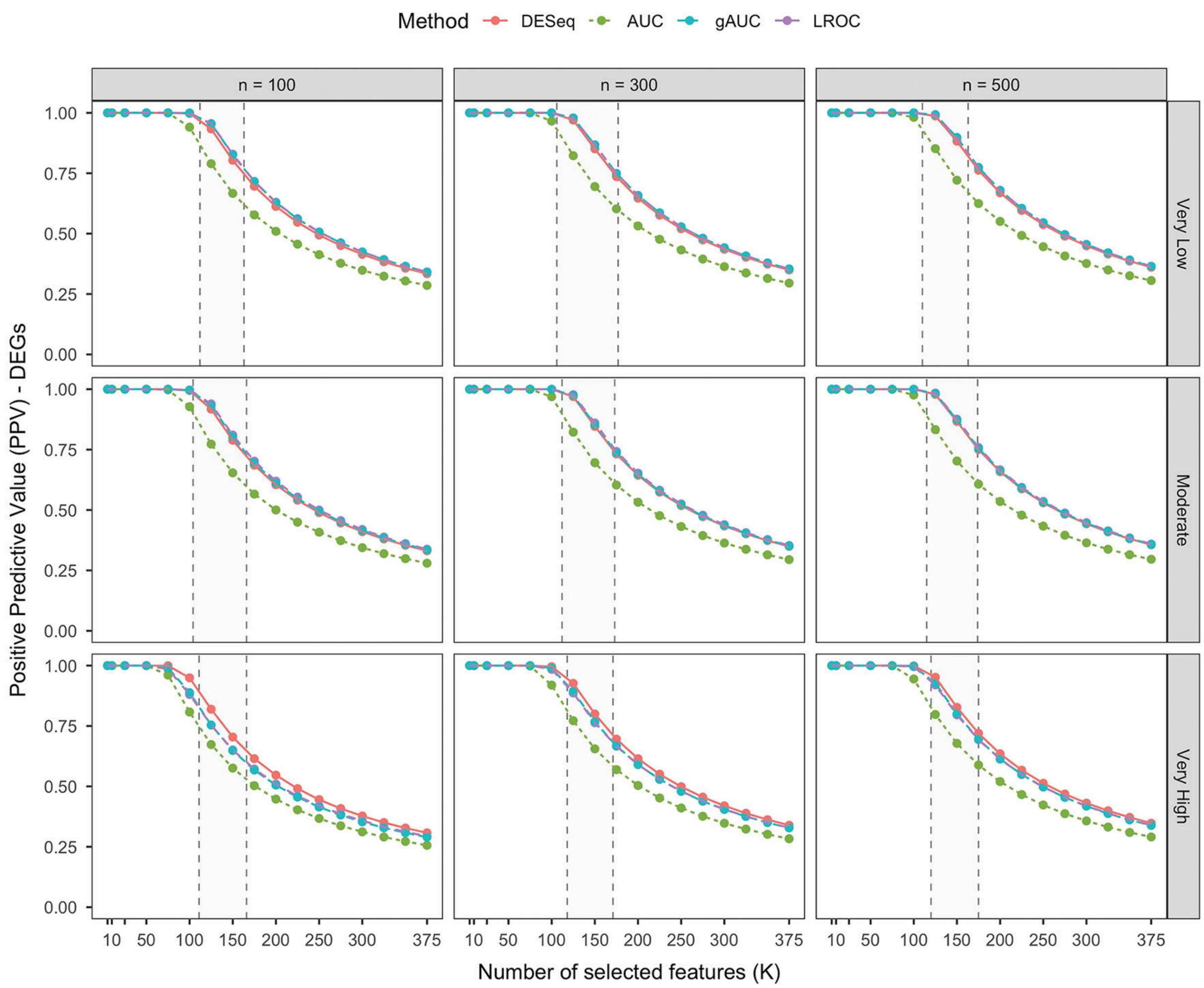

FIG. 5. Sensitivity analysis of the methods across simulations under the low-differential expression scenario—positive predictive value for DEGs. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve; AUC, area under the curve; DEGs, differentially expressed genes.

positive and ER-negative subtypes. In contrast, tumor-suppressor pathways involving genes such as PTEN may be disrupted by both loss-of-function and aberrant overexpression.[35,36] The activation marker TNFRSF9 also shows a bimodal expression pattern in tumor-infiltrating regulatory T-cells in non-small-cell lung cancer.[17] Together, these examples illustrate that transcriptomic heterogeneity can manifest as bimodality in both mRNA and miRNA expression, underscoring the need for approaches that can flag non-monotonic expression–disease relationships as candidates for further investigation.

Conventional DE tools primarily detect mean-shift signals and may miss improper expression patterns. Earlier microarray-based studies sought to address this limitation. Parodi et al.[37] introduced “not proper ROC curves” and the ABCR statistic to detect genes whose ROC curves cross the random expectation line, indicating subgroup effects. Silva-Fortes et al.[38] proposed the “arrow plot,” which combines ROC AUC and overlap measures to identify bimodal or subgroup-specific expression patterns missed by standard approaches. Building on these conceptually related ROC-based ideas, we use gAUC and LROC as screening indices and illustrate their behavior through simulations and a real dataset application.

Insights from simulation study*:* In our simulations, ranking behavior was primarily influenced by sample size and overdispersion. At the one-shot benchmark (available only in simulations), gAUC and LROC generally recovered more IGs than DESeq2 under low-to-moderate overdispersion, whereas LROC performance deteriorated under extreme overdispersion. The sensitivity analysis summarizes the trade-off between true positives and false positives under ranking-based screening. This limitation likely reflects LROC’s reliance on a parametric binormal assumption, which may be violated in RNA-Seq data. Future work may address this limitation by developing non-parametric or regularized variants of LROC to improve robustness while preserving its geometric interpretability.

Although we did not explicitly simulate an IG-generating mechanism driven solely by variance inflation, ROC-derived screening may,

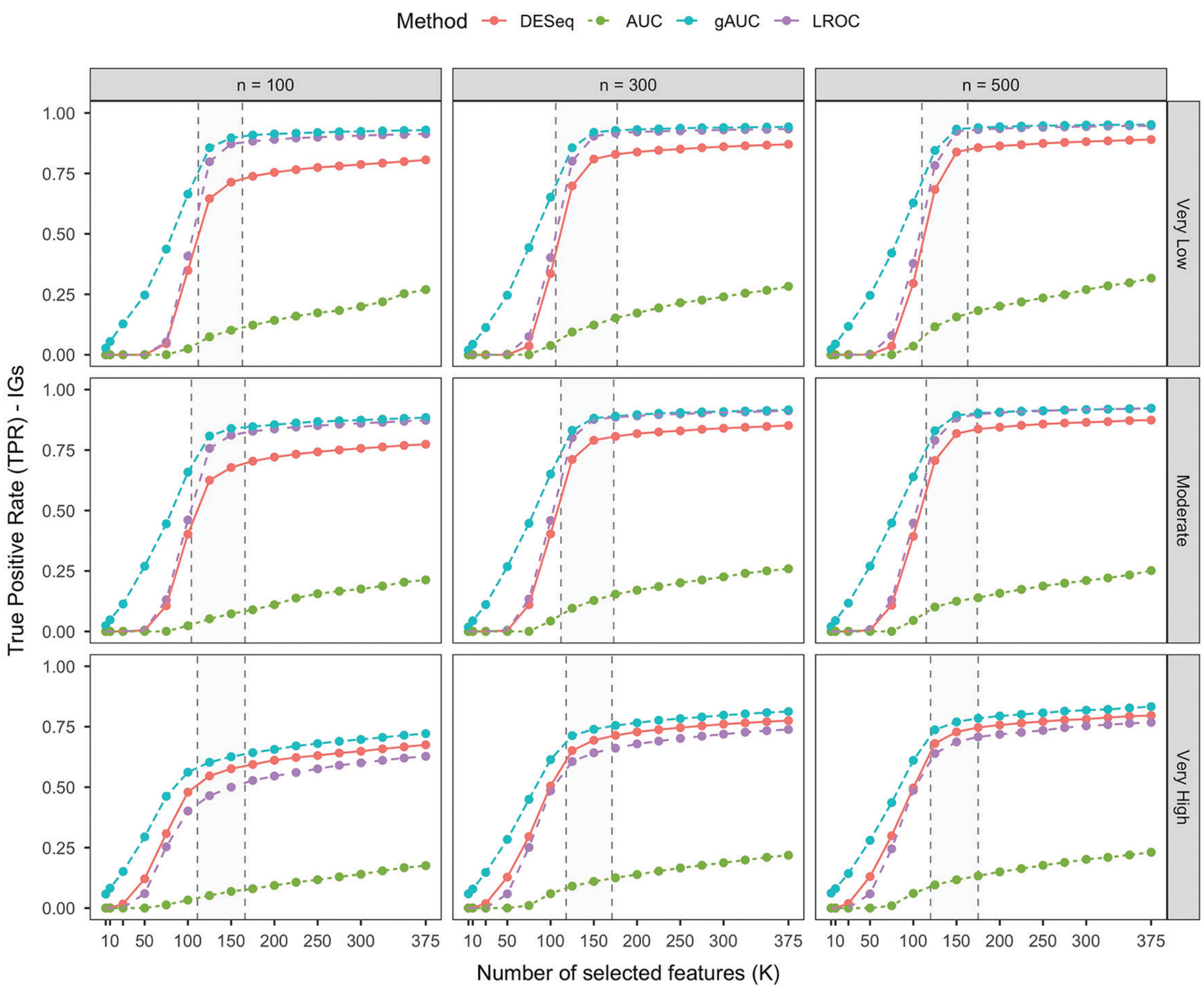

FIG. 6. Sensitivity analysis of the methods across simulations under the low-differential expression scenario — true positive rate for improper genes. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve; AUC, area under the curve; IGs, improper genes.

in principle, detect IGs arising from variance inflation–driven differences; however, this scenario was not evaluated in the present study. We also observed that DESeq2 recovered a larger fraction of simulated IGs under very high overdispersion. One plausible explanation is that, as dispersion increases, some mixture-generated IGs may resemble variance-inflation–like profiles or exhibit stronger mean-shift components, thereby becoming more detectable by DESeq2. Consequently, under a dedicated variance-inflation simulation design, the unimodal structure of a variance-inflation-only mechanism may lead DESeq2 and ROC-based screening to produce more similar IG recovery patterns. This hypothesis should be evaluated in future work.

Visual diagnostics of improper profiles: Beyond numerical performance metrics, we propose visual inspection of the AUC space (e.g., plotting cAUC against gAUC and/or LROC) as a diagnostic tool. In our simulated data, DEGs, IGs, and non-significant genes occupied distinct regions: DEGs showed high cAUC and gAUC, IGs showed moderate cAUC but high gAUC, and non-DE genes clustered near baseline. Such plots are descriptive and do not quantify statistical uncertainty; therefore, they are intended as exploratory diagnostics to guide screening and threshold selection.

In real-data applications, we used heuristic cut-offs, which should be interpreted as post-hoc screening rules. Because these cut-offs depend on data structure and analyst objectives, they are inherently subjective: more permissive thresholds increase sensitivity but also raise FPRs, whereas more stringent thresholds reduce false positives at the cost of increased false negatives. Accordingly, threshold tuning should reflect the intended trade-off between sensitivity and the burden of false discoveries.

Application to CC miRNA data: CC remains a significant global health burden. In the CC miRNA dataset, ROC-derived screening indices identified candidate improper miRNAs that were not prioritized by DESeq2 (Figures 10-13). Using the heuristic rule described in the materials and methods section, we identified candidate improper miRNAs for downstream analysis and summarized their patterns

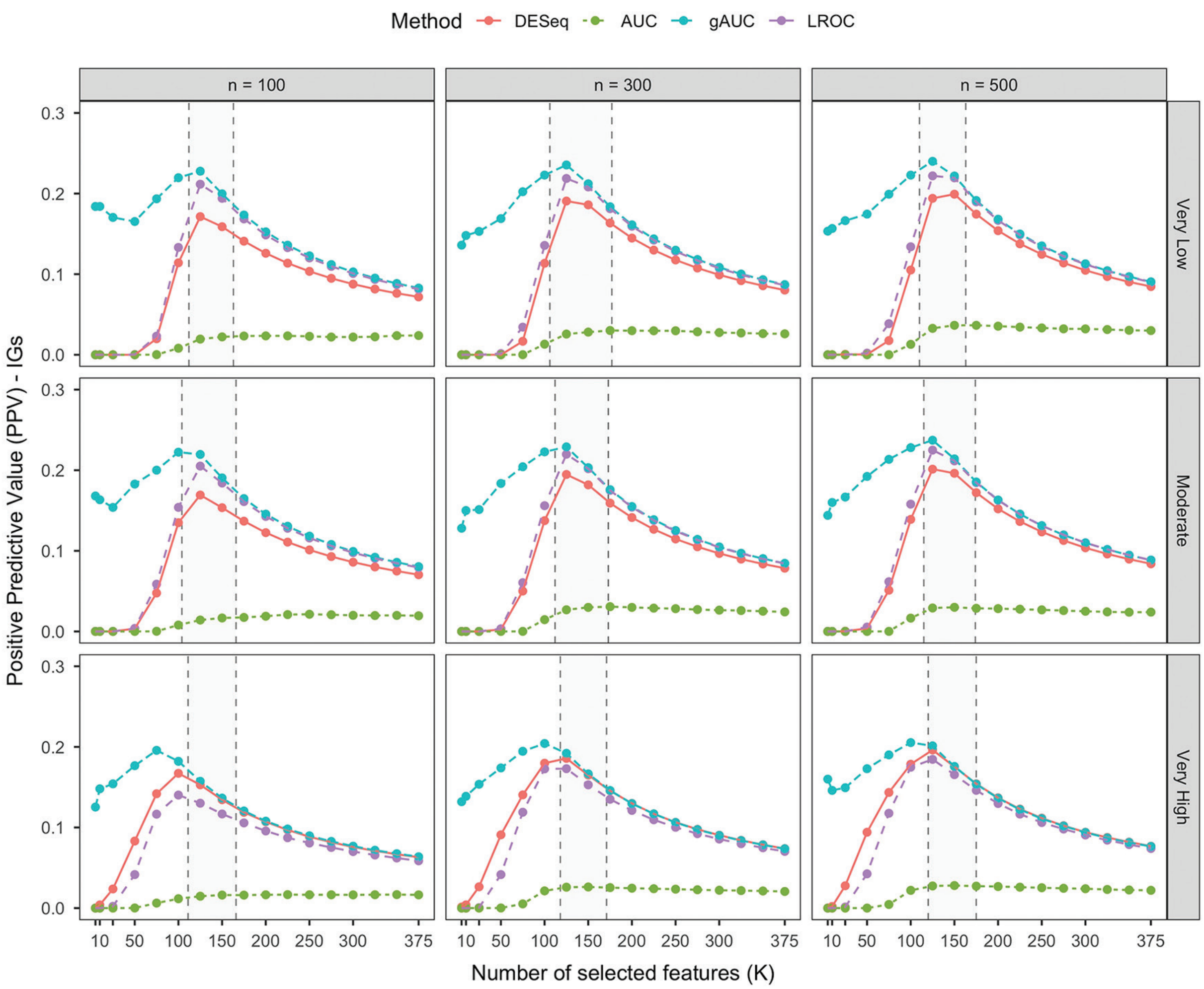

FIG. 7. Sensitivity analysis of the methods across simulations under the low-differential expression scenario—positive predictive value for improper genes. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve; AUC, area under the curve; IGs, improper genes.

using diagnostic plots. These findings are exploratory and intended for prioritization rather than confirmation.

Improper or bimodal miRNA expression patterns may reflect underlying biological heterogeneity. ROC-derived indices can therefore help prioritize such candidates for downstream evaluation, including literature-based support and independent validation. For example, miR-15a-5p has been associated with tumor-suppressive activity when underexpressed, whereas miR-93-5p has been linked to oncogenic activity when overexpressed; their appearance among candidate improper miRNAs is consistent with these dual functional roles.

Literature-based plausibility assessment of miRNAs in CC data

To evaluate the biological relevance of improper miRNAs identified in the CC dataset, we conducted a literature-based plausibility assessment of the top candidates. Several highly ranked miRNAs identified by our approach have well-established roles in CC progression, supporting the biological plausibility of the ROC-based findings. Notably, tumor-suppressor miRNAs such as let-7d, miR-30c-5p, miR-382-5p, and miR-145 are consistently reported to be downregulated in cervical tumors.[39-42] These miRNAs regulate key oncogenic pathways involved in cell-cycle control, survival signaling, angiogenesis, and metastasis; thus, their reduced expression is consistent with loss of tumor-suppressive function.

In contrast, several oncogenic miRNAs highlighted by our analysis, including miR-301a, miR-19a/b, miR-93-5p, miR-106b, and miR-15a-5p, have been repeatedly implicated in CC as drivers of proliferation, invasion, and resistance to apoptosis.[43-47] These miRNAs typically target key tumor-suppressor pathways, including PTEN/PI3K, TGF-β signaling, and p53-related stress responses, thereby promoting epithelial-mesenchymal transition, angiogenesis, and metastatic potential. Being classified as improper miRNAs suggests that their dysregulation—particularly deviations toward both low and high expression levels—may contribute to malignant behavior, consistent

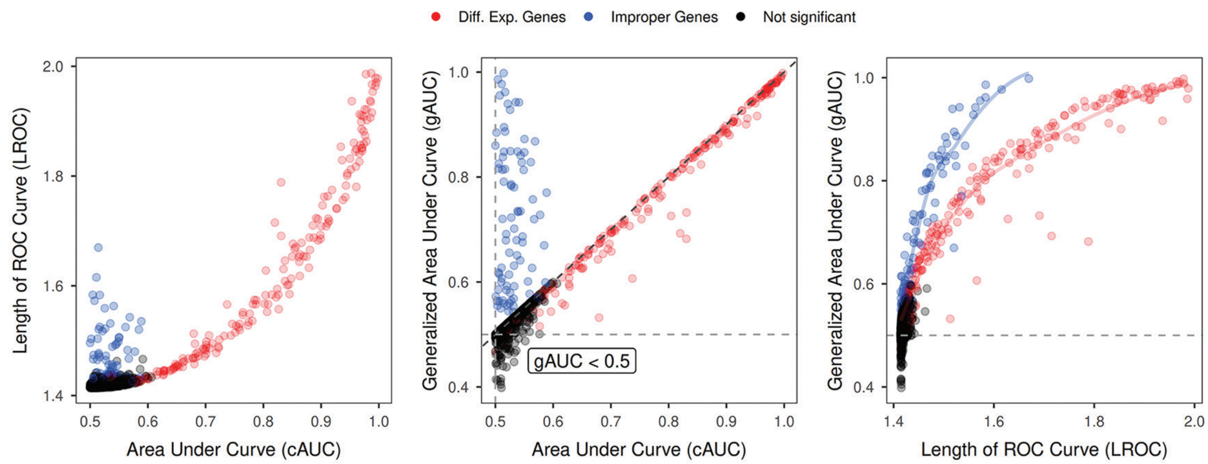

FIG. 8. Scatterplot matrix for the relationships among three receiver operating characteristic-derived diagnostic indices of simulated gene expression data. Points are colored according to the true status of the genes in the simulated dataset. ROC, receiver operating characteristic.

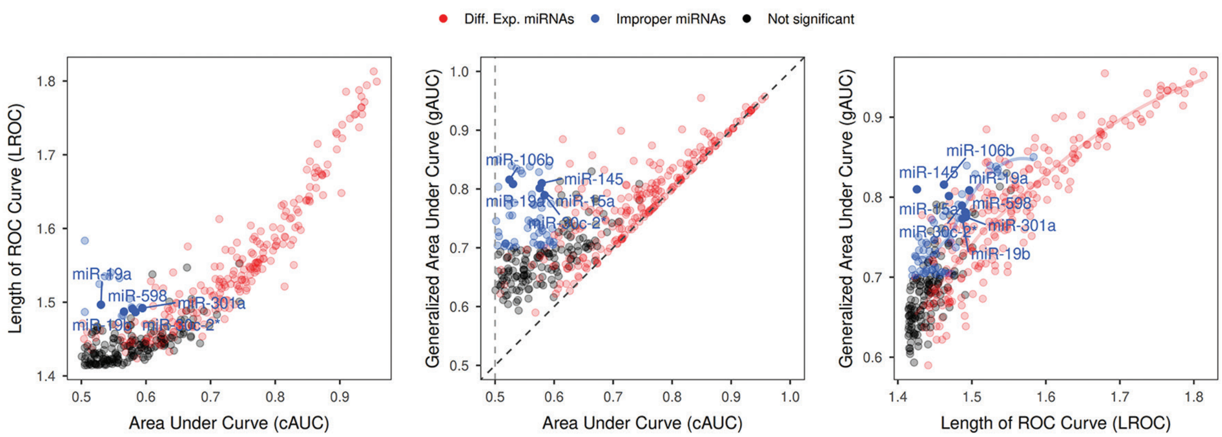

FIG. 9. Scatterplot relationships among ROC-based metrics for the cervical cancer dataset. Points are colored according to the differential expression results of DESeq2 and heuristic thresholds for ROC-derived metrics. ROC, receiver operating characteristic. cAUC, classical area under the curve.

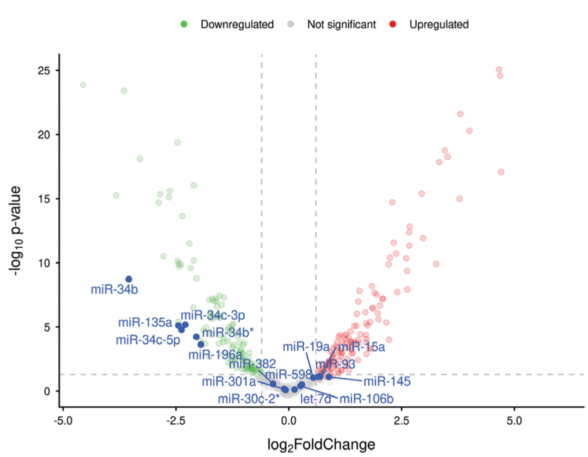

FIG. 10. Volcano plot of DESeq2 results for the cervical cancer dataset. The top-ranked candidate miRNAs exhibiting improper expression profiles, as identified by the gAUC and LROC metrics, are annotated in the figure. Some of these improper miRNA profiles fall within the upregulated or downregulated regions, which are also detected by the DESeq2 framework. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve.

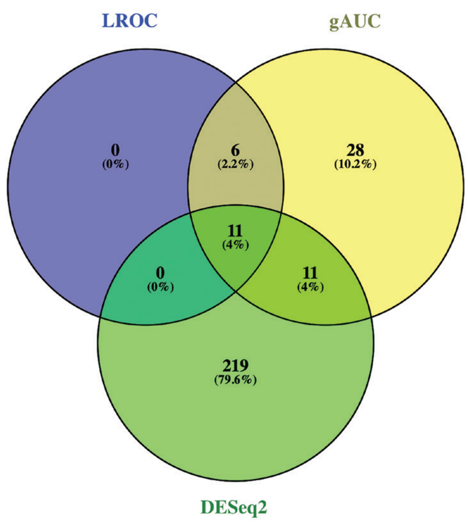

FIG. 11. Venn diagram showing the overlap between miRNAs identified by DESeq2, LROC, and gAUC in the cervical cancer dataset. Number of selected features by DESeq2 is defined using the criteria, e.g., BH-adjusted p value < 0.05 and absolute $\log_2$ fold-change > 0.6. LROC, length of the receiver operating characteristic; gAUC, generalized area under the curve.

with experimentally observed context-dependent or dual functional roles. Together with additional less-characterized candidates (e.g., miR-598), these findings suggest that improper miRNA profiles may reflect biologically meaningful heterogeneity in CC and warrant further functional investigation.

We acknowledge several limitations of this work. The primary limitation is that gAUC and LROC are used as heuristic screening scores without p values or formal type I error control; therefore, results depend on post-hoc threshold selection and do not support strict inferential claims. Additional limitations include the restriction to binary outcomes (two-group comparisons) and the lack of an inferential calibration framework for the ROC-derived indices (e.g., null-only behavior, specificity, and nominal type I error or FPR control), beyond the empirical FPR sensitivity benchmarks provided in the [Supplementary Materials (Section B)]. These issues should be addressed in future methodological work, for example, by developing data-driven thresholding strategies or formal test statistics with known sampling distributions.

In conclusion, improper, non-monotonic expression profiles in RNA-Seq data can be difficult to prioritize using mean-shift–focused DE testing alone. In both the simulation studies and the CC case

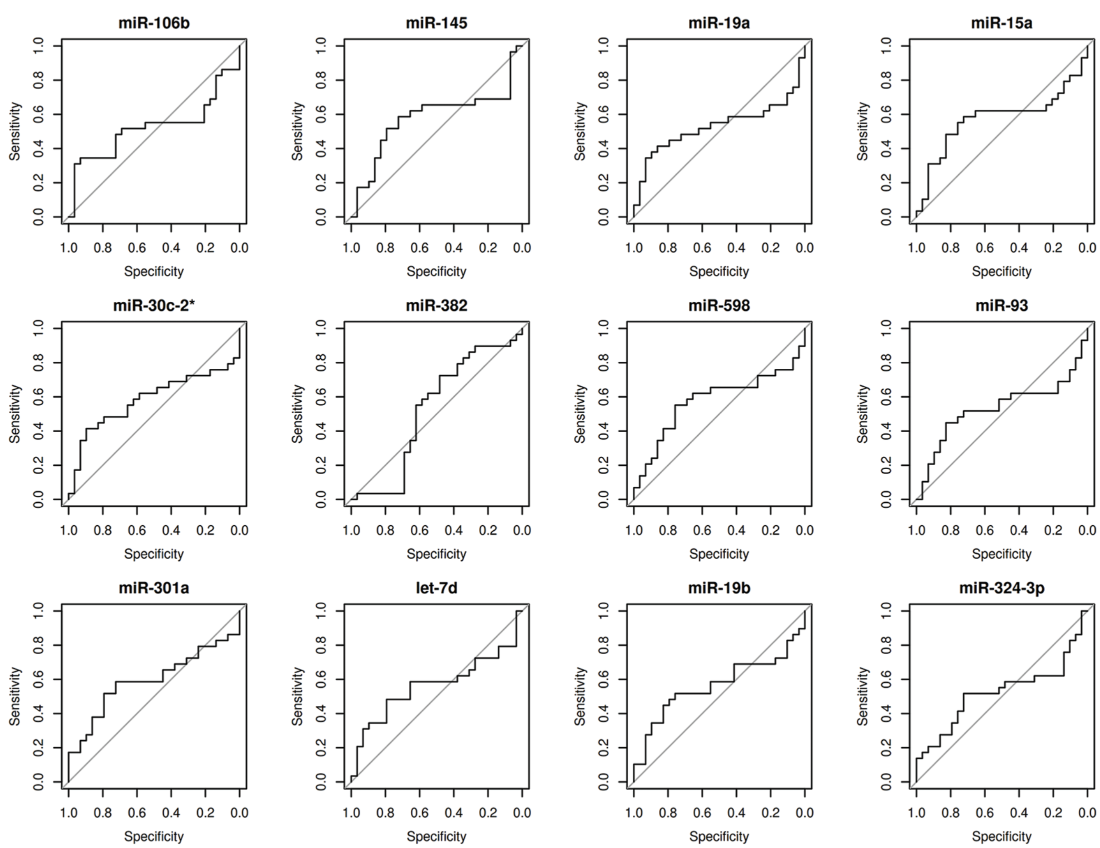

FIG. 12. Receiver operating characteristic (ROC) curves for the top improper miRNAs identified in the cervical cancer dataset. Each panel represents an individual miRNA exhibiting a non-monotonic expression profile, with both low and high expression levels contributing to discrimination between tumor and control samples.

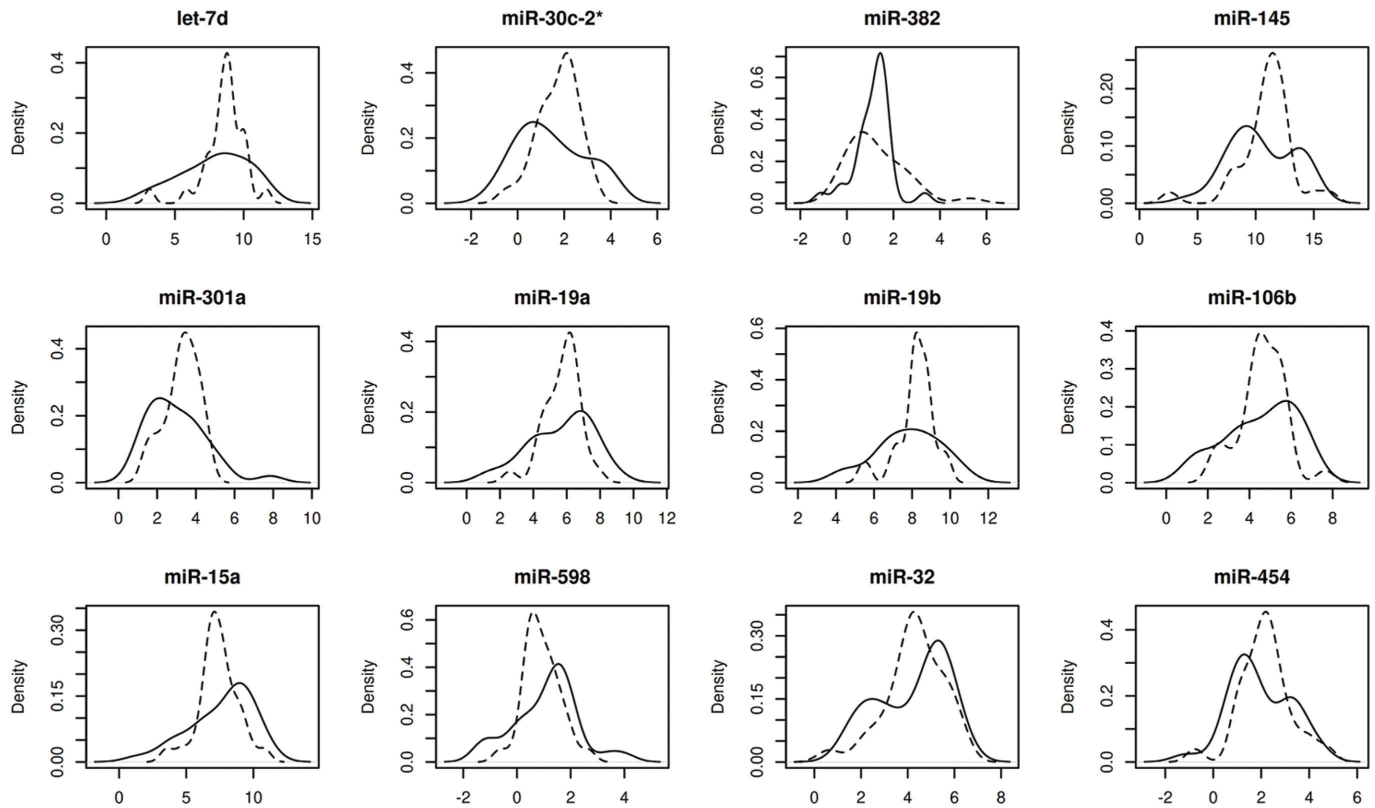

FIG. 13. Density plot of distributions for improper miRNA expression profiles detected by ROC-derived indices. These miRNAs are a subset of the top candidates supported by the literature.
ROC, receiver operating characteristic.

study, gAUC (and, in some settings, LROC) provided complementary screening rankings that may help flag candidate non-monotonic features for follow-up analysis. Because these indices rely on heuristic thresholds and lack error control, the findings should be interpreted as exploratory and require independent validation.

Ethics Committee Approval: Not applicable.

Informed Consent: Not applicable.

Data Sharing Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

Authorship Contributions: Concept- D.G.; Design- M.B.G., E.Ö., D.G.; Supervision- D.G.; Funding- M.B.G., Data Collection or Processing- M.B.G., A.D.Ö., H.F.K.; Analysis and/or Interpretation- M.B.G., E.Ö., Ü.E., H.F.K., D.G.; Literature Review- M.B.G., E.Ö., Ü.E., A.D.Ö., H.F.K., D.G.; Writing- M.B.G., E.Ö., Ü.E., A.D.Ö., H.F.K., D.G.; Critical Review- M.B.G., E.Ö., A.D.Ö., D.G.

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Supplementary Materials Section A and B: https://balkanmedicaljournal.org/img/files/balkan-2026-1-309-supplementary.pdf

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Balkan Med J 2026;43:343-344

Half-and-Half Nails and Terry's Nails: Distinct Nail Manifestations of the Same Underlying Chronic Kidney Disease

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A 68-year-old man undergoing maintenance hemodialysis, who was routinely screened for skin abnormalities, presented with nail discoloration. He had developed chronic renal failure secondary to type II diabetes mellitus and had been on hemodialysis for the past 3 years. Examination revealed that all fingernails and toenails were affected. The distal portion of the nails (50–60% of the nail length) showed a sharply demarcated brownish band. In most nails, particularly the fingernails, the lunula was clearly visible. Additionally, mild longitudinal ridging was observed. A diagnosis of half-and-half nails was made (Figure 1a). The patient was unaware of any nail abnormalities.

Another 53-year-old man had a long history of glomerulonephritis that had progressed to chronic renal insufficiency. Renal replacement therapy had been initiated 42 months before presentation, and he had been maintained on hemodialysis up to the time of examination. All of the patient's fingernails and toenails were affected. The nails appeared white and pale, with only a narrow distal band of normal pinkish-brown coloration. The nail plates exhibited a ground-glass appearance. The lunula was absent in all nails (Figure 1b). A diagnosis of Terry's nails was established. Although the patient could not specify the onset of the nail changes, he associated their appearance with the period of hemodialysis treatment.

Cutaneous manifestations are common in patients with chronic kidney disease (CKD). These include CKD-associated pruritus, xerosis, skin discoloration, calciphylaxis, acquired perforating dermatosis, pseudoporphyria, bullous disease of hemodialysis, and nephrogenic systemic fibrosis.[1] In contrast, nail abnormalities associated with CKD

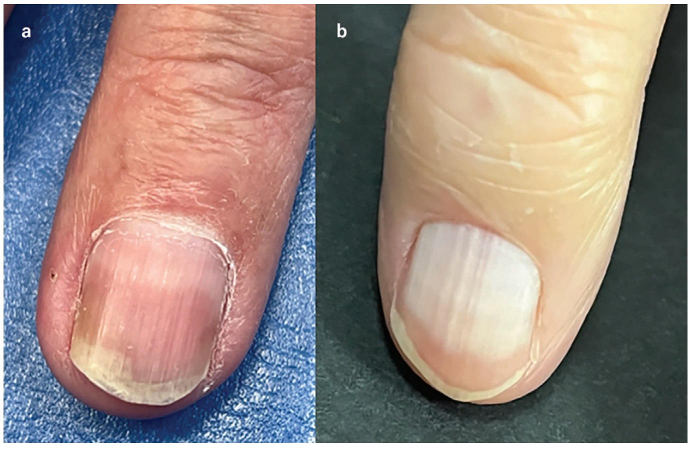

FIG. 1. Nail abnormalities in maintenance hemodialysis patients: half-and half nails (a) and Terry's nails (b).

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have received relatively little attention in the literature.[2] The most frequently reported CKD-related nail abnormalities include half-and-half nails, absent lunula, Terry's nails, splinter hemorrhages, and Muehrcke's nails.[2,3]

Half-and-half nails, also known as Lindsay's nails, are characterized by a sharp chromatic demarcation between the proximal whitish nail bed and a distinctly demarcated reddish-brown to brown discoloration of the distal nail bed, typically involving 20–60% of the nail length.[3,4] Although strongly associated with renal disease, half-and-half nails are not specific to CKD and have been observed in other systemic conditions and in healthy older adults.[3] The pathogenesis of half-and-half nails remains incompletely understood. However, the distal nail bed discoloration is believed to be predominantly pigmentary in nature, resulting from increased melanin deposition rather than purely vascular changes. This hypothesis is supported by the clinical observation that the distal reddish-brown to brown band does not blanch under pressure, indicating a lack of significant vascular contribution. Uremic toxins and metabolic disturbances associated with CKD may stimulate melanocyte activity in the nail matrix or nail bed, leading to melanin accumulation in the distal portion of the nail.[3,4] Terry's nails present as bilaterally symmetrical, diffuse whitening of the nail bed involving all fingernails, with a narrow distal band measuring approximately 1–2 mm that retains a pinkish color. The lunula is absent.[3] The characteristic diffuse white, ground-glass appearance of the nail bed in Terry's nails is thought to result primarily from vascular and hemodynamic alterations, including reduced perfusion of the nail bed, increased connective tissue, and changes in nail bed vascularity. These hemodynamic alterations are consistent with the systemic conditions most commonly associated with Terry's nails, including CKD, chronic liver disease, congestive heart failure, and aging.[3,5,6] In conclusion, careful nail examination may provide valuable clinical information in patients with CKD. The present cases illustrate that the same underlying disease, CKD, can manifest with different nail abnormalities, reflecting distinct pathophysiological mechanisms. Awareness of this variability may help clinicians recognize subtle signs of renal disease and underscores the importance of nail examination in patients with CKD.

Informed Consent: Informed consent was obtained from both patients for the presentation and publication of their clinical data and photographs.

Authorship Contributions: Concept- M.S., B.L., J.C.S.; Design- M.S., B.L., J.C.S.; Supervision- A.K.J. M.K. J.C.S.; Data Collection or Processing- M.S., B.L.; Analysis and/or Interpretation- M.S., B.L.; Literature Review- M.S., B.L.; Writing- M.S.; Critical Review- M.S., B.L., A.K.J. M.K., J.C.S.

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Massive Resting Shunt on c-TCD: A Clue to Pulmonary Arteriovenous Malformation in a Young Stroke Patient

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A 17-year-old female presented with migraine with visual aura. Brain magnetic resonance imaging demonstrated a hyperintense area on diffusion-weighted imaging with corresponding hypointensity on the ADC adjacent to the right cerebral aqueduct [also hyperintense on fluid-attenuated inversion recovery, indicating an acute ischemic infarction (Figure 1a-c)]. No traditional cerebrovascular risk factors were identified. A right-to-left shunt was considered owing to the combination of early-onset cryptogenic stroke, migraine with aura, and a family history of persistent epistaxis affecting her mother, grandfather, and all four maternal aunts.[1,2]

Contrast-enhanced transcranial Doppler (c-TCD) was performed following a standardized protocol.[3] Briefly, an agitated saline contrast agent (9 mL of isotonic saline, 1 mL of air, and a drop of the patient's blood) was administered intravenously. The bilateral middle cerebral arteries were monitored for 25 s after injection, both at rest and during a Valsalva maneuver. The right-to-left shunt was graded based on the maximum microbubble count using a standard four-level classification: small (≤ 10 microbubbles); moderate, (11–20 microbubbles); large, (> 20 microbubbles); and "curtain-like" pattern (uncountable microbubbles).[4] c-TCD examination revealed a "curtain-like" pattern of microbubbles in bilateral middle cerebral arteries immediately and persistently after contrast injection at rest, which was unaffected during the Valsalva maneuver (Figure 1d-i). This finding is highly suggestive of an extracardiac shunt, typically pulmonary arteriovenous malformation (PAVM).[5] Subsequent computed tomography (CT) angiography confirmed a PAVM in the right lower lobe (Figure 1j and k). Genetic testing identified a pathogenic frameshift variant in *ENG* (c.1097_1100delACGC), confirming the diagnosis of hereditary hemorrhagic telangiectasia (HHT) type 1.[2]

Paradoxical embolism through the PAVM is the most plausible mechanism for cerebral infarction in this young patient. First, her risk of paradoxical embolism score of 9 indicates a high-risk of stroke from paradoxical embolism.[5] c-TCD showed immediate, persistent "curtain-like" microbubbles without Valsalva augmentation, a pattern characteristic of PAVM, which was confirmed by CT angiography.[6] Second, although she had migraine with aura, the periaqueductal infarct location was atypical for primary migrainous infarction.[7] Third, small vessel disease was highly unlikely given her age and absence of cerebrovascular risk factors or corresponding imaging results.[8]

The confirmation of a pathogenic *ENG* variant and diagnosis of HHT type 1 have direct and profound implications for the patient and her family. Patient management includes treating beyond the symptomatic PAVM to encompass systematic screening for other potential vascular malformations (e.g., cerebral and hepatic) and lifelong multidisciplinary follow-up according to HHT guidelines.[2] Importantly, this genetic diagnosis requires a change in the family care model, allowing for genetic testing for at-risk first-degree relatives. Given the 50% transmission risk of this autosomal dominant disorder, preconception genetic counseling is imperative.[9] This highlights the essential role of genetic testing in shifting HHT management from reactive to preventive.

After diagnostic assessment, the patient underwent embolization of the PAVM in the right lower lobe. After the procedure, her migraine with visual aura disappeared completely, and she did not experience any neurological symptoms. The one-year radiological follow-up confirmed complete occlusion of the malformation.

HHT, an autosomal dominant genetic disorder, affects approximately 1 in 5000 individuals and is defined by vascular malformations that might lead to complications, such as paradoxical embolization.[2] PAVM screening in patients and their first-degree relatives is crucial because untreated PAVMs are associated with risk of stroke and other severe complications.[10] Early detection allows for procedures such as embolization to prevent recurrent paradoxical emboli. This case highlights the significance of using c-TCD for PAVM screening in young patients experiencing a stroke, particularly those with migraines, unexplained neurological symptoms, or a relevant family history.

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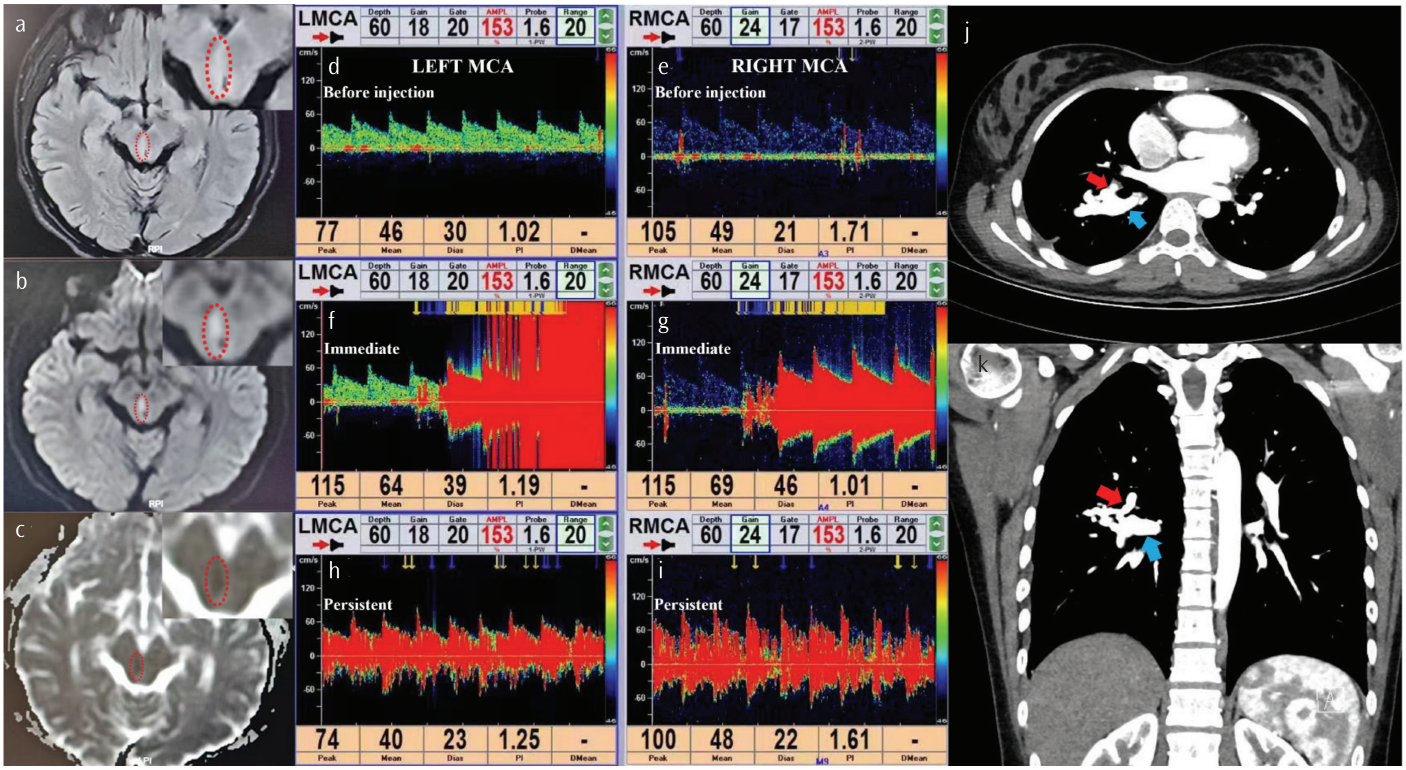

FIG. 1. Brain MRI, c-TCD, and CT angiography. (a-c) Brain MRI of a right periaqueductal acute infarct (outlined by red dashed lines; magnified views in insets). Signal characteristics: subtle hyperintensity on fluid-attenuated inversion recovery (a), hyperintensity on diffusion-weighted imaging (b), and hypointensity on ADC (c). (d-i). Bilateral MCA c-TCD waveforms at baseline (d, e), immediately after (f, g), and persistently after contrast injection at rest (h, i), showing an immediate and persistent "curtain-like" microbubble pattern. (j, k) CT angiography confirms a right lower lobe pulmonary arteriovenous malformation, showing early contrast filling within the malformation (red arrow) and dilated draining pulmonary veins (blue arrow). MRI, magnetic resonance imaging; CT, computed tomography; c-TCD, contrast-enhanced transcranial Doppler.

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Digital Health and Remote Patient Monitoring in Heart Failure: From Pathophysiology to Value-Based Care in Emerging Health Systems

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Heart failure (HF) is one of the most complex, dynamic, and resource-intensive cardiovascular syndromes worldwide. It is associated with high morbidity, frequent hospitalizations, and substantial healthcare expenditures.[1] Despite major advances in pharmacological and device-based therapies, HF management remains largely reactive and symptom-driven rather than proactive. Moreover, adherence to guideline-directed therapy by both physicians and patients remains suboptimal.

Pathophysiological processes, including neurohormonal activation, often precede increases in intracardiac filling pressures and ultimately lead to congestion. Gradual alterations in autonomic tone may manifest as changes in heart rate, heart rate variability, blood pressure, sleep quality, and physical activity well before clinical deterioration becomes apparent. In addition, atrial fibrillation and malignant ventricular arrhythmias may occur abruptly in the setting of hemodynamic and autonomic stress.[2] Early detection of these physiological changes, prior to the onset of symptoms, could potentially prevent hospital admissions. However, current clinical practice remains predominantly symptom-driven and does not adequately reflect the dynamic biological progression of HF, resulting in delayed intervention and preventable decompensation.

HF is particularly well suited for longitudinal, physiology-driven surveillance. Accordingly, digital health technologies and remote patient monitoring (RPM) have emerged as promising strategies to identify early signs of deterioration—such as changes in hemodynamics, cardiac rhythm, autonomic balance, activity patterns, or patient-reported symptoms—and to enable timely intervention. When supported by adequate technical infrastructure and HF-trained healthcare professionals, these approaches may increase time spent at home and reduce HF-related hospitalizations.[3]

A growing body of evidence indicates that technology alone does not improve outcomes in HF; rather, clinical benefit depends on integrating remote monitoring into structured care pathways that clearly define responsibilities and enable proactive intervention. Earlier physician-led telemonitoring systems, such as Telemedical Interventional Monitoring in HF (TIM-HF), did not demonstrate a significant mortality benefit in ambulatory HF, highlighting the limitations of remote monitoring when workflow integration and response intensity are insufficient.[3] In contrast, TIM-HF2 incorporated daily transmission of home-based vital parameters and symptom assessments to a centralized telemedical center with predefined escalation pathways. This approach reduced the primary composite endpoint—days lost due to unplanned cardiovascular hospitalization or all-cause death—and was associated with lower all-cause mortality.[4]

Unlike earlier telemonitoring studies characterized by passive data transfer and heterogeneous clinical response pathways, TIM-HF2 evaluated a structured "remote patient management" model that combined daily physiological and symptom surveillance with a clearly accountable clinical process. Patients transmitted home-based measurements (e.g., body weight, blood pressure, and heart rate) along with standardized symptom and health status data to a central telemedical center staffed by trained healthcare professionals under physician supervision. The telemedical team systematically reviewed incoming data, contacted patients when alerts or concerning trends were identified, coordinated medication adjustments and care modifications with treating physicians, and escalated cases to HF specialists according to predefined criteria. In TIM-HF2, remote patient management significantly reduced the primary composite endpoint (percentage of days lost due to unplanned cardiovascular

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hospitalization or all-cause death) compared with usual care. All-cause mortality was numerically lower (hazard ratio approximately 0.70), although cardiovascular mortality did not reach conventional statistical significance. These findings underscore a key principle: remote monitoring improves outcomes when embedded within an integrated, connected-care model in which continuous interaction between patients and healthcare professionals transforms physiological signals into timely, actionable therapeutic decisions. In contrast, deploying technology without a clearly defined response system is unlikely to yield meaningful clinical benefit.[4]

Among patients with limited digital literacy, regular telephone-based follow-up achieved outcomes comparable to—or in some cases better than—those observed with app-based system.[5] This observation further emphasizes the essential role of human interaction in digitally enabled care models.

Patients with implantable devices, including pacemakers, implantable cardioverter-defibrillators (ICDs), cardiac resynchronization therapy (CRT) devices, and pulmonary artery pressure (PAP) sensors, can benefit from high-resolution, device-based monitoring.

Invasive, device-based diagnostic techniques provide precise longitudinal data, enabling detailed correlations with HF pathophysiology, particularly congestion. Continuous monitoring of intracardiac or pulmonary pressures via implantable PAP sensors allows clinicians to adjust diuretic and vasodilator therapy—either escalating or de-escalating treatment—before overt symptoms arise. In the monitoring of PAP in patients with chronic-HF randomized trial, PAP-guided care significantly improved health-related quality of life and reduced HF hospitalizations compared with standard care in a contemporary European cohort.[6] Importantly, the benefit of structured monitoring is not limited to invasive devices. The fully published TIM-HF2 demonstrated that a structured remote patient management program, combining daily non-invasive home measurements (e.g., body weight, blood pressure, and heart rate) with standardized symptom and status assessments reviewed by a dedicated telemedical center with predefined escalation pathways, significantly reduced the primary composite endpoint (days lost due to unplanned cardiovascular hospitalization or all-cause death) and was associated with a lower all-cause mortality signal than usual care.[7] Collectively, these studies support a core principle: technology delivers clinical value only when integrated into a clear workflow that transforms physiological signals into timely therapeutic interventions and fosters sustained patient–clinician interaction.

Implanted electronic cardiac devices, including ICDs and CRT systems, also provide extensive diagnostic data on cardiac rhythm, physical activity, thoracic impedance, and autonomic tone. Detection of atrial fibrillation, ventricular arrhythmias, or device alerts necessitates timely clinical action to mitigate HF progression. When combined, these parameters can be monitored longitudinally to predict impending deterioration.[8] Effective utilization of these techniques relies on expert receiving centers, ideally employing interdisciplinary teams with expertise in both device management and HF care. These centers must maintain patient accessibility during the day and provide targeted therapeutic guidance.

For patients without implantable devices, PAP sensors, or deviceless remote monitoring. However, individual, single metrics are insufficient on their own; clinical utility depends on interpreting multiparametric data within a structured workflow and linking it to timely therapeutic interventions. Simple, cost-effective measures such as daily body weight, pulse, blood pressure, and symptom scores remain valuable. Remote consultations via platforms such as Teams or Zoom can further aid assessment of congestion through visual inspection, artificial intelligence (AI)–based voice analysis, and dyspnea scoring.

Wearable devices and smartphone cameras can be used for photoplethysmography (PPG) to estimate heart rate, detect irregular rhythms, evaluate peripheral perfusion, and assess surrogate markers of autonomic balance. Advanced camera-based PPG applications allow facial analysis without additional hardware, reducing barriers to participation and enhancing patient engagement. Bed– or smartphone–based ballistocardiography provides additional insights into cardiac mechanical activity and volume status. Voice-based biomarkers also offer a promising avenue; subtle changes in speech characteristics, vocal endurance, and respiratory modulation may indicate pulmonary congestion or fatigue before clinical symptoms are apparent.[9,10] Early data suggest that voice analysis may enable remote detection of HF worsening, providing a low-burden, patient-friendly signal that is particularly advantageous for elderly patients or those with limited digital literacy. These approaches may be especially relevant in middle-income countries or rural settings.[11] Importantly, some algorithms can also detect atrial fibrillation, a common trigger of decompensation and HF hospitalization.

Consequently, non-invasive, “deviceless” monitoring technologies are playing an increasingly important complementary role in the management of HF. A low-cost, multimodal strategy represents the near-future standard, while more sophisticated approaches to digital signal analysis are becoming progressively important for early detection and intervention.

Techniques in AI and machine learning can facilitate the integration of large numbers of heterogeneous physiological signals into individualized risk trajectories, enabling personalized HF management. One promising paradigm is “management by exclusion,” in which patients who remain within their unique physiological limits can be safely deprioritized, allowing clinical attention to focus on those who fall outside these limits. This approach is particularly advantageous in staff-limited settings.[12] Properly designed digital systems, rather than adding cognitive burden, can effectively triage risk, minimize clinician workload, and support proactive decision-making. Crucially, AI should function as a clinical decision-support tool while maintaining clinician oversight and accountability.

Evidence from both clinical trials and real-world implementations indicates that technology alone does not improve outcomes. Meaningful clinical benefit arises from integrating digital tools into connected-care models, where patient–healthcare professional

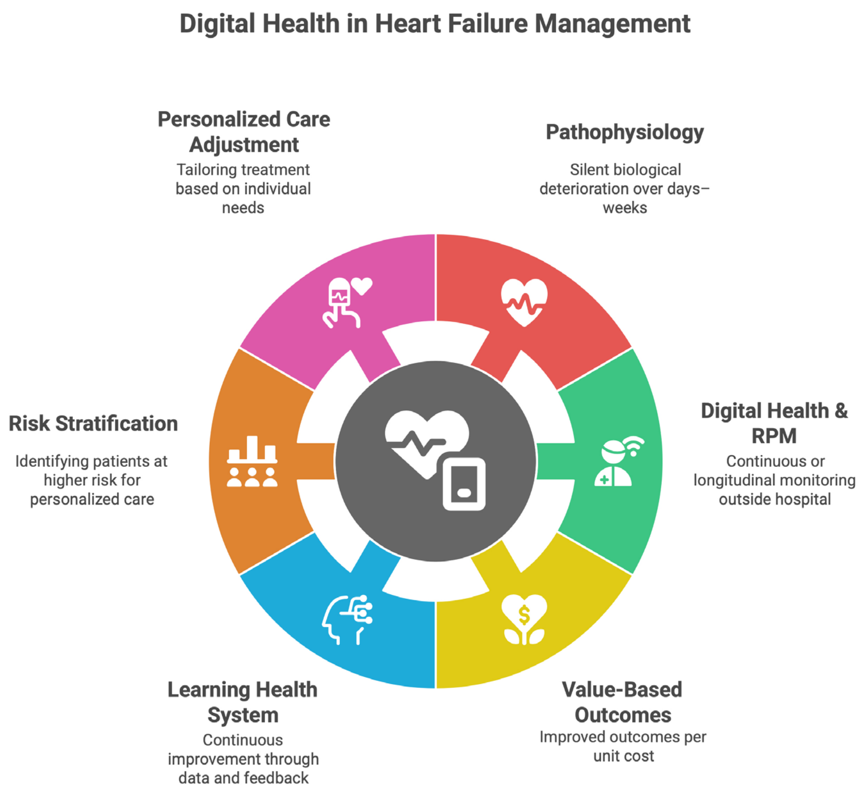

FIG. 1. Digital health–enabled heart failure management framework. The schematic depicts how digital health and remote patient monitoring (RPM) translate heart failure pathophysiology into value-based care. Silent biological deterioration occurring over days to weeks (hemodynamic congestion, autonomic imbalance, arrhythmia risk) is documented through ongoing or longitudinal monitoring outside the hospital. Digital data integration facilitates risk stratification, personalized treatment adjustments, and connected care models that enable earlier intervention. This is an iterative process that leads to a learning health system in which clinical outcomes and healthcare efficiency are enhanced and value-based outcomes are maximized through reduced hospitalizations and improved resource use.

interaction is timely, continuous, guided by defined escalation pathways, and grounded in shared decision-making. The human relationship remains essential: continuous communication, patient and family education, and trust are critical for engagement and adherence, particularly in chronic conditions such as HF (Figure 1).

The future of HF care lies in tailoring interventions to individual patients, delivered through variable intensity modalities rather than a single universal monitoring model. In low-resource settings, a practical initial approach is a cost-effective (RPM) template that incorporates widely available, non-invasive parameters such as daily body weight, blood pressure, heart rate/pulse, symptom scores, and systematic phone-based follow-up, with occasional video consultations when feasible. This model is scalable, suitable for elderly patients with limited digital literacy, and capable of detecting early signs of congestion or rhythm instability before hospitalization becomes necessary. Successful scaling of digital HF care in regions such as Eastern Europe, the Caucasus, and Central Asia requires prioritization of clinician and HF nurse training, clear escalation pathways, and integration of (RPM) into routine workflows rather than operating as parallel “technology-only” processes. Ideally, a pragmatic randomized controlled trial comparing this simplified remote monitoring approach with standard episodic care should be conducted, with outcomes aligned with value-based healthcare metrics, including HF hospitalizations, total bed days, quality of life, and cost-effectiveness. From a policy perspective, this strategy targets the primary cost driver in HF—hospital admissions—by shifting care upstream from reactive inpatient rescue to proactive HF prevention. Ultimately, digital health should be viewed not as an expensive technological tool, but as a scalable, connected-care method capable of enhancing quality and efficiency in short-staffed or rural settings, while preserving the fundamental human elements of continuous dialogue and shared decision-making.

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Real-World Diagnostic and Therapeutic Insights in Transthyretin Cardiac Amyloidosis: Experience from the Black Sea Region of Türkiye

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Transthyretin cardiac amyloidosis (ATTR-CM) is an increasingly recognized cause of heart failure with preserved ejection fraction (HFpEF) in older adults.[1,2] However, real-world data from middle-income countries remain limited.[3] In Türkiye, diagnostic pathways generally follow international guideline recommendations, but access to disease-modifying therapies is strongly influenced by national reimbursement rules, which require specific biomarker thresholds and defined disease stages.[4] Tafamidis, the first approved transthyretin stabilizer, has demonstrated benefit in clinical trials,[5] yet its real-world use and impact under these constraints have not been fully described. In this letter, we report the diagnostic characteristics and twelve-month clinical outcomes of patients with ATTR-CM in the Black Sea region of Türkiye, comparing those treated with tafamidis with untreated patients in routine clinical practice.

We prospectively followed 120 consecutive adults with ATTR-CM diagnosed and managed at two tertiary centers in the Black Sea region between 2023 and 2025. Diagnosis was based on a guideline-aligned non-invasive algorithm, which included echocardiography with strain analysis, serum and urine immunofixation with free light chain assays to exclude light chain amyloidosis, and technetium-labeled bone scintigraphy. Genetic testing was performed to distinguish wild-type from hereditary transthyretin disease.[6] All patients were discussed in a multidisciplinary team. Tafamidis eligibility and reimbursement followed National Social Security Institution criteria for ATTR-CM, which require patients to have NT-proBNP levels > 600 pg/mL, New York Heart Association functional class I–III with a six-minute walk distance > 100 m, left ventricular wall thickness ≥ 12 mm, no prior heart or liver transplantation, preserved renal function (estimated glomerular filtration rate ≥ 30 mL/min/1.73 m^2), and a modified body mass index (BMI) (BMI × serum albumin) > 600.

Among the cohort, 57 patients received tafamidis, while 63 remained untreated despite confirmed ATTR-CM diagnosis. Baseline clinical characteristics were largely comparable between the two groups, including age, sex distribution, functional status, comorbidities, and echocardiographic findings, although untreated patients tended to have higher natriuretic peptide concentrations (Table 1). Several diagnostic "red flag" features were frequently observed, such as atypical HFpEF presentations, discordance between electrocardiographic voltage and left ventricular wall thickness, and intolerance to commonly used rate-limiting agents. A geographical pattern was also evident: patients residing farther from tertiary centers generally experienced longer delays in diagnosis, highlighting the impact of regional accessibility on timely disease recognition.

The primary outcome was the composite of cardiovascular death or heart failure hospitalization over twelve months. Tafamidis use was associated with a lower risk of this composite endpoint compared with no tafamidis (log-rank $p = 0.021$; hazard ratio, 0.66; 95% confidence interval: 0.46–0.95), as illustrated in the Kaplan–Meier curve in Figure 1. Secondary outcomes, including heart failure hospitalization and intensification of outpatient diuretic therapy, also favored the tafamidis group. These associations remained consistent in prespecified adjusted analyses using inverse probability of treatment weighting and in sensitivity analyses that accounted for competing risks and restricted mean survival time at twelve months. Over 12 months of follow-up, the composite endpoint occurred in 28.6% of untreated patients and 14.0% of those receiving tafamidis, corresponding to an absolute risk reduction of 14.5% and a number needed to treat of 7 to prevent one additional composite event.

These findings align with emerging international real-world data from European, Japanese, and Hellenic cohorts, which consistently demonstrate that tafamidis treatment is associated with fewer clinical events and more stable functional status in patients with ATTR-CM.[7-10] At the same time, our results highlight several context-specific challenges in Türkiye. First, the positive correlation between distance to the hospital and time to diagnosis suggests that patients in rural areas reach specialist care later, despite presenting with characteristic

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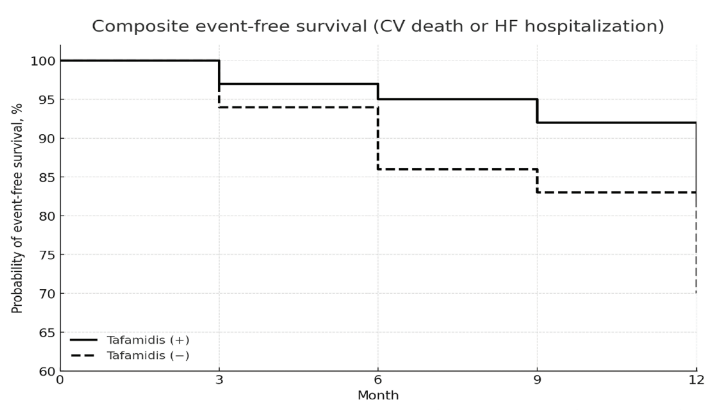
TABLE 1. Baseline Clinical and Echocardiographic Characteristics of Patients with ATTR-CM.

Parameters	All patients (n = 120)	Tafamidis (-) (n = 63)	Tafamidis (+) (n = 57)	*p* value
Age, years	77.1 ± 7.7	78.7 ± 7.5	76.8 ± 8.1	0.051
Gender; male, n (%)	77 (65)	40 (63)	37 (64)	0.614
Residence: urban, n (%)	85 (70)	46 (73)	39 (69)	0.208
HFpEF, n (%)	99 (83)	51 (81)	48 (85)	0.201
wt-ATTR, n (%)	114 (95)	59 (94)	55 (96)	0.472
hATTR, n (%)	6 (5)	2 (3)	4 (7)	0.067
Body mass index, kg/m^2	26.1 (24.8–28.1)	25.9 (24.1–29.3)	26.2 (23.8–28.1)	0.318
NYHA class, n (%)				
I-II	89 (74)	46 (73)	43 (75)	0.336
III	29 (24)	15 (24)	14 (25)	0.659
IV	2 (1)	2 (3)	0	0.101
Six-Minute Walk test, mt	189 (149–221)	192 (152–252)	187 (137–231)	0.061
Ischemic heart disease, n (%)	31 (26)	17(27)	14 (25)	0.517
Hypertension, n (%)	75 (62)	40 (64)	35 (61)	0.258
Diabetes mellitus, n (%)	19 (16)	10 (16)	9 (15)	0.604
Hyperlipidemia, n (%)	33 (28)	16 (25)	17 (30)	0.172
Stroke, n (%)	35 (29)	20 (31)	15 (26)	0.084
Malignancy, n (%)	5 (4)	3 (4)	2 (3)	0.378
Carpal tunnel syndrome, n (%)	42 (35)	20 (31)	22 (38)	0.103
Lumbar canal stenosis, n (%)	9 (7)	5 (8)	4 (7)	0.303
Peripheral neuropathy, n (%)	10 (8)	5(8)	5 (9)	0.541
Rhythm, n (%)				
Sinus rhythm	42 (35)	24 (38)	18 (32)	0.083
Atrial fibrillation	72 (60)	39 (62)	33 (57)	0.152
Pacemaker	13 (11)	8 (12)	5 (9)	0.138
QRS duration, ms	130 ± 12	132 ± 17	128 ± 16	0.205
Laboratory parameters				
Hemoglobin, gr/dL	13.3 (12.5–14.3)	13.1 (11.9–14.2)	13.2 (12.1–14.4)	0.092
NT-pro-BNP, pg/mL	3502 (1194–4891)	3854 (3059–5003)	3280 (1202–4483)	0.003
High-sensitivity troponin T, ng/L	59 (31–79)	61 (45–108)	58 (40–77)	0.059
eGFR, mL/min/1.73 m^2	46 (33–57)	43 (31–52)	47 (36–55)	0.061
Serum albumin	3.83 ± 0.61	3.95 ± 0.46	3.74 ± 0.61	0.059
Echocardiography				
LV end-diastolic diameter, mm	45 (39–51)	46 (41–53)	43 (39–48)	0.528
Interventricular septum thickness, mm	15 (13–18)	16 (14–20)	15(14–19)	0.269
Ejection fraction, %	50 (44–57)	50 (41–58)	51 (45–57)	0.674
Absolute global longitudinal strain (%)	11.6 (9.0–14.0)	10.9 (7.6–11.7)	11.9 (9.4–14.4)	0.107
Pulmonary artery pressure, mm/hg	31 (22–36)	33 (26–39)	30 (21–35)	0.058
Aortic stenosis	16 (13)	9 (14)	7 (12)	0.136

TABLE 1. Continued.

Parameters	All patients (n = 120)	Tafamidis (-) (n = 63)	Tafamidis (+) (n = 57)	*p* value
Medication, n (%)				
RAAS blockers	41 (34)	20 (32)	21 (36)	0.162
Beta-blocker	18 (15)	9 (14)	9 (16)	0.783
Calcium channel blocker	39 (33)	21 (34)	18 (31)	0.479
Diuretics	110 (91)	58 (92)	52 (91)	0.375
SGLT-2 inhibitors	88 (73)	47 (74)	41 (72)	0.571
Antiplatelet	39 (32)	20 (32)	19 (33)	0.394
Oral anticoagulant	84 (70)	44 (69)	40 (70)	0.475
Indicators of clinical worsening during follow-up				
Inability to continue tafamidis therapy, n (%)	20 (17)	14 (22)	6 (10)	0.003
Hospitalization for cardiovascular causes, n (%)	9 (7)	6 (9)	3 (5)	0.025
All-cause mortality, n (%)	19 (16)	14 (22)	5 (8)	0.001
Composite endpoint (CV death or HF hospitalization), n (%)	19 (16)	13 (20)	6 (9)	0.030
Clinical outcomes during 12-month follow-up				
Inability to continue tafamidis therapy, n (%)	5 (4)	-	5 (9)	NA
Hospitalization for cardiovascular causes, n (%)	13 (11)	9 (14)	4 (7)	0.003
All-cause mortality, n (%)	15 (13)	10 (16)	5 (9)	0.005
Composite endpoint (CV death or HF hospitalization), n (%)	26 (22)	18 (28)	8 (14)	0.001

Data are presented as mean ± standard deviation, median with interquartile range (IQR), or number (percentage), as appropriate. Comparisons between groups were performed using the χ^2 test. The composite endpoint includes cardiovascular death or hospitalization for heart failure. ATTR-CM, transthyretin cardiac amyloidosis; hATTR, hereditary ATTR; wt-ATTR, wild-type ATTR; HFpEF, heart failure with preserved ejection fraction; eGFR, estimated glomerular filtration rate; NA, not applicable; NT-proBNP, N-terminal pro–B-type natriuretic peptide; NYHA, New York Heart Association; RAAS, renin–angiotensin–aldosterone system; LV, left ventricle; CV, cardiovascular death; HF, heart failure.



	Months				
Numbers at risk	0	3	6	9	12
Tafamidis (+)	57	54	52	51	50
Tafamidis (−)	63	59	56	53	48

FIG. 1. Kaplan-Meier curves for composite event-free survival (cardiovascular death or heart failure hospitalization) in patients with and without tafamidis.

CV, cardiovascular death; HF, heart failure; HR, hazard ratio; CI, confidance interval.

diagnostic red flags.[4,11,12] Second, a substantial proportion of patients who fulfilled diagnostic criteria could not initiate tafamidis because they did not meet strict reimbursement thresholds based on natriuretic peptide levels and disease stage, or because they had advanced heart failure, active malignancy, or preferred not to start long-term therapy.[13] Taken together, these observations indicate that both system-level and patient-level factors may restrict timely initiation of disease-modifying treatment, even when clinical suspicion and diagnostic pathways are appropriately established.

This letter has several limitations. The findings are based on two tertiary centers in a single region, with a modest sample size and a twelve-month follow-up. Treatment allocation was non-randomized and partly influenced by reimbursement criteria; thus, residual confounding may persist. Therefore, the observed associations should not be interpreted as evidence of causal benefit.

Nonetheless, this regional experience provides a concise snapshot of how ATTR-CM is diagnosed and managed in routine practice in Türkiye. Tafamidis treatment was associated with fewer early composite clinical events over twelve months in patients who could initiate therapy, whereas many others were denied access due to strict biomarker thresholds or late presentation. These findings underscore a nationwide need to reduce diagnostic delays, expand access to non-invasive diagnostic tools, and revise reimbursement criteria to align treatment availability with guideline-based recommendations. Simultaneously, ongoing clinician education and structured collaboration between tertiary centers and peripheral hospitals will be crucial to ensure that advances in amyloidosis care translate into equitable outcomes across the country.

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Ethics Committee Approval: The study protocol was approved by the Ordu University Non-Interventional Scientific Research Ethics Committee (approval number: 296, date: 12.09.2025).

Informed Consent: Written informed consent to participate was obtained from all patients included in this study.

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Conflict of Interest: The authors declare that they have no conflict of interest.

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A Case of Submandibular Ectopic Parathyroid Adenoma: Diagnostic Challenges and Clinical Insights

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We report a case of a left submandibular ectopic parathyroid adenoma (EPA) with a tortuous diagnostic course, highlighting potential misdiagnoses and key diagnostic strategies. A 51-year-old male office worker presented with an incidentally discovered left submandibular mass persisting for 6 months and a newly detected thyroid nodule for 1 month. Initially, the submandibular mass was misdiagnosed as reactive lymphadenopathy at a local clinic, where only physical examination was performed without imaging or serological tests. Five months later, another hospital misdiagnosed the same mass as a schwannoma based on neck ultrasound (Figure 1a), during which a TI-RADS 4A nodule in the upper pole of the right thyroid was incidentally identified. Contrast-enhanced computed tomography (CT) of the neck (Figure 1b) prompted referral for surgical evaluation.

The patient was referred to our thyroid surgery clinic due to the right upper thyroid TI-RADS 4A nodule, seeking simultaneous resection of both lesions. Comprehensive evaluation revealed elevated serum calcium (2.79 mmol/L; reference range, 2.10–2.55 mmol/L), decreased serum phosphorus (0.70 mmol/L; reference range, 0.81–1.45 mmol/L), and markedly elevated intact parathyroid hormone (iPTH, 547 pg/mL; reference range, 15–65 pg/mL), suggestive of primary hyperparathyroidism (PHPT). ^{99}PTH, methoxyisobutylisonitrile (MIBI) scintigraphy showed no abnormal uptake (Figures 1c, d). Ultrasound-guided fine-needle aspiration biopsy (FNAB) revealed epithelioid cells arranged in nested and sheet-like patterns, consistent with parathyroid adenoma, confirming left submandibular EPA. Genetic testing for *MEN1*, *CDC73*, and *AIP* genes was negative. Because FNAB for parathyroid lesions is rarely performed in our practice, we did not measure needle washout PTH, relying solely on cytological examination.

The patient underwent resection of the left submandibular EPA combined with right thyroid nodule excision. A 6-cm transverse incision was made along the suprasternal skin crease to facilitate exploration of remaining parathyroid glands and thyroid nodule removal. After incising the skin and subcutaneous tissue, the left sternocleidomastoid muscle was retracted laterally to expose the left internal jugular vein and common carotid artery. Dissection along the common carotid artery revealed a mass measuring approximately 2 cm in diameter inferior to the left carotid bifurcation, adjacent to the left submandibular gland (Figures 1e, f). The mass was carefully mobilized to avoid injury to the vagus nerve, carotid artery, and internal jugular vein, while preserving the vascular pedicle of the upper thyroid pole. The mass was completely resected without iatrogenic injury. Three normal parathyroid glands (right superior, right inferior, and left inferior) were identified; no left superior parathyroid tissue was found in its normal anatomical location, indicating that the ectopic mass originated from the left superior parathyroid gland. The resected specimen appeared grayish-red and well-circumscribed (Figures 1g, h). Intraoperative frozen section suggested “parathyroid hyperplastic tissue, adenoma not excluded”; the right thyroid nodule was confirmed as a benign nodular goiter. Postoperatively, serum calcium and iPTH rapidly normalized (2.33 mmol/L and 13 pg/mL, respectively) without hypocalcemic symptoms. Final histopathology confirmed EPA (Figure 1i) and a benign thyroid nodule. Two-year follow-up showed no recurrence.

PHPT is a metabolic disorder caused by excessive PTH secretion. EPA accounts for 10–22%[1] of PHPT cases, commonly occurring in intrathyroidal and mediastinum;[2,3] mediastinal sites; submandibular ectopy is extremely rare (4–6% of EPA cases,[1,2] rendering), making preoperative diagnosis challenging. In this case, the sequential misdiagnoses were due to limited awareness of this rare ectopic site and over-reliance on empirical judgment, compounded by the patient’s asymptomatic presentation (without typical PHPT manifestations such as bone pain).[4] MIBI scintigraphy, the first-line modality for localizing hyperfunctioning parathyroid tissue, may yield false-negative results in rare ectopic locations due to low tracer uptake or reduced lesion activity.[5] For MIBI-negative cases, ^{18}F-fluorocholine positron emission tomography (PET)-CT is a valuable adjunct, offering a reported detection sensitivity of 93.7% and superior target-to-background ratio.[5] Although FNAB combined with serological tests confirmed the diagnosis in this patient, PET-

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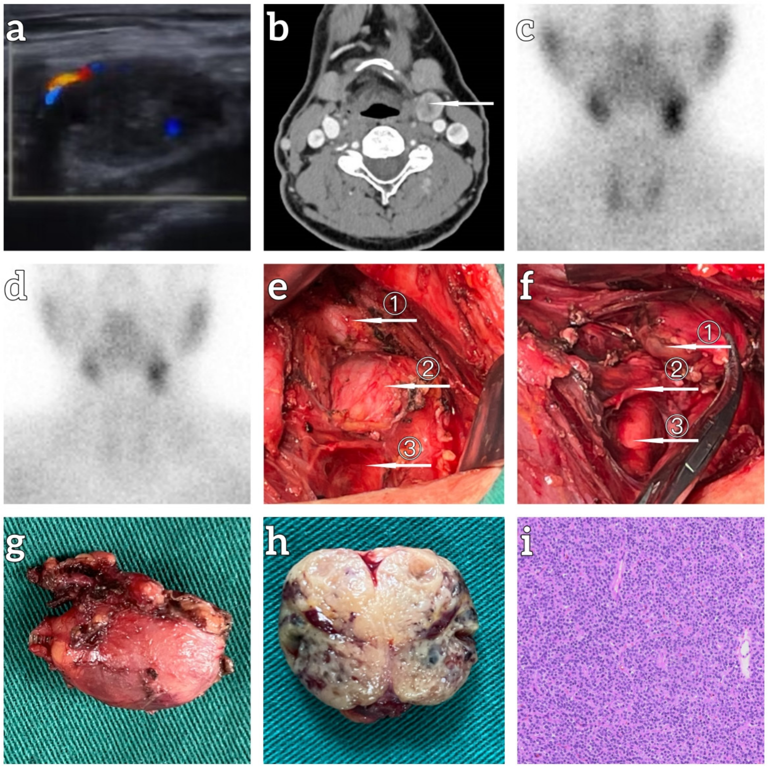

FIG. 1. Radiological, intraoperative, and histopathological findings of the case. Color Doppler ultrasound image showing a hypoechoic nodule adjacent to the left submandibular gland and medial to the left common carotid artery (a). Contrast-enhanced neck CT image: the arrow indicates the mass with moderately heterogeneous enhancement (b). ^{99mTc}-MIBI scintigraphy images (c, d): Early phase (30 minutes post-injection) (c); delayed phase (2 hours post-injection) (d); no abnormal radioactive tracer accumulation was detected in the neck or mediastinum. Intraoperative exposure of the ectopic parathyroid adenoma: arrow ① = submandibular gland; arrow ② = ectopic parathyroid adenoma; arrow ③ = left common carotid artery (e). Intraoperative mobilization of the adenoma: arrow ① = ectopic parathyroid adenoma; arrow ② = superior thyroid vein; arrow ③ = left common carotid artery (f). Gross appearance of the resected adenoma specimen (g). Cut surface of the adenoma showing grayish-red solid tissue (h). Postoperative histopathological image (hematoxylin and eosin staining) (i). CT, computed tomography, MIBI, methoxyisobutylisonitrile.

CT can provide precise anatomical localization and metabolic information for lesions with occult locations or indeterminate FNAB results, guiding surgical planning.

For neck masses in special locations (e.g., carotid sheath, submandibular region), FNAB effectively differentiates parathyroid lesions from neurogenic tumors, lymph node disorders, and salivary gland neoplasms, significantly improving diagnostic accuracy.[6] When ectopic parathyroid tissue is suspected. PTH measurement in needle washout fluid[1] can further confirm ectopic parathyroid tissue; in this case, cytology alone was sufficient due to our insufficient awareness of this approach. Delayed diagnosis of EPA may lead to complications such as osteoporosis, pathological fractures, and nephrocalcinosis, complicating surgical[4] management. Therefore, comprehensive evaluation—including serology, FNAB, and appropriate imaging—is crucial for diagnosing neck masses in rare locations.

Informed Consent: Written informed consent was obtained from the patient for the publication of this case report and any accompanying images. The patient has been informed that their personal identifying information will be anonymized to protect privacy.

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